

# Online Appendix

## Collected and Discarded: Family Guarantees in U.S. Mortgage Underwriting

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This appendix is referenced from the main paper and is not meant to be read on its own. Section and exhibit numbers are prefixed OA throughout, so a reference to Table OA1 is always to this document and a reference to Table B1 is always to the printed appendix.

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## OA1 Proofs of Theoretical Results

*Proof of Proposition 1 (Which Margin the Guarantee Reaches).* Both parts rest on one feature of (2): the equity test's only menu-dependent object is  $\overline{\text{ltv}}^k$ , an institutional constant that is a function of neither rate nor income, so the guarantee cannot slacken the equity test and weakly tightens it.

(i) Fix a household in  $\mathcal{E}(\overline{\text{ltv}}^{co})$ , i.e. with  $a < (1 - \overline{\text{ltv}}^{co})P$ . Its equity test fails identically in  $r_{co}$ , so by (2)  $\mathcal{F}_{co} = \emptyset$  for every  $r_{co} \geq 0$  and its contribution to  $A_{co}$  is zero at every rate. Differentiating the zero function gives  $\partial A_{co}/\partial r_{co}|_{\mathcal{E}} = 0$ . The economics is that the two tests are conjunctive: a price operates inside the feasible set and cannot enlarge it along a dimension the price does not enter.

(ii) Raising  $\overline{\text{ltv}}^{co}$  to  $\overline{\text{ltv}}^{co} + \varepsilon$  lowers the deposit threshold to  $(1 - \overline{\text{ltv}}^{co} - \varepsilon)P$ , so the households newly passing the equity test are those with  $a \in [(1 - \overline{\text{ltv}}^{co} - \varepsilon)P, (1 - \overline{\text{ltv}}^{co})P]$ , a set of positive measure for every  $\varepsilon > 0$  whenever  $F_a(\cdot | y)$  has positive density there. They are admitted if they also clear the income test at the enlarged loan  $L^* = \overline{\text{ltv}}^{co}P$ , which is the qualification in the statement; the mass of admissions is then strictly increasing in  $\overline{\text{ltv}}^{co}$ .  $\square$

*Proof of Proposition 2 (The Unpriced Guarantee).* Write  $K \equiv 1 + \rho + g > 0$ ,  $\mu \equiv \mu_{co}(y)$  and  $D \equiv \mu + (1 - \mu)\bar{p} = \mu(1 - \bar{p}) + \bar{p}$ , so that  $1 + r_{co} = K/\mu$  by (3) and  $1 + \tilde{r}_{co} = K/D$  by (6). Since  $\bar{p} > 0$  and  $\mu < 1$ ,  $D > \mu$ .

(i) Directly,

$$W(y) = \frac{K}{\mu} - \frac{K}{D} = K \frac{D - \mu}{\mu D} = K \frac{(1 - \mu)\bar{p}}{\mu[\mu + (1 - \mu)\bar{p}]} > 0, \quad (\text{OA1})$$

which is (7). Monotonicity in coverage: with  $u \equiv (1 - \mu)\bar{p}$ , (OA1) reads  $W = (K/\mu) u/(\mu + u)$ , and  $u \mapsto u/(\mu + u)$  is strictly increasing, while  $u$  is strictly increasing in  $\bar{p}$ . Monotonicity in risk: from  $W = K(1/\mu - 1/D)$  and  $dD/d\mu = 1 - \bar{p} \geq 0$ ,

$$\frac{dW}{d\mu} = K \left[ -\frac{1}{\mu^2} + \frac{1 - \bar{p}}{D^2} \right] < 0,$$

because  $D \geq \mu$  and  $1 - \bar{p} \leq 1$  with at least one inequality strict for  $\bar{p} > 0$ . Hence  $W$  is strictly decreasing in  $\mu_{co}$ , i.e. strictly increasing in the pool's default risk  $1 - \mu_{co}$ . Finally, substituting the posted rate  $K/\mu$  into (5) gives the lender's realised profit on a cosigned loan,

$$\Pi_{co} = D \frac{K}{\mu} L^* - K L^* = K L^* \frac{D - \mu}{\mu} = W(y) D L^* > 0,$$

so the rule-constrained price is not a break-even price on the cosigned menu, and the rent it collects is proportional to the wedge.

(ii)  $\tilde{r}_{co}(y) < r_{so}(y)$  iff  $K/D < K/\mu_{so}$ , i.e. iff  $D > \mu_{so}$ , i.e. iff  $\mu + (1 - \mu)\bar{p} > \mu_{so}$ , i.e. iff  $\bar{p} > (\mu_{so} - \mu)/(1 - \mu) = \bar{p}^*$ , which is (8). Under  $\mu < \mu_{so} < 1$  the numerator is positive and strictly smaller than the denominator, so  $\bar{p}^* \in (0, 1)$  and the threshold is interior: some but not all coverage levels deliver a negative fair premium.

(iii) The posted premium is  $r_{co} - r_{so} = K(1/\mu - 1/\mu_{so}) > 0$  under  $\mu < \mu_{so}$ , while by (ii) the fair premium is negative for  $\bar{p} > \bar{p}^*$ . The two therefore carry opposite signs on that range: the rule charges a positive premium on a pool that, netting the guarantee the lender actually receives, warrants a negative one.  $\square$

*Proof of Lemma 1 (Informative Participation).* By (9),  $q_\theta(y, y_p) = F_\eta(\beta \Delta V(y, \theta) - (1 - \pi_\theta) \Delta u_p)$ . Since  $F_\eta$  is strictly increasing on its support ( $f_\eta > 0$  on  $\mathbb{R}$ ),  $q_G > q_B$  holds if and only if the argument is larger for the Good type:

$$\beta[\Delta V(y, G) - \Delta V(y, B)] > (1 - \pi_G)\Delta u_p - (1 - \pi_B)\Delta u_p = -\Delta\pi \Delta u_p. \quad (\text{OA2})$$

From the decomposition (B.13),  $\Delta V(y, \theta) = \pi_\theta G + (1 - \pi_\theta)\Delta \underline{u}$ , so

$$\Delta V(y, B) - \Delta V(y, G) = (\pi_B - \pi_G)G + (\pi_G - \pi_B)\Delta \underline{u} = \Delta\pi (\Delta \underline{u} - G). \quad (\text{OA3})$$

Substituting (OA3) into (OA2) and dividing by  $\Delta\pi > 0$ , the condition  $q_G > q_B$  becomes  $\Delta u_p > \beta(\Delta \underline{u} - G)$ . Writing  $G = u_{co} - u_{so}$  from (B.14) and  $\Delta \underline{u} = \underline{u}_{co} - \underline{u}_{so}$ , I have  $\Delta \underline{u} - G = (u_{so} - u_{co}) + (\underline{u}_{co} - \underline{u}_{so})$ , which is exactly the bracket in Assumption 1. Hence  $q_G(y, y_p) > q_B(y, y_p)$  for all  $(y, y_p)$  is equivalent to Assumption 1, and holds under it.

Integrating the pointwise inequality over  $F(y_p | y)$  preserves it, so  $\bar{q}_G(y) > \bar{q}_B(y)$ . By (10),  $\Lambda_1(y)$  is strictly increasing in the likelihood ratio  $\bar{q}_G/\bar{q}_B > 1$ , giving  $\Lambda_1(y) > \lambda(y)$ ; the analogous argument with complements  $1 - \bar{q}_\theta$  gives  $\Lambda_0(y) < \lambda(y)$ . Therefore  $\bar{\pi}_1(y) > \bar{\pi}(y) > \bar{\pi}_0(y)$  for all  $y$ .  $\square$

*Proof of Lemma 2 (Wealth Effect).* Hold the rate schedule fixed (its income variation is second order under (A3c)). With  $c_{co}(y) = y - (1 + r_{co})L^*$ ,  $H^* = \alpha L^*$  fixed,  $c_{so}(y) = (1 - \overline{\text{dti}})y$  and  $H_{so}(y) = \alpha \overline{\text{dti}}y / (1 + r_{so})$ , the repayment surplus  $G(y) = u(c_{co}(y), H^*) - u(c_{so}(y), H_{so}(y))$  has derivative

$$\frac{dG}{dy} = u_c(c_{co}, H^*) - \left[ (1 - \overline{\text{dti}}) u_c(c_{so}, H_{so}) + \frac{\alpha \overline{\text{dti}}}{1 + r_{so}} u_h(c_{so}, H_{so}) \right]. \quad (\text{OA4})$$

At  $y^{**} = (1 + r_{co})L^* / \overline{\text{dti}}$  the consumption levels coincide,  $c_{so}(y^{**}) = c_{co}(y^{**})$ , because  $c_{so} - c_{co} = (1 + r_{co})L^* - \overline{\text{dti}}y = \overline{\text{dti}}(y^{**} - y)$ ; the same identity shows  $c_{co}(y) > c_{so}(y)$  for all  $y > y^{**}$ . Moreover  $H_{so}(y^{**}) = \alpha(1 + r_{co})L^* / (1 + r_{so}) > \alpha L^* = H^*$  since  $r_{co} > r_{so}$ , and  $H_{so}(y)$  is strictly increasing while  $H^*$  is constant, so  $H_{so}(y) > H^*$  for all  $y \geq y^{**}$ , with the gap diverging.

Fix  $y > y^{**}$ , so  $c_{co} > c_{so}$  and  $H_{so} > H^*$ . By strict concavity in  $c$  ( $u_{cc} < 0$ ) and the complementarity  $u_{ch} > 0$ ,

$$u_c(c_{co}, H^*) < u_c(c_{so}, H^*) < u_c(c_{so}, H_{so}), \quad (\text{OA5})$$

the first inequality because  $c_{co} > c_{so}$  and the second because  $H_{so} > H^*$ . Substituting (OA5) into (OA4),

$$\frac{dG}{dy} < u_c(c_{so}, H_{so}) - (1 - \overline{\text{dti}})u_c(c_{so}, H_{so}) - \frac{\alpha \overline{\text{dti}}}{1 + r_{so}} u_h(c_{so}, H_{so}) = \overline{\text{dti}} \left[ u_c - \frac{\alpha}{1 + r_{so}} u_h \right]_{(c_{so}, H_{so})}. \quad (\text{OA6})$$

The bracket is the solo borrower's net marginal valuation of an extra dollar of income at the DTI margin: a unit of income buys  $(1 - \overline{\text{dti}})$  units of consumption and  $\alpha \overline{\text{dti}} / (1 + r_{so})$  units of housing, so the bracket is non-positive exactly when

$$\frac{u_h(c_{so}, H_{so})}{u_c(c_{so}, H_{so})} \geq \frac{1 + r_{so}}{\alpha},$$

which is the housing-valuation condition maintained in (A1). (Under Cobb–Douglas preferences with housing utility share  $\theta_u$ , substituting  $u_h/u_c = \frac{\theta_u}{1-\theta_u} c/h$  at the solo bundle  $c_{so} = (1 - \overline{\text{dti}})y$ ,  $H_{so} = \alpha \overline{\text{dti}} y / (1 + r_{so})$  reduces the condition to exactly  $\theta_u \geq \overline{\text{dti}}$ : the utility share of housing weakly exceeds the DTI cap.) Hence the bracket is non-positive for  $y > y^{**}$  and  $dG/dy < 0$ . Equivalently,  $H_{so}(y) \rightarrow \infty$  with  $H^*$  fixed makes the solo bundle strictly dominate the cosign bundle in the housing dimension at a rate that, under (A1), drives  $G$  strictly down. This is the local content of the boundary condition (A2),  $\lim_{y \rightarrow \infty} \Delta V(y, \theta) < 0$ .  $\square$

*Proof of Lemma 3 (Single-Crossing and Cutoff Ordering).* (i) *Single-crossing.* From (B.13),  $\partial \Delta V / \partial \pi_\theta = G(y) - \Delta \underline{u}(y)$ , so

$$\frac{\partial^2 \Delta V}{\partial y \partial \pi_\theta} = \frac{dG}{dy} - \frac{d\Delta \underline{u}}{dy}. \quad (\text{OA7})$$

By Lemma 2,  $dG/dy < 0$  for  $y > y^{**}$ ; and since  $\Delta \underline{u}(y) = u(y - \phi, h_{co}) - u(y - \phi, 0)$  with  $u_{ch} > 0$  and  $h_{co} > 0$ , I have  $d\Delta \underline{u}/dy = u_c(y - \phi, h_{co}) - u_c(y - \phi, 0) \geq 0$ . Hence the cross-partial is strictly negative, as claimed in Lemma 3 (with equality to  $dG/dy$  in the case  $d\Delta \underline{u}/dy = 0$ ). The net gain from cosigning is more steeply decreasing in income for high- $\pi$  (Good) types.

(ii) *Cutoff ordering.* Let  $\tilde{y}_G$  solve  $\Delta V(\tilde{y}_G, G) = 0$ . From (B.13),  $0 = \pi_G G(\tilde{y}_G) + (1 - \pi_G) \Delta \underline{u}$  forces  $G(\tilde{y}_G) < 0 < \Delta \underline{u}$ , hence  $\Delta \underline{u} - G(\tilde{y}_G) > 0$ . By (OA3),  $\Delta V(\tilde{y}_G, B) - \Delta V(\tilde{y}_G, G) = \Delta \pi (\Delta \underline{u} - G(\tilde{y}_G)) > 0$ , so  $\Delta V(\tilde{y}_G, B) > 0$ : at the Good-type cutoff the Bad type strictly prefers cosigning. Since  $\Delta V(y, B)$  is continuous and strictly decreasing through its zero (by (A2) and part (i)), its root  $\tilde{y}_B$  satisfies  $\tilde{y}_B > \tilde{y}_G$ .  $\square$

*Proof of Lemma 4 (Belief-rate fixed point).* *Self-map.*  $\Lambda_1(y; r) \in [0, 1]$  as a probability, so  $\mu_{co} = \pi_B + \Delta \pi \Lambda_1 \in [\pi_B, \pi_G]$  and, by (B.21),  $T_y(r) \in \mathcal{R}$  for every  $r \in \mathcal{R}$ : beliefs are bounded whatever parents do, so the posted rate cannot leave the interval.

*Continuity.* By (A1)  $u$  is  $C^2$ , so  $\Delta V(y, \theta; r, r_{so})$  in (B.13) is continuous in  $r$ ;  $F_\eta$  is a distribution function with a density, so  $q_\theta(y, y_p; r) = F_\eta(a_\theta)$  is continuous in  $r$ ; and since  $q_\theta \in [0, 1]$ , dominated convergence carries continuity to  $\bar{q}_\theta(y; r) = \int q_\theta dF(y_p | y)$ . The denominator of (10) is  $\lambda \bar{q}_G + (1 - \lambda) \bar{q}_B$ , strictly positive because  $F_\eta$  has full support (A3) and, being continuous on the compact set  $\mathcal{R}$ , bounded below by some  $\underline{q} > 0$ . Hence  $\Lambda_1(y; \cdot)$  and therefore  $T_y$  are continuous on  $\mathcal{R}$ .

*Existence.*  $T_y(\cdot) - \text{id}$  is continuous on  $\mathcal{R}$ , weakly positive at  $\min \mathcal{R}$  and weakly negative at  $\max \mathcal{R}$  by the self-map property, so the intermediate value theorem gives an  $r$  with  $T_y(r) = r$ . That  $r$  is (B.16) evaluated at consistent beliefs.

*Uniqueness and convergence.* Differentiating (B.21),

$$T'_y(r) = - \frac{(1 + \rho + g) \Delta \pi}{\mu_{co}(r)^2} \frac{\partial \Lambda_1(y; r)}{\partial r}.$$

From (10),  $\partial \Lambda_1 / \partial r$  is a rational function of  $(\bar{q}_G, \bar{q}_B, \bar{q}'_G, \bar{q}'_B)$  whose denominator is bounded below by  $\underline{q}^2$ , and  $\bar{q}'_\theta(r) = \int f_\eta(a_\theta) (\partial a_\theta / \partial r) dF$  with  $|\partial a_\theta / \partial r| = \beta \pi_\theta u_c(y - (1 + r)L^*, H^*)L^*$  bounded on  $\mathcal{R}$  by (A1). Hence  $|\partial \Lambda_1 / \partial r| \leq C \bar{f}$  for a constant  $C$  depending only on primitives, and  $|T'_y| \leq C' \bar{f}$  with  $C' = (1 + \rho + g) \Delta \pi C / \pi_B^2$ . Under the dispersion condition (A3b),  $\bar{f}$  is small enough that  $C' \bar{f} < 1$ , so  $T_y$  is a contraction on the complete metric space  $\mathcal{R}$ ; Banach's theorem then gives a unique fixed point and convergence of  $r_{n+1} = T_y(r_n)$  from any  $r_0 \in \mathcal{R}$ .  $\square$

*Proof of Proposition 3 (Three-Region PBE).* *Existence and uniqueness of  $y_G^{pool}$ .* Along the pooling schedule  $r_{co}^{pool}(y) = (1 + \rho + g) / \bar{\pi}_1(y) - 1$ , the map  $y \mapsto \Delta V(y, G; r_{co}^{pool}(y), r_{so}(y))$  is continuous (by (A1) and continuity of  $\bar{\pi}_1$ ). By (A2) it is strictly positive as  $y \rightarrow y_F^+$  and strictly negative as  $y \rightarrow \infty$ , so a root exists; by Lemma 3(i) the function is strictly decreasing where it crosses zero, so the root  $y_G^{pool}$  defined in (B.19) is unique.

*Region 1 ( $y < y_G^{pool}$ ).* Here  $\Delta V(y, G) > 0$ , and by Lemma 3(ii),  $\Delta V(y, B) > \Delta V(y, G) > 0$ , so both types choose *co* when  $o = 1$ . The pool contains both types in proportions set by  $\bar{q}_\theta(y)$ , so Bayes' rule (10) gives  $\mu_{co}(y) = \bar{\pi}_1(y)$  and the pooling rate (B.16); this is the rate used to define the region, so beliefs, pricing and sorting are mutually consistent. Parental screening is active with  $q_G > q_B$  by Lemma 1.

*Exit of Good types and the separating belief.* For  $y \geq y_G^{pool}$ , Good types choose *so*. Any borrower observed choosing *co* is therefore a Bad type, so Bayes' rule pins  $\mu_{co}(y) = \pi_B$  and the constant separating rate  $r_{co}^{sep}$  in (B.17); since  $\bar{\pi}_1(y) > \pi_B$ ,  $r_{co}^{sep} > r_{co}^{pool}(y)$ .

*Existence and uniqueness of  $y_B^{sep}$ .* Under the survival condition,  $\Delta V(y_G^{pool}, B; r_{co}^{sep}, r_{so}) > 0$ , while by (A2) the same object is negative as  $y \rightarrow \infty$ ; continuity and the strict monotonicity of  $\Delta V(\cdot, B)$  (Lemma 3(i)) deliver a unique root  $y_B^{sep} > y_G^{pool}$ , as in (B.20). For  $y \in [y_G^{pool}, y_B^{sep})$  Bad types strictly prefer *co* at  $r_{co}^{sep}$  (Region 2); for  $y \geq y_B^{sep}$  even Bad types prefer *so*, so no cosign pool forms (Region 3).

*Off-path beliefs and equilibrium.* Assign the worst-case belief  $\mu_{co} = \pi_B$  to any out-of-equilibrium choice of *co* in Regions 2–3; this is admissible and makes deviation to the cosign menu unprofitable for Good types, so no profitable deviation exists. The cutoffs are uniquely determined by the two monotone crossings, and the belief and rate systems are then pinned by the cutoffs, giving a unique PBE with  $y_F < y_G^{pool} < y_B^{sep} < \infty$  and the schedule (B.22).  $\square$

*Proof of the Two-Region Case (Remark following Proposition 3).* Suppose the survival condition fails, i.e.  $\Delta V(y_G^{pool}, B; r_{co}^{sep}, r_{so}) \leq 0$ . By Lemma 3(i),  $\Delta V(\cdot, B; r_{co}^{sep}, r_{so})$  is strictly decreasing, so for every  $y > y_G^{pool}$  I have  $\Delta V(y, B; r_{co}^{sep}, r_{so}) < \Delta V(y_G^{pool}, B; r_{co}^{sep}, r_{so}) \leq 0$ . Thus once Good types exit at  $y_G^{pool}$ , Bad types are unwilling to cosign at the separating rate they would face, so the cosign pool is empty for all  $y \geq y_G^{pool}$ : Region 2 has zero Lebesgue measure and  $y_B^{sep} = y_G^{pool}$ . The

equilibrium collapses to two regions separated by the single cutoff  $y_G^{pool}$ : both types cosign below it (priced at  $\bar{\pi}_1(y)$ ) and all borrowers go solo above it. Existence and uniqueness of  $y_G^{pool}$  follow exactly as in the proof of Proposition 3.  $\square$

*Proof of Proposition 4 (Pool Quality Decomposition).* Equations (B.24)–(B.25) are the mass-weighted average repayment probabilities of the two pools under the three-region sorting of Proposition 3: the cosign pool is Region 1 (priced at  $\bar{\pi}_1$ ) plus Region 2 (priced at  $\pi_B$ ); the solo pool is the Region-2 block ( $y_G^{pool} \leq y < y_B^{sep}$ , priced at  $\bar{\pi}_2$ , since every Good type and only the *unoffered* Bad types sort there), plus the Region-3 block ( $y \geq y_B^{sep}$ , priced at the unconditional  $\bar{\pi}$  because no cosign menu exists there), plus the low-income solo sub-pool ( $y < y_G^{pool}$ , priced at  $\bar{\pi}_0$ ). Writing  $B \equiv \omega_{so}^2 \int_{y_G^{pool}}^{y_B^{sep}} \bar{\pi} f_{so}^2 + \omega_{so}^3 \int_{y_B^{sep}}^{\infty} \bar{\pi} f_{so}^3$  for the high block evaluated at the benchmark and differencing,

$$\bar{\mu}_{so} - \bar{\mu}_{co} = B + \underbrace{\omega_{so}^2 \int_{y_G^{pool}}^{y_B^{sep}} (\bar{\pi}_2 - \bar{\pi}) f_{so}^2}_{=S} + \omega^{lo} \int_0^{y_G^{pool}} \bar{\pi}_0 f_{so}^{lo} - \omega_{co}^1 \int_0^{y_G^{pool}} \bar{\pi}_1 f_{co}^1 - \omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} \pi_B f_{co}^2. \quad (\text{OA8})$$

On the three interior integrals add and subtract the unconditional benchmark  $\bar{\pi}(y)$ :

$$\omega^{lo} \int_0^{y_G^{pool}} \bar{\pi}_0 f_{so}^{lo} = \omega^{lo} \int_0^{y_G^{pool}} \bar{\pi} f_{so}^{lo} - \underbrace{\omega^{lo} \int_0^{y_G^{pool}} (\bar{\pi} - \bar{\pi}_0) f_{so}^{lo}}_{=D_2}, \quad (\text{OA9})$$

$$-\omega_{co}^1 \int_0^{y_G^{pool}} \bar{\pi}_1 f_{co}^1 = -\omega_{co}^1 \int_0^{y_G^{pool}} \bar{\pi} f_{co}^1 - \underbrace{\omega_{co}^1 \int_0^{y_G^{pool}} (\bar{\pi}_1 - \bar{\pi}) f_{co}^1}_{=D_1}, \quad (\text{OA10})$$

$$-\omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} \pi_B f_{co}^2 = -\omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} \bar{\pi} f_{co}^2 + \underbrace{\omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} (\bar{\pi} - \pi_B) f_{co}^2}_{=A}. \quad (\text{OA11})$$

The three retained  $\bar{\pi}$  integrals combine into the reweighting term  $R = \omega^{lo} \int_0^{y_G^{pool}} \bar{\pi} f_{so}^{lo} - \omega_{co}^1 \int_0^{y_G^{pool}} \bar{\pi} f_{co}^1 - \omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} \bar{\pi} f_{co}^2$ . Collecting gives the identity  $\bar{\mu}_{so} - \bar{\mu}_{co} = B + D_1 + D_2 + S + R + A$ , establishing claim (i).<sup>1</sup>

*Signs.*  $B > 0$  as  $\bar{\pi} > 0$ .  $S > 0$  because  $\Lambda_2(y) > \lambda(y)$  by (B.23): in Region 2 the Bad types holding an offer exit to the cosign menu, so the solo pool they leave behind is strictly better than the unconditional benchmark.  $S$  and  $A$  are the two sides of one exit, and both widen the gap. By Lemma 1,  $\bar{\pi}_1 > \bar{\pi} > \bar{\pi}_0$ , so the integrands of  $D_1$  and  $D_2$  are positive and both terms are negative.  $R$  is *not* signed by the maintained primitives: it compares *mass-weighted* integrals of the same benchmark  $\bar{\pi}$  under the two densities. Although higher-income parents cosigning more readily

<sup>1</sup>  $A$  centres the Region-2 cosign block on the unconditional benchmark  $\bar{\pi}$ , as the other two interior integrals are centred. Centring instead on the pooling belief  $\bar{\pi}_1(y)$  those borrowers would have carried had the Good types not left gives the strictly larger trap loss  $\omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} (\bar{\pi}_1 - \pi_B) f_{co}^2 > A > 0$ , the difference  $\omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} (\bar{\pi}_1 - \bar{\pi}) f_{co}^2$  being a belief-level term that then moves into  $R$ . Neither centring changes the identity or the sign of the trap channel.

tilts the normalised cosign density  $f_{co}^1$  rightward relative to  $f_{so}^{lo}$  (which, with  $\bar{\pi}$  increasing, would sign the comparison of the *normalised* integrals), the masses satisfy  $\omega^{lo} \neq \omega_{co}^1$  in general, and no restriction among the maintained assumptions pins their ratio. The decomposition is an accounting identity; the sign of the overall gap is an empirical question, settled by the NSMO pool-quality evidence rather than by primitives. Finally  $A > 0$  because the Region-2 cosign pool is priced at the worst case  $\pi_B < \bar{\pi}(y)$ : the trap holds those borrowers below the benchmark and so widens the gap. Claim (ii) is immediate from the identity:  $\bar{\mu}_{co} < \bar{\mu}_{so} \iff B + D_1 + D_2 + S + R + A > 0$ , for which  $B + S + A > |D_1| + |D_2| + |R|$  is sufficient. Claim (iii) is (11) evaluated at a given income: where  $\mu_{co}(y) < \mu_{so}(y)$ , the pricing condition (3) gives  $\partial r_k / \partial g = 1/\mu_k(y)$  and hence  $\beta_{rate}^{co}(y) > \beta_{rate}^{so}(y)$  directly. The claim is pointwise and does not extend to the pool averages: Jensen's inequality bounds  $\mathbb{E}[1/\mu_k \mid k]$  below by  $1/\bar{\mu}_k$  for *each pool separately*, and two lower bounds do not order the two expectations. Since  $\mu_{co}$  is degenerate at  $\pi_B$  over Region 2 while  $\mu_{so}$  is not, the pools differ in the dispersion of  $\mu$  and not only in its mean, so  $\bar{\mu}_{co} < \bar{\mu}_{so}$  is consistent with  $\mathbb{E}[1/\mu_{co} \mid co] < \mathbb{E}[1/\mu_{so} \mid so]$ .  $\square$

*Proof of Proposition 5 (Sensitivity Decoupling). Derivation of the rejection-pass-through formula (B.28).* Write  $p_k(g) = \mathbb{E}^k[H_k(DTI_k(g))]$ . A marginal increase  $dg$  shifts every borrower's DTI upward by  $\sigma_k dg$  via (12). Since  $H_k$  is differentiable with bounded derivative and  $\sigma_k$  is integrable (guaranteed by  $F_L(\cdot \mid y)$  admitting a density with finite mean), differentiation under the expectation gives

$$\beta_{rej}^k = \frac{\partial p_k}{\partial g} = \mathbb{E}^k[H'_k(DTI_k) \sigma_k], \quad (\text{OA12})$$

which is (B.28). Under the sharp-cap special case  $H_k = \mathbb{I}\{DTI_k > \overline{\text{dti}}\}$ ,  $H'_k$  is a Dirac mass at  $\overline{\text{dti}}$  and the expression collapses to the classical threshold-density formula  $\mathbb{E}^k[\sigma_k \mid DTI_k = \overline{\text{dti}}] \cdot f_{DTI}^k(\overline{\text{dti}}; g)$ .

*Decoupling.* Dividing the two pools' expressions,

$$\beta_{rej}^{co} < \beta_{rej}^{so} \iff \frac{\mathbb{E}^{co}[H'_{co}(DTI_{co}) \sigma_{co}]}{\mathbb{E}^{so}[H'_{so}(DTI_{so}) \sigma_{so}]} < 1, \quad (\text{OA13})$$

which is the leniency ratio (B.29) that (C2') delivers. It remains to show that the restriction is not vacuous, i.e. that with a *common* enforcement hazard the ranking would run the other way. Each borrower's DTI sensitivity is  $\sigma_k = L/(Y_k \mu_k(y))$ . Under (C1), Proposition 4 gives  $\bar{\mu}_{co} < \bar{\mu}_{so}$ , so the risk-pricing channel  $1/\mu_{co} > 1/\mu_{so}$  raises  $\sigma_{co}$  relative to  $\sigma_{so}$ : with  $H_{co} = H_{so}$  and comparable positioning of the two DTI distributions,  $\mathbb{E}[\sigma_{co}] > \mathbb{E}[\sigma_{so}]$  pushes (OA13) *above* one and  $\beta_{rej}^{co} > \beta_{rej}^{so}$ .

Condition (C2') overturns this through the enforcement margin, and under (B.27) it does so as a threshold in one primitive. Write

$$\Phi(\tau) \equiv \mathbb{E}^{co}[H'(DTI_{co} - \tau \gamma(y_p)) \sigma_{co}], \quad (\text{OA14})$$

so that  $\beta_{rej}^{co} = \Phi(\tau)$  and  $\beta_{rej}^{so} = \Phi_{so} \equiv \mathbb{E}^{so}[H'(DTI_{so}) \sigma_{so}]$ , the latter free of  $\tau$ . Since  $\gamma \geq 0$  and

$H$  is convex on the range of effective ratios,  $H'$  is non-decreasing there, so  $\tau \mapsto H'(DTI_{co} - \tau\gamma)$  is non-increasing pointwise and strictly decreasing wherever  $\gamma > 0$  and  $H$  is strictly convex. As  $\sigma_{co} > 0$ ,  $\Phi$  is continuous and strictly decreasing on  $\{\tau : \gamma > 0\}$ , so

$$\beta_{rej}^{co} < \beta_{rej}^{so} \iff \Phi(\tau) < \Phi_{so} \iff \tau > \tau^*,$$

with  $\tau^*$  the unique root of  $\Phi(\tau) = \Phi_{so}$  when one exists. At  $\tau = 0$  the two hazards coincide, and the previous paragraph showed that (C1) then gives  $\Phi(0) > \Phi_{so}$ ; hence  $\tau^* > 0$  and the threshold is interior. If in addition  $H'(d) \rightarrow 0$  as  $d$  falls below the cap, then  $\Phi(\tau) \rightarrow 0 < \Phi_{so}$ , so  $\tau^*$  is finite and decoupling obtains for every large enough weight. Because  $\Phi$  is strictly decreasing, the leniency ratio (B.29) is strictly decreasing in  $\tau$  and crosses one exactly at  $\tau^*$ , which is what makes the empirical slope comparison of Appendix OA4.9 an estimate of  $\tau$  against  $\tau^*$  rather than a restatement of the conclusion. Note that the sparsity route to (OA13) under the sharp cap (a small cosigned threshold density  $f_{DTI}^{co}(\overline{\text{dti}}; g)$ ; Appendix OA2) is *not* supported by the data, which show cosigned applications concentrated at and above the threshold; leniency, not distance, delivers the insulation.  $\square$

*Proof of Proposition 6 (The Cost of the Split).* (i) From (OA3),  $\Delta V(y, B) - \Delta V(y, G) = \Delta\pi(\Delta u - G)$ , so

$$a_G - a_B = \beta[\Delta V(y, G) - \Delta V(y, B)] + \Delta\pi \Delta u_p(y_p) = \Delta\pi[\Delta u_p(y_p) - \beta(\Delta u - G)], \quad (\text{OA15})$$

positive by Assumption 1. Only the first bracketed term varies with  $y_p$ , and  $\Delta u'_p(y_p) = u'(y_p) - u'(y_p - m) < 0$  by the strict concavity of (A1) — on the saturated region  $\bar{p}(y_p) = 1$ , so the parent's outlay is the whole payment  $m$  and is constant in  $y_p$  — so  $a_G - a_B$  is strictly decreasing in  $y_p$ . Write  $s \equiv -\Delta u'_p(y_p) > 0$ ; then  $a'_\theta(y_p) = (1 - \pi_\theta) s > 0$ , so both arguments rise with parental income, and  $a'_G < a'_B$  because  $\pi_G > \pi_B$ .

Let  $\varphi \equiv f_\eta/F_\eta$ , strictly decreasing by the log-concavity of (A3d). Differentiating  $\log(q_G/q_B) = \log F_\eta(a_G) - \log F_\eta(a_B)$ ,

$$\frac{d}{dy_p} \log \frac{q_G}{q_B} = \varphi(a_G) a'_G - \varphi(a_B) a'_B < 0,$$

since  $a_G > a_B$  gives  $\varphi(a_G) < \varphi(a_B)$  while  $0 < a'_G < a'_B$ : both factors are smaller and all four quantities positive. The two channels reinforce rather than offset — the gap between the types' thresholds narrows, and both sit further into the upper tail of  $F_\eta$  where the screen discriminates less. Finally  $\Lambda_1(y, y_p)$  in (B.30) is a strictly increasing transformation of  $q_G/q_B$ , hence strictly decreasing in  $y_p$ .

(ii)  $\mu_{co}(y, y_p) = \pi_B + \Delta\pi\Lambda_1(y, y_p)$  inherits the strict monotonicity, so  $r_{co}(y, y_p) = (1 + \rho + g)/\mu_{co}(y, y_p) - 1$  is strictly increasing in  $y_p$ . By the law of total probability  $\Lambda_1(y) = \mathbb{E}[\Lambda_1(y, y_p) \mid y, o = 1]$ , so  $\mu_{co}(y) = \mathbb{E}[\mu_{co}(y, y_p) \mid y, o = 1]$  lies strictly between the infimum and supremum of  $\mu_{co}(y, \cdot)$  whenever the conditional distribution of  $y_p$  is non-degenerate. Continuity and strict

monotonicity then give a unique  $y_p^0(y)$  with  $\mu_{co}(y, y_p) \geq \mu_{co}(y)$  as  $y_p \leq y_p^0(y)$ , and hence  $r_{co}(y) \geq r_{co}(y, y_p)$  on the same partition: cosigners below  $y_p^0$  pay more than the rule would charge them on the fuller information set, and those above pay less.

(iii) The cosigned rate enters the parent's problem only through  $\Delta V(y, \theta; r_{co}, r_{so})$ , and  $\partial \Delta V / \partial r_{co} = -\pi_\theta u_c(y - (1 + r_{co})L^*, H^*)L^* < 0$  by (B.13). Since  $F_\eta$  is strictly increasing,  $\partial q_\theta / \partial r_{co} < 0$ . Consider the counterfactual in which the rule is permitted to read  $y_p$  and compare it with pooled pricing at the initial schedule. By (ii) pooled pricing raises the rate faced by every family with  $y_p < y_p^0$  and lowers it for every family with  $y_p > y_p^0$ , and the rate difference  $r_{co}(y) - r_{co}(y, y_p)$  is itself strictly decreasing in  $y_p$ . The ratio of post- to pre-change participation,  $q_\theta(y, y_p; r_{co}(y)) / q_\theta(y, y_p; r_{co}(y, y_p))$ , is therefore strictly increasing in  $y_p$ , crossing one at  $y_p^0$ . A density reweighted by a strictly increasing likelihood ratio dominates the original in the monotone-likelihood-ratio order and hence first-order stochastically, so the conditional distribution of  $y_p$  in the cosign pool shifts to the right. By (i)  $\Lambda_1(y, \cdot)$  is strictly decreasing, so its expectation under the shifted distribution is strictly smaller:  $\Lambda_1(y)$  under pooled pricing is strictly below its  $y_p$ -conditional counterpart. The statement is first-order, evaluated at the initial schedule; it does not require the belief-rate fixed point, which Proposition 3 takes as given.  $\square$

*Proof of Corollary 1 (Screen reversal).* On the saturated region the outlay is  $m$ , fixed, so with  $u$  concave  $\Delta u_p(y_p) = u(y_p) - u(y_p - m) \leq m u'(y_p - m)$ , which tends to 0 as  $y_p \rightarrow \infty$  whenever  $u'(c) \rightarrow 0$ . Taking the limit in (OA15),  $a_G - a_B \rightarrow -\beta \Delta \pi (\Delta \underline{u} - G)$ , which is strictly negative wherever  $\Delta \underline{u} > G$ . That inequality holds at the good-type cutoff: the argument in part (ii) of the proof of Lemma 3 shows  $G(\tilde{y}_G) < 0 < \Delta \underline{u}$ , and it persists above  $\tilde{y}_G$  because  $dG/dy < 0$  there by Lemma 2. Since  $a_G - a_B$  is continuous and strictly decreasing in  $y_p$  by Proposition 6(i), and strictly positive at the lower end of the support by Assumption 1, there is a unique  $\bar{y}_p(y)$  at which it vanishes; above it  $q_G < q_B$ , the likelihood ratio falls below one, and  $\Lambda_1(y, y_p) < \lambda(y)$ .  $\square$

## OA2 Threshold Sparsity under a Sharp Cap: A Lognormal Benchmark

*Status of this appendix.* Under the sharp-cap special case  $H_j = \mathbb{K}\{DTI_j > \kappa\}$ , the leniency condition (C2') of Proposition 5 reduces to a *threshold-sparsity* condition on the density of cosigned applications at the cap. This appendix develops that benchmark analytically. The direct empirical test (Appendix OA4.9) *rejects* sparsity — cosigned applications concentrate at and above the threshold — so the operative route to insulation is the enforcement-leniency channel of (C2'), and the analysis below is retained as the sharp-cap benchmark rather than as the maintained mechanism.

I verify the sparsity condition under a lognormal specification for the joint distribution of  $(L, y, y_p)$ , showing that the extensive-margin insulation channel would dominate the intensive-margin quality channel for a sufficiently large parental income buffer.

**Specification.** Let  $(\log L, \log y, \log y_p)$  be jointly normal, and approximate joint qualifying income  $Y_{co} = y + y_p$  as lognormal, so that  $\log Y_j$  is normal for both menus with  $Y_{so} = y$ .<sup>2</sup> From (12),  $DTI_j = (1 + r_j)L/Y_j$ , hence

$$\log DTI_j = \log(1 + r_j) + \log L - \log Y_j \sim \mathcal{N}(m_j, s_j^2), \quad (\text{OA16})$$

with  $m_j = \log(1 + r_j) + \mu_{\log L} - \mu_{\log Y_j}$  and  $s_j^2 = \text{Var}(\log L - \log Y_j)$ . The lognormal density of  $DTI_j$  at the rejection threshold  $\kappa$  is

$$f_{DTI}^j(\kappa) = \frac{1}{\kappa s_j} \phi(d_j), \quad d_j \equiv \frac{\log \kappa - m_j}{s_j}, \quad (\text{OA17})$$

where  $\phi$  is the standard normal density.

**Closed form for the rejection pass-through.** Because  $\sigma_j = L/(Y_j \mu_j) = DTI_j/[(1 + r_j)\mu_j]$  is proportional to  $DTI_j$ , the sensitivity *conditional on being at the margin* is the constant

$$\mathbb{E}^j[\sigma_j \mid DTI_j = \kappa] = \frac{\kappa}{(1 + r_j)\mu_j}. \quad (\text{OA18})$$

Substituting this and (OA17) into the rejection-pass-through formula (B.28) gives the closed form

$$\beta_{rej}^j = \mathbb{E}^j[\sigma_j \mid DTI_j = \kappa] f_{DTI}^j(\kappa) = \frac{\phi(d_j)}{(1 + r_j)\mu_j s_j}. \quad (\text{OA19})$$

**Verification of (C2).** Dividing the two pools' expressions in (OA19),

$$\frac{\beta_{rej}^{co}}{\beta_{rej}^{so}} = \underbrace{\frac{(1 + r_{so})\mu_{so}s_{so}}{(1 + r_{co})\mu_{co}s_{co}}}_{\text{intensive factor} = O(1)} \cdot \underbrace{\frac{\phi(d_{co})}{\phi(d_{so})}}_{\text{extensive factor}} = O(1) \cdot \exp\left(-\frac{1}{2}(d_{co}^2 - d_{so}^2)\right). \quad (\text{OA20})$$

The intensive factor is bounded: although  $\mu_{co} < \mu_{so}$  raises  $1/\mu_{co}$  (the quality channel of (C1), which on its own would push  $\beta_{rej}^{co}$  up), the ratio remains  $O(1)$  in the parental buffer. The extensive factor is governed by the location shift. Since  $\mu_{\log Y_{co}} = \mathbb{E}[\log(y + y_p)] > \mathbb{E}[\log y] = \mu_{\log Y_{so}}$ , I have

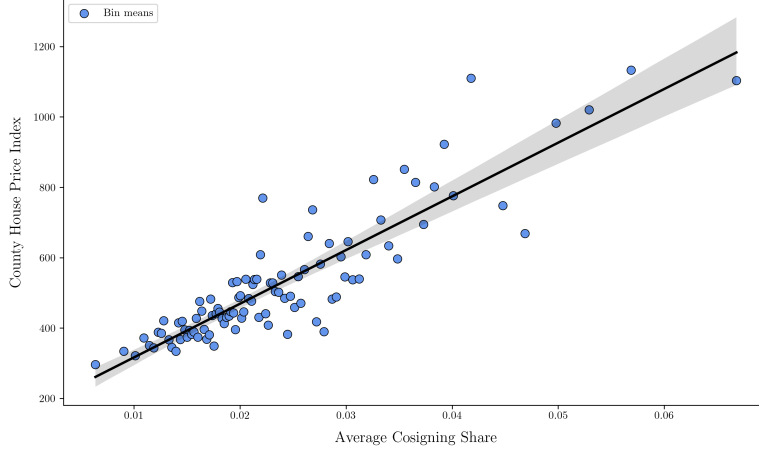
$$m_{so} - m_{co} = (\mu_{\log Y_{co}} - \mu_{\log Y_{so}}) - (\log(1 + r_{co}) - \log(1 + r_{so})) > 0 \quad (\text{OA21})$$

whenever the parental income buffer exceeds the rate wedge, i.e.  $\mu_{\log Y_{co}} - \mu_{\log Y_{so}} > \log \frac{1+r_{co}}{1+r_{so}}$  — which holds for any non-trivial  $y_p$  because the rate wedge is second order. Hence  $m_{co} < m_{so}$ : the cosign DTI distribution is shifted left. With  $\kappa$  in the upper tail ( $\log \kappa > m_j$ , so  $d_j > 0$ ) it follows

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<sup>2</sup>The lognormal closure for the sum  $y + y_p$  is the standard Fenton–Wilkinson approximation; it is exact in the limit where parental income dominates,  $y_p \gg y$ , which is the empirically relevant case for the cosign margin.

FIGURE OA1. Cosigning prevalence and local house prices



County-level cosigning prevalence against the FHFA All-Transactions house price index, binned. Fact 1 of Section 4 states the gradient against price relative to income (Figure 2); this price-index version is the benchmark for the cross-county validation of Section OA3.5. Prevalence is the age-gap measure, so the gradient is in intergenerational cosigning rather than in multi-party borrowing.

that  $d_{co} > d_{so} > 0$ , and therefore

$$\frac{\phi(d_{co})}{\phi(d_{so})} = \exp\left(-\frac{1}{2}(d_{co}^2 - d_{so}^2)\right) < 1, \quad (\text{OA22})$$

decreasing in the buffer  $m_{so} - m_{co}$ . Because the Gaussian tail decays faster than any bounded factor, there exists a buffer threshold  $\underline{\Delta}$  such that whenever  $\mu_{\log Y_{co}} - \mu_{\log Y_{so}} > \underline{\Delta}$  the product (OA20) is strictly below one:

$$\beta_{rej}^{co} < \beta_{rej}^{so}. \quad (\text{OA23})$$

Equivalently, in the sharp-cap threshold-density form of (C2'), the threshold-density ratio  $f_{DTI}^{co}(\kappa)/f_{DTI}^{so}(\kappa) = \frac{s_{so}}{s_{co}} \phi(d_{co})/\phi(d_{so})$  falls below the sensitivity ratio  $\mathbb{E}[\sigma_{so}]/\mathbb{E}[\sigma_{co}]$  once the parental buffer is large enough, which is exactly Condition (C2). This confirms that the insulation channel dominates the quality channel on the rejection margin under lognormality.

## OA3 Data Construction

### OA3.1 HMDA Sample Construction

The estimation sample is built from the HMDA Loan Application Register (LAR) snapshot files published by the CFPB for 2018–2024, read directly from the pipe-delimited annual archives. Construction proceeds in two stages. A first pass (notebook `001_lar_preparation`) applies the structural product and scope filters and writes an intermediate file; the income, down-payment, and outcome filters follow (`002_lar_preparation_2`); and the conventional, fixed-rate restriction that defines the analysis sample is applied in the analysis notebooks (003–007). Table OA1 reports every filter in sequence, its effect on the row count, and is regenerated end-to-end from the public

FIGURE OA2. Cosigning and house prices relative to income, single years

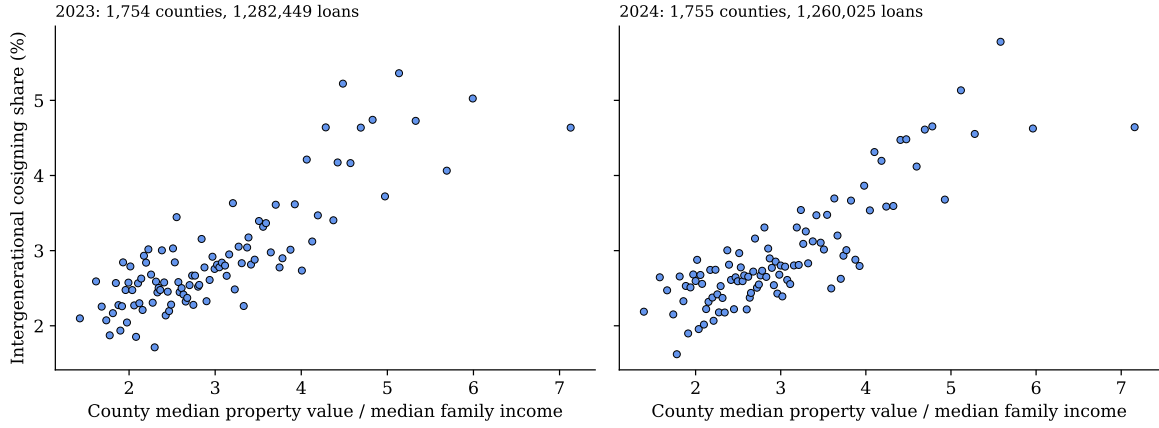


Figure 2 of Section 4 for a single year of originations at a time, 2023 (left) and 2024 (right), on the same county set: county intergenerational cosigning share against the county’s median property value among purchase applications divided by its ACS median family income, in one hundred equal-count bins of counties weighted by loans. The 2024 income denominator uses the latest available ACS vintage where the 2024 release does not cover the county.

LAR by `empirics/sample_construction_funnel.py`.

TABLE OA1. HMDA sample construction. Each row applies one filter sequentially to the 2018–2024 LAR; “Remaining” is the count after the filter and “Dropped” the records it removes. Intermediate drops are order-dependent; the final count is not.

| Filter step                            | Remaining   | Dropped    |
|----------------------------------------|-------------|------------|
| RAW (all LAR rows read)                | 124,615,183 |            |
| Sex reported (drop Sex-Not-Available)  | 103,350,736 | 21,264,447 |
| County & state non-missing             | 101,418,323 | 1,932,413  |
| Home purchase (loan_purpose=1)         | 42,502,777  | 58,915,546 |
| Principal residence (occupancy=1)      | 38,437,780  | 4,064,997  |
| Single unit (total_units=1)            | 37,828,586  | 609,194    |
| Consumer (business_purpose≠1)          | 37,785,492  | 43,094     |
| Closed-end (open_end_loc=2)            | 36,563,979  | 1,221,513  |
| First lien (lien_status=1)             | 35,686,317  | 877,662    |
| Conventional+FHA (loan_type in 1,2)    | 31,437,797  | 4,248,520  |
| Exclude territories (PR/GU/VI)         | 31,360,113  | 77,684     |
| Forward mortgage (reverse≠1)           | 31,352,922  | 7,191      |
| Applicant age <45                      | 20,341,531  | 11,011,391 |
| In an MSA/MD (derived_msa_md≠0)        | 20,341,513  | 18         |
| Loan amount ≥ \$20,000                 | 20,263,206  | 78,307     |
| Property value ≥ \$25,000              | 16,745,236  | 3,517,970  |
| Loan term ≥ 5 months                   | 16,722,221  | 23,015     |
| Income in (0, \$10M]                   | 16,443,150  | 279,071    |
| Downpayment ≤ \$500,000                | 16,249,160  | 193,990    |
| Definitive outcome (action in 1,3,7)   | 14,944,588  | 1,304,572  |
| Conventional first lien only           | 11,305,506  | 3,639,082  |
| Fixed rate (intro_rate-period missing) | 10,534,919  | 770,587    |

**Product and structural filters.** I keep first-lien, purchase-money mortgages on owner-occupied (principal-residence), single-unit properties: HMDA `loan_purpose = 1`, `occupancy_type = 1`, `total_units = 1`, and `lien_status = 1`. I exclude open-end lines of credit (`open_end_line_of_credit = 2` retains closed-end loans), business- or commercial-purpose loans (`business_or_commercial_purpose ≠ 1`), and reverse mortgages (`reverse_mortgage ≠ 1`). The first stage retains loan types 1 and 2 (conventional and FHA); the analysis sample is then restricted to *conventional first liens* (`derived_loan_product_type = "Conventional:First Lien"`), which drops FHA. FHA loans are insured outside the GSE channel, are not priced through the risk-based loan-level price-adjustment grid, and are unaffected by the 2022 guarantee-fee reform, so they are inappropriate for both the pricing mechanism and the natural experiment. Finally, the analysis sample retains only *fixed-rate* loans (`intro_rate_period` missing, which excludes adjustable-rate mortgages).

**Scope filters.** I restrict to the fifty states and the District of Columbia (dropping Puerto Rico, Guam, and the U.S. Virgin Islands) and to properties in a metropolitan area (`derived_msa_md ≠ 0`). Two scope choices are specific to this study. First, the sample is limited to a **primary applicant under age 45** (`applicant_age ∈ {< 25, 25–34, 35–44}`), the population for whom first-time purchase and parental support are relevant. Second, I keep applications whose reported applicant sex is male, female, or *joint* (two applicants of different sex), dropping only records where sex is not reported. Retaining the “joint” category is essential for the cosigning question: an intergenerational cosign in which parent and child are of opposite sex (a daughter and her father, a son and his mother) is coded “joint” in HMDA, and 47.2% of the intergenerational pairs the age-gap rule identifies are in fact of opposite sex (Section 4.4), so excluding the category would discard nearly half of them and bias the cosigning sample toward same-sex pairs. The cosigner is identified from the applicant–co-applicant age gap (Section 4.1), not from sex. The primary applicant’s sex, used as a control, is taken from `applicant_sex` rather than the derived field, which is undefined for joint applications.

**The age-gap rule, bin by bin.** HMDA reports applicant and co-applicant age in bins, so the twenty-year gap of Section 4.1 is applied bin to bin. Table OA2 lists the admissible pairs. The rule is conservative in one direction: a genuine parent–child pair whose reported bins imply a gap of, say, eighteen years is never flagged, so the measure understates intergenerational cosigning rather than overstating it.

TABLE OA2. Age-bin pairs flagged as intergenerational cosigning. A pair is flagged when the reported bins imply an age gap of at least twenty years between the co-applicant and the primary applicant. Primary applicants aged 45 and over are outside the estimation sample.

| Primary applicant age | Co-applicant age (cosigner) |
|-----------------------|-----------------------------|
| < 25                  | 45–54, 55–64, 65–74, > 74   |
| 25–34                 | 55–64, 65–74, > 74          |
| 35–44                 | 65–74, > 74                 |

**Range, quality, and outcome filters.** I drop records with non-positive or implausibly large income (outside (0, \$10,000,000]; HMDA income is reported in thousands), implied down payments above \$500,000, loan amounts below \$20,000, property values below \$25,000, or loan terms under five months — predominantly data errors or non-residential loans that survive the product filters. I keep only applications with a definitive outcome: originated, denied, or pre-approval denied (`action_taken`  $\in \{1, 3, 7\}$ ); applications approved but not accepted, withdrawn, incomplete, or purchased from another institution are excluded.

For each county, a dedicated Selenium-based scraper navigates the county clerk’s web portal as websites change by county. The webscraped data always contains the official information on mortgagors, mortgagees, loan amount, municipality and day of recording. Some website offer more detailed information such as property address and day of contract. Day of recording and day of contract are usually close but differ around holidays such as Christmas or New Year’s. Whereas sometimes possible to filter directly for residential mortgages, many counties do not allow such targeted searches, so any mortgages with corporate identifiers (for example, LLC or INC) are dropped. All loan amounts below \$20,000 are dropped as junior liens or home equity lines are sometimes included when scraped.

**Full applicant and loan profile by application type.** Table OA3 is the unabridged version of Table 4: the pooled means of every applicant and loan characteristic in the estimation sample, separately for solo, same-generation joint and intergenerationally cosigned applications. Section 4.4 reports the three rows that carry its argument; the remainder are here. Two rows need reading with care. Reported income is the *combined* gross income the underwriter relied on, so on a cosigned application it already includes the parent’s earnings: that row measures the admission channel rather than the child’s own earnings, and for the same reason the housing-payment-to-income row is deflated for cosigned and joint applications. The interest-rate and rejection rows are pooled means over the whole sample, and so answer a different question from Facts 2 and 3 of Section 4, which compare cosigned and solo loans within property-value-by-loan-amount cells and hold the size of the transaction fixed.

**The cosigner’s age profile.** Figure OA3 plots graphically what Table OA4 reports in levels: the distribution of cosigner age, and how rejection and the contract rate vary with it *within* each primary-applicant age bin. The gradient is reported only here: an older guarantor contributes less qualifying income and less credit enhancement, and the market prices the difference, but no later section cuts on guarantor age. The 45–54 row contains only applicants under 25, because the twenty-year age-gap rule admits that pairing for no older applicant. The marginal distribution therefore mixes cosigner age with applicant age and the rows are not comparable across bins; the gradient has to be read within a primary-applicant bin.

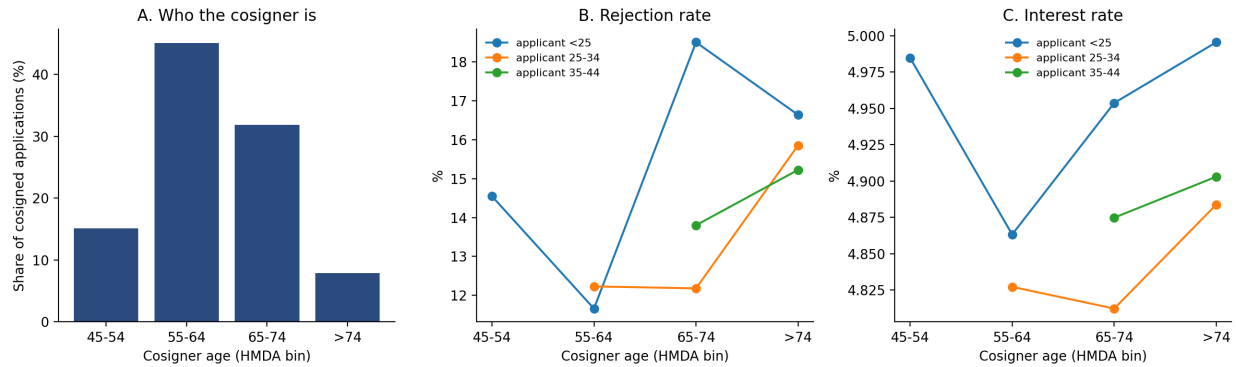
**Where cosigned loans go.** Table OA5 gives the purchaser of record in the year of origination, by application type. It is the evidence behind the claim of Section 4.4 that cosigned lending sits

TABLE OA3. Applicant and loan profile by application type, full

|                                      | Mean by application type |                |                 | Difference       |                   |
|--------------------------------------|--------------------------|----------------|-----------------|------------------|-------------------|
|                                      | Solo<br>(1)              | Joint<br>(2)   | Cosigned<br>(3) | Cos.—Solo<br>(4) | Cos.—Joint<br>(5) |
| Reported income (combined, \$000s)   | 94.2 (108.5)             | 143.1 (112.4)  | 172.6 (225.4)   | 78.4*** (0.4)    | 29.5*** (0.4)     |
| Property value (\$000s)              | 332.9 (240.2)            | 443.5 (286.8)  | 355.7 (225.9)   | 22.9*** (0.4)    | −87.8*** (0.4)    |
| Loan amount (\$000s)                 | 280.4 (198.3)            | 367.9 (226.1)  | 290.1 (175.9)   | 9.6*** (0.3)     | −77.9*** (0.3)    |
| Down payment (\$000s)                | 52.4 (85.5)              | 75.6 (89.3)    | 65.7 (79.7)     | 13.3*** (0.2)    | −9.9*** (0.2)     |
| Loan-to-value (%)                    | 87.19 (42.09)            | 85.65 (14.69)  | 84.25 (15.11)   | −2.94*** (0.03)  | −1.40*** (0.03)   |
| Combined LTV, reported (%)           | 86.93 (12.03)            | 85.29 (11.78)  | 83.67 (13.12)   | −3.26*** (0.03)  | −1.62*** (0.03)   |
| Share with CLTV > 95 (%)             | 20.30 (40.22)            | 11.60 (32.02)  | 5.42 (22.64)    | −14.88*** (0.05) | −6.18*** (0.05)   |
| Share with DTI ≥ 43% (%)             | 32.35 (46.78)            | 24.30 (42.89)  | 39.74 (48.94)   | 7.39*** (0.09)   | 15.44*** (0.09)   |
| Share with DTI ≥ 45% (%)             | 23.28 (42.26)            | 17.06 (37.61)  | 30.15 (45.89)   | 6.87*** (0.09)   | 13.09*** (0.09)   |
| Housing payment / income (%)         | 21.07 (33.69)            | 17.98 (31.88)  | 14.89 (28.92)   | −6.19*** (0.06)  | −3.09*** (0.06)   |
| Interest rate (%)                    | 4.618 (1.606)            | 4.633 (1.603)  | 4.869 (1.631)   | 0.251*** (0.003) | 0.236*** (0.003)  |
| Rate spread over APOR (pp)           | 0.449 (0.979)            | 0.403 (22.514) | 0.504 (0.856)   | 0.055*** (0.002) | 0.101*** (0.011)  |
| Discount points (\$)                 | 2,633 (3, 370)           | 3,224 (4, 665) | 2,975 (3, 559)  | 342*** (11)      | −249*** (11)      |
| Share 30-year term (%)               | 93.19 (25.19)            | 92.61 (26.16)  | 94.11 (23.55)   | 0.92*** (0.05)   | 1.50*** (0.05)    |
| Rejection rate (%)                   | 13.30 (33.96)            | 9.57 (29.42)   | 13.18 (33.83)   | −0.12* (0.06)    | 3.61*** (0.06)    |
| Share female primary applicant (%)   | 38.64 (48.69)            | 34.15 (47.42)  | 49.82 (50.00)   | 11.18*** (0.09)  | 15.68*** (0.09)   |
| Share White (%)                      | 71.63 (45.08)            | 78.52 (41.07)  | 74.24 (43.73)   | 2.61*** (0.08)   | −4.28*** (0.08)   |
| Share Black (%)                      | 7.99 (27.11)             | 3.90 (19.35)   | 5.34 (22.47)    | −2.65*** (0.04)  | 1.44*** (0.04)    |
| Share Asian (%)                      | 10.44 (30.58)            | 8.59 (28.02)   | 10.02 (30.03)   | −0.42*** (0.06)  | 1.43*** (0.06)    |
| Share race not provided (%)          | 8.81 (28.34)             | 8.01 (27.14)   | 9.25 (28.97)    | 0.44*** (0.05)   | 1.24*** (0.05)    |
| Share Hispanic or Latino (%)         | 14.34 (35.05)            | 11.42 (31.80)  | 14.69 (35.40)   | 0.34*** (0.07)   | 3.27*** (0.07)    |
| Share above the conforming limit (%) | 2.64 (16.02)             | 6.06 (23.86)   | 1.41 (11.78)    | −1.23*** (0.02)  | −4.65*** (0.02)   |
| County house price index             | 749.2 (427.5)            | 800.1 (462.4)  | 866.3 (477.2)   | 117.1*** (0.9)   | 66.2*** (0.9)     |
| Applications                         | 5,837,560                | 4,403,026      | 294,333         |                  |                   |

Notes: HMDA LAR 2018–2024, estimation sample of this appendix; the three application types are defined in Section 4.1. Columns (4) and (5) are differences in means, Welch  $t$ -test, though with millions of applications per cell the magnitudes and not the stars carry the content. Rate, spread, points, term and payment-to-income are measured on originations, every other row on applications; combined LTV is winsorized to (1, 150].

FIGURE OA3. The cosigner's age and the terms the family gets



Notes: Panel A is the distribution of cosigner age; Panels B and C plot rejection and the mean contract rate against cosigner age *within* each primary-applicant age bin. The age-gap rule makes the series different lengths by construction — a 45–54 cosigner appears only for an applicant under 25.

TABLE OA4. Who the cosigner is, and how outcomes vary with cosigner age

| Cosigner age<br>(HMDA bin) | Share of<br>cosigned (%) | Rejection<br>rate (%) | Interest<br>rate (%) | Property<br>value (\$000s) | Combined<br>LTV (%) | Share<br>DTI $\geq 43$ (%) |
|----------------------------|--------------------------|-----------------------|----------------------|----------------------------|---------------------|----------------------------|
| 45-54                      | 15.2                     | 14.55                 | 4.985                | 300.4                      | 86.01               | 32.36                      |
| 55-64                      | 45.1                     | 12.10                 | 4.835                | 368.9                      | 83.90               | 34.15                      |
| 65-74                      | 31.9                     | 13.45                 | 4.854                | 405.3                      | 81.66               | 41.10                      |
| $\geq 74$                  | 7.9                      | 15.63                 | 4.906                | 375.9                      | 82.79               | 41.84                      |

*Notes:* Cosigned applications only, by the co-applicant’s reported HMDA age bin; column 1 is the share of all cosigned applications. Rejection is measured on applications; interest rate, property value, combined LTV and the DTI share on originations.

further inside the loan-level price-adjustment grid than ordinary lending does: cosigned loans are more likely to be sold to Fannie Mae or Freddie Mac and less likely to be retained on balance sheet.

TABLE OA5. Where cosigned loans go: purchaser in the year of origination

| Purchaser in year of origination | Solo      | Joint     | Cosigned |
|----------------------------------|-----------|-----------|----------|
| Not sold in calendar year        | 17.48     | 19.18     | 14.25    |
| Fannie Mae                       | 29.62     | 28.33     | 32.41    |
| Ginnie Mae                       | 0.01      | 0.01      | 0.01     |
| Freddie Mac                      | 23.48     | 22.76     | 26.31    |
| Farmer Mac                       | 0.00      | 0.00      | 0.00     |
| Private securitizer              | 0.87      | 0.89      | 0.71     |
| Commercial bank / thrift         | 10.52     | 10.95     | 9.54     |
| Affiliate institution            | 1.44      | 1.60      | 1.59     |
| Other purchaser                  | 5.10      | 4.45      | 3.44     |
| Credit union / mortgage co.      | 10.99     | 11.35     | 11.31    |
| Life insurance company           | 0.49      | 0.47      | 0.43     |
| Originations                     | 5,060,885 | 3,981,507 | 255,540  |

*Notes:* Column percentages of originations, by application type. HMDA records a purchaser only for loans sold in the same calendar year, so “not sold in calendar year” pools loans genuinely retained with loans sold the following January; the comparison across columns is unaffected so long as that timing is common to the three application types.

**Why applications are denied.** The descriptive facts report whether an application is denied; HMDA also records why. Table OA6 gives the decomposition by application type, and it separates the two margins Section 2 distinguishes. Cosigned denials are the most debt-to-income-coded of the three columns and the least coded for insufficient cash, which is the pattern the underwriting rules imply: joint income is what a cosigner supplies, so it is the constraint a cosigned file is still pressing against when it fails, while the down payment is the one margin no cosigner reaches. The collateral and cash codes are reported separately but are substitutes for one event: an appraisal below the contract price is a failure of the security and, for that reason, leaves the buyer short of funds to close, and nothing in the reporting rules dictates which of the two the lender records. The

table therefore also reports their union, which is invariant to that choice. It changes nothing: the cosigned deficit on the equity margin is  $-3.6$  percentage points measured on the cash code alone and  $-3.6$  measured on the union, because the collateral code is flat across the three columns. Two cautions apply to reading the credit-history row as evidence on the pricing axis. The shares are conditional on denial, so a column that is heavier on one reason is mechanically lighter on the rest; and HMDA carries no credit score, so the table cannot hold the applicant’s own score fixed. The weakest-link claim of Appendix OA3.8 rests on the Selling Guide, not on this table.

TABLE OA6. Reasons reported for denied applications, by application type

| Reason reported for the denial                  | Solo    | Joint   | Cosigned | All       |
|-------------------------------------------------|---------|---------|----------|-----------|
| Debt-to-income ratio                            | 38.91   | 30.69   | 41.43    | 36.19     |
| Credit history                                  | 39.79   | 43.01   | 38.55    | 40.85     |
| Collateral                                      | 11.81   | 10.03   | 11.77    | 11.20     |
| Insufficient cash (down payment, closing costs) | 13.39   | 12.90   | 9.77     | 13.11     |
| Employment history                              | 4.07    | 5.50    | 4.43     | 4.57      |
| Mortgage insurance denied                       | 0.20    | 0.21    | 0.23     | 0.20      |
| Collateral <i>or</i> insufficient cash          | 24.68   | 22.46   | 21.13    | 23.81     |
| of which both cited                             | 0.53    | 0.48    | 0.41     | 0.51      |
| Any recorded reason                             | 81.36   | 78.98   | 83.90    | 80.63     |
| Recorded reasons per denial                     | 1.08    | 1.02    | 1.06     | 1.06      |
| Denied applications                             | 776,675 | 421,519 | 38,793   | 1,236,987 |
| Denial rate (% of applications)                 | 13.30   | 9.57    | 13.18    | 11.74     |

*Notes:* Percent of *denied* applications flagged with each reason, on the sample of Table OA3. A denial is HMDA `action_taken` 3 (application denied) or 7 (preapproval request denied), the two actions for which a reason must be reported. Columns do not sum to 100: a lender may report several reasons for one denial, and a denial may carry none of the reasons listed. Lenders may report up to four reasons and the sample construction reads the first three; of the ten HMDA reason codes it carries the six shown, omitting unverifiable information, incomplete application, and other. “Any recorded reason” is the share of denials with at least one of the six, and so bounds how much the two omissions remove. Neither omission disturbs the comparison across columns unless it sorts by application type. “Collateral *or* insufficient cash” is the union of those two codes, not their sum. The co-citation row is reported for transparency rather than as a test of whether lenders treat the two codes as interchangeable: at 1.06 recorded reasons per denial, denials citing any two reasons are rare enough that the overlap is uninformative about coding practice.

**Prevalence under both filing directions.** Table OA7 gives, year by year and age bin by age bin, the two measures the filing-order paragraph of Appendix OA3.6 compares in the pooled sample: the published measure, which requires the child to be the primary applicant, and the measure that also counts pairs filed with the parent first. The gap widens over the sample period and is widest among the youngest buyers, so the published series understates both the level of cosigning and its growth.

**Geographic reference: town to census-tract mapping.** Because HMDA records the census tract of the property while some deed records report only the municipality, I construct a crosswalk between New York jurisdictions and 2020 census tracts using the TIGER (FHFA Tracts & Blocks

TABLE OA7. Cosigning prevalence under both filing directions

|                                           | Applications | Cosigning rate (%) |                 | $\Delta$ (pp) |
|-------------------------------------------|--------------|--------------------|-----------------|---------------|
|                                           |              | Child filed first  | Both directions |               |
| <i>Panel A: by age of the young party</i> |              |                    |                 |               |
| Under 25                                  | 932,799      | 8.89               | 12.10           | 3.21          |
| 25–34                                     | 5,332,525    | 2.81               | 3.53            | 0.71          |
| 35–44                                     | 4,269,595    | 1.44               | 1.91            | 0.47          |
| <i>Panel B: by application year</i>       |              |                    |                 |               |
| 2018                                      | 1,392,389    | 2.43               | 3.11            | 0.68          |
| 2019                                      | 1,453,679    | 2.32               | 3.05            | 0.73          |
| 2020                                      | 1,760,139    | 2.45               | 3.17            | 0.72          |
| 2021                                      | 1,973,341    | 2.73               | 3.54            | 0.81          |
| 2022                                      | 1,542,093    | 3.20               | 4.19            | 0.99          |
| 2023                                      | 1,201,065    | 3.32               | 4.42            | 1.10          |
| 2024                                      | 1,212,213    | 3.34               | 4.44            | 1.10          |

*Notes:* “Child filed first” is the published measure, the age-gap rule of Section 4.1 on the estimation sample. “Both directions” adds, to numerator and denominator alike, the applications in which the older party was filed first, which the estimation sample excludes. In Panel A the “young party” is the primary applicant under the first measure and the co-applicant under the second.

and Places/County Subdivisions layers). For each county, spatial intersection of the county subdivision layer with the census tract layer recording all tracts that fall within each town boundary.

**Lender identification and LEI matching.** The lender name as it appears in the deed may differ from the institution’s legal name in HMDA. To link deed records to the HMDA LAR lenders, each unique mortgagee string is matched to a Legal Entity Identifier (LEI) via the GLEIF API, using two search strategies: (i) an exact `legalName` field lookup, and (ii) a full-text search across GLEIF’s name index (covering legal names, trade styles, and prior names). MERS (Mortgage Electronic Registration Systems) entries and government agency entries (HUD, USDA Rural Housing, state housing authorities) are flagged as ineligible and excluded from the LEI-match pipeline. Unmatched lenders that appear frequently are resolved manually.

**Merging deed records to HMDA.** The merge procedure links deed records to originated HMDA loans using a hierarchical key: records are first matched on (activity year, LEI, loan amount, town) simultaneously. Mortgage amounts are rounded to the nearest \$5,000 as HMDA is also rounded and the inconsistent denomination of fees associated with the documentation of the deed. Only exact matches are considered. This comes at an obvious caveat that occasionally, the same loan amount is requested in a given town, at a given bank in a given year. These deeds are dropped.

### OA3.2 NSMO: Provenance and Analysis Samples

The NSMO supplies the default wedge, the typed default ratio ( $M_5$ ), the pre-cure wedge ( $M_6$ ), the parental-coverage benchmark and the cosign income gradient, and three different analysis samples appear in the text. This subsection states the file, the coverage, and how those samples relate.

**The survey.** The National Survey of Mortgage Originations (Federal Housing Finance Agency and Consumer Financial Protection Bureau, 2026) samples borrowers from the National Mortgage Database, a stratified random sample of closed-end first-lien residential mortgages, and surveys them shortly after origination. Loan performance is appended from the NMDB rather than self-reported, which is what makes the delinquency variables administrative and pre-cure (Appendix OA4.11). I use the Public Use File, which covers 62,359 respondents across survey waves 1 to 46 and origination years 2013 to 2024. The PUF is disclosure-protected: several variables, the origination credit score among them, are released binned rather than continuous, which is why the score enters every specification below as a categorical rather than a linear control.

TABLE OA8. Construction of the NSMO analysis samples

| Step                                                        | Respondents | % of file |
|-------------------------------------------------------------|-------------|-----------|
| (1) NSMO PUF respondents, all waves                         | 62,359      | 100.0     |
| (2) Conventional ( <code>loan_type = 1</code> )             | 48,129      | 77.2      |
| (3) + first-time buyer ( <code>first_mort.r = 1</code> )    | 8,249       | 13.2      |
| (4) + cosigning question answered ( <code>x20a</code> )     | 8,249       | 13.2      |
| (5) + fixed rate, single-family detached, primary residence | 4,663       | 7.5       |
| (6) + origination credit score present                      | 4,663       | 7.5       |

*Notes:* From `nsmo.v70.1324.puf.csv`, the NSMO Public Use File, survey waves 1–46 and origination years 2013–2024; survey weights (`analysis.weight`) applied to all reported shares. Step (3) is the universe the paper calls conventional  $\times$  first-time buyer; step (5) adds the product filters used for the default and pricing work. The survey and its disclosure protection are described in the text above.

**Three samples, and why they differ.** Table OA8 gives the funnel; the three samples the text reports sit at different points on it, and the differences are substantive rather than incidental. The cosign income gradient uses conventional first-time buyers (8,249 respondents), which is the paper’s stated universe. The rate-by-FICO schedule adds the product filters and restricts to solo borrowers (4,280), since it is the solo pricing schedule the model calibrates against. The default-wedge controls ladder, by contrast, runs on the 42,631 fixed-rate, single-family, primary-residence loans with complete controls in Table OA27, with *no* conventional-loan and *no* first-time-buyer restriction.

That last point is worth stating plainly rather than leaving to be inferred from a table note. The wedge is therefore measured on a broader population than the gradient, than the paper’s stated universe, and than the model object  $M_6$  targets. Widening the population should if anything attenuate a wedge driven by first-time buyers, since it adds repeat buyers with longer credit histories, so the direction works against the paper rather than for it. But the reader should know

the populations differ, and any comparison between the wedge and the model moment inherits that difference.

**Data availability.** The HMDA Loan Application Register is public and is obtained from the Consumer Financial Protection Bureau. The NSMO Public Use File is public and is obtained from the Federal Housing Finance Agency. The Fannie Mae Single-Family Loan Performance Data are obtained from Fannie Mae after free registration. The PSID is public and is obtained from the University of Michigan. The deed records are collected from New York county clerk land-record portals that provide free public access to recorded instruments; the scraping and cleaning code is part of the replication package, and because redistribution rights for the recorded instruments themselves rest with the individual county clerks, the package ships the code and the derived county-level aggregates rather than the raw scraped records. All code required to reproduce every table and figure is included; every number in the text is emitted by one of the scripts listed in `writing/stats.tex` rather than typed by hand.

### OA3.3 Fannie Mae Loan Performance and the Link to HMDA

Facts 3 and 4 follow loans after origination in the Fannie Mae Single-Family Loan Performance Data (Fannie Mae, 2026b). Fannie Mae publishes one file per acquisition quarter holding every monthly report on the fixed-rate, fully documented single-family loans it acquires: the origination terms, both borrowers' credit scores, and each month's delinquency status, forbearance, modification and deferral flags, and the reason the loan ends. I keep purchase loans on one-unit principal residences. The files used here cover acquisitions from 2018Q1 to 2024Q4.

**The link to HMDA.** Fannie Mae records no borrower ages, so the parent-child flag comes from HMDA. I match each Fannie Mae loan to the HMDA originations of the same year on state, county (through the loan's three-digit ZIP code), exact note rate, loan amount in \$10,000 bands, term, combined loan-to-value ratio and debt-to-income band within one step, and the presence of a co-borrower, and keep only loans with exactly one HMDA candidate. Of the 6.85 million one-unit Fannie Mae loans originated in 2018–2024, 4.56 million are matched, between 55% and 84% by origination year. Rerunning the match with every Fannie Mae rate shifted by an eighth of a point puts false matches at 0.3–1.4% of matches. Unique matches are harder in dense cells, so first-time buyers, high loan-to-value loans and the 2020–21 vintages match less often, which is why every comparison is made within the score band or the price cell. The analysis sample is 2,295,805 matched first-time purchases: 71,819 cosigned, 929,696 other two-borrower and 1,294,290 solo-borrower loans.

**Definitions.** Delinquency is Fannie Mae's month-end status. Its lowest category covers current loans and loans less than 30 days late together, so a payment made late within the month is not observed. Months in forbearance count neither as delinquency nor as exposure, and neither do

months of arrears carried out of forbearance until the loan is next current. A forced exit is a foreclosure or deed-in-lieu, a short sale, a third-party sale or a sale of the loan as non-performing. The last rows of Table OA9 follow each loan’s first 30-day delinquency for twelve months, excluding delinquencies with less than twelve months of follow-up, with forbearance in the window, or ended by a repurchase.

**Results.** Table OA9 reports the full conditioning ladder. The 30-day gap turns from positive to slightly negative once the score band is held fixed. The gaps in serious delinquency, forced exits and recovery keep their sign and most of their size as leverage, origination quarter, debt-to-income and state are added. At the same score, 0.7% of cosigned loans that fall 30 days behind end in a forced exit, against 2.4% of solo-borrower loans, and a year after a first 30-day delinquency 75.7% are current without a servicer workout, against 66.4%. The NSMO ladder of Table OA27 records a raw delinquency gap of similar size in percentage points but cannot tell a parent from a spouse; Fact 4 uses these records for that reason.

**The note rate, and Fact 3 in other data.** In the linked loans, cosigned loans pay +0.23 percentage points more than solo-borrower loans. Holding the score band or the loan-to-value ratio fixed leaves the gap about where it is; holding the whole price cell fixed, the origination quarter together with the score band and the loan-to-value ratio, cuts it to +0.08. Two other sources give the same reading. In the National Survey of Mortgage Originations (NSMO; [Federal Housing Finance Agency and Consumer Financial Protection Bureau, 2026](#)), which records the score, the cosigned premium is +0.01 pp once FICO and the combined loan-to-value ratio are held fixed. HMDA records no score, and there cosigned loans price +0.251 percentage points above solo-borrower loans, 4.869% against 4.618%; the gap survives property-value, loan-amount, year, state and county cells (Table OA17).

### OA3.4 Deed Sample: Construction, Validation, and Provenance

The HMDA sample is documented by the twenty-two step funnel of Table OA1. The deed records, which supply the augmented prevalence measure of Section 4.2 and the cosign-share moment  $M_3$ , had no equivalent. This appendix supplies one, adds the first validation of the > 2-signatory rule, and corrects the provenance of the intensive-margin estimates. Everything below is emitted by scripts under `empirics/utils/` (`deed_construction_funnel.py`, `deed_signatory_audit.py`, `deed_intensive_margin.py`); no number here is typed by hand. Table OA10 is that funnel, from raw scrape to matched sample.

**Coverage, and a correction.** The scrape reaches 40 county directories, of which 27 clear the cleaning stage and 27 produce a merged file. One of those bundles the five New York City boroughs, so the sample covers 31 counties rather than the thirty-four previously reported in Section 4.2; that figure was not reproducible from the pipeline and has been corrected. Fourteen counties were

TABLE OA9. Delinquency and Its Resolution: Cosigned vs. Solo Loans (Fannie Mae)

| Outcome                                                                        | Solo   | Difference to solo (pp), conditioning ladder |                    |                    |                    |                    |
|--------------------------------------------------------------------------------|--------|----------------------------------------------|--------------------|--------------------|--------------------|--------------------|
|                                                                                |        | Raw                                          | Score              | +LTV               | Price cell         | +Full              |
| <i>Ever 30+ days past due</i>                                                  |        |                                              |                    |                    |                    |                    |
| Cosigned (parent-child)                                                        | 11.54% | +2.32***<br>(0.20)                           | -0.68***<br>(0.14) | -0.36**<br>(0.15)  | -0.24<br>(0.15)    | -0.35**<br>(0.16)  |
| Other two-borrower                                                             |        | -3.48***<br>(0.12)                           | -5.33***<br>(0.11) | -5.23***<br>(0.11) | -5.14***<br>(0.11) | -4.68***<br>(0.13) |
| <i>Ever 90+ days past due</i>                                                  |        |                                              |                    |                    |                    |                    |
| Cosigned (parent-child)                                                        | 2.59%  | -0.31***<br>(0.07)                           | -1.23***<br>(0.07) | -1.00***<br>(0.07) | -1.00***<br>(0.06) | -1.03***<br>(0.06) |
| Other two-borrower                                                             |        | -1.18***<br>(0.06)                           | -1.74***<br>(0.06) | -1.68***<br>(0.05) | -1.70***<br>(0.05) | -1.58***<br>(0.05) |
| <i>Forced exit, loans ever 30+ days past due</i>                               |        |                                              |                    |                    |                    |                    |
| Cosigned (parent-child)                                                        | 2.43%  | -1.48***<br>(0.15)                           | -1.68***<br>(0.16) | -1.32***<br>(0.14) | -1.36***<br>(0.14) | -1.28***<br>(0.14) |
| Other two-borrower                                                             |        | -1.09***<br>(0.11)                           | -1.29***<br>(0.12) | -1.26***<br>(0.12) | -1.33***<br>(0.12) | -1.02***<br>(0.09) |
| <i>Current without a workout, 12 months after the first 30-day delinquency</i> |        |                                              |                    |                    |                    |                    |
| Cosigned (parent-child)                                                        | 66.36% | +6.27***<br>(0.58)                           | +9.34***<br>(0.56) | +7.47***<br>(0.54) | +7.58***<br>(0.54) | +7.83***<br>(0.50) |
| Other two-borrower                                                             |        | +2.19***<br>(0.35)                           | +5.21***<br>(0.30) | +5.03***<br>(0.27) | +4.92***<br>(0.27) | +5.33***<br>(0.29) |

*Notes:* Linear-probability estimates on Fannie Mae Single-Family Loan Performance loans matched one-to-one to HMDA: first-time buyers, purchase, principal residence, one unit, originated 2018–2024 ( $N = 2,295,805$  loans: 71,819 cosigned, 929,696 other two-borrower, 1,294,290 solo). Cosigned is the HMDA age-gap flag for a parent-child pair of borrowers. Months in forbearance, and arrears carried out of forbearance until the loan is next current, count neither as delinquency nor as exposure. The forced-exit rows use the 234,299 loans ever 30 days past due (forced exit: foreclosure, short sale, deed-in-lieu, third-party or non-performing note sale); the last rows use the 176,191 first 30-day delinquencies with 12 months of follow-up and no forbearance or repurchase in that window. **Solo** is the solo mean; the other columns are differences to solo in percentage points. Ladder: **Score** fixed effects for the lower of the borrowers' credit scores in nine bands (the score the GSE grid prices); **+LTV** adds LTV bands; **Price cell** origination quarter  $\times$  score band  $\times$  LTV band; **+Full** adds DTI-band and state fixed effects, log loan amount and the high-balance flag (and the calendar quarter of the delinquency in the last rows). Standard errors clustered by state. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

TABLE OA10. Construction of the web-scraped deed sample

| Step                                                           | Records      | Counties |
|----------------------------------------------------------------|--------------|----------|
| (1) Instruments scraped                                        | 4,307,527    | 40       |
| (2) After cleaning (corporate borrowers, loans below \$20,000) | 1,035,671    | 27       |
| (3) Lender resolved to a GLEIF entity identifier               | 773,005      | 27       |
| (4) Matched to an originated HMDA loan                         | 36,958       | 27       |
| of which flagged by the deed rule ( $> 2$ signatories)         | 1,286        |          |
| of which flagged by the HMDA age-gap rule                      | 1,304        |          |
| flagged by both                                                | 178          |          |
| <b>flagged by either (the augmented measure)</b>               | <b>2,412</b> |          |

*Notes:* Step (2) to step (4) is a 3.6% match rate: the scrape is every recorded mortgage instrument, the HMDA sample only conventional fixed-rate first-lien purchase loans to owner-occupiers under 45 (Table OA1). The county column counts merged *files*, one of which bundles the five New York City boroughs. The two flags overlap on 7.4% of flagged loans (Jaccard).

scraped but never cleared cleaning, in most cases because the portal returned partial records. The step (2) to step (4) match rate of 3.6% is not a measure of merge failure: the scraped universe is every recorded mortgage instrument, while the HMDA sample is restricted to conventional, fixed-rate, first-lien purchase loans to owner-occupiers under 45, so most scraped instruments have no eligible counterpart. Table OA11 names every county in the sample and reports the instruments scraped in each recording year, so the coverage behind the prevalence measure can be read directly rather than inferred. Coverage is not uniform: portals differ in how far back their free full-text access reaches, and the resulting county–year pattern, not any sampling design, is what determines which markets the deed measure observes.

**Validating the  $> 2$ -signatory rule.** The rule cannot, on its own, separate a child and a parent from a couple and a sibling, three co-investors, or a trust. Borrower names survive only in the raw scrapes, as the **OR Party** field, so the checks below run there and are joined back to the merged sample on county and instrument identifier; they reach 53% of flagged loans, three merged files having been written without an identifier column.

Two mechanical contaminants are bounded and small. County clerks record aliases as separate party rows, which could inflate a two-party instrument into a “cosigned” three-party one; collapsing alias rows moves only 0.7% of instruments below three, so the practice adds rows to instruments that already have three distinct people rather than manufacturing them. Non-person parties (limited companies, trusts, estates) appear on 11.2% of all 3+ instruments but only 2.4% of those that reach the merged sample, the corporate filter and the purchase-loan restriction having removed most of them.

The third check fails, and the failure is worth reporting because it bounds what the measure can be asked to do. Surname structure looks like a natural way to separate families from co-investors, but it cannot: a daughter who has taken her husband’s name, her husband, and her father produce three distinct surnames, and so do three unrelated parties. Testing this directly,

TABLE OA11. Web-scraped deed coverage by county and year

| County       | 2018         | 2019         | 2020         | 2021         | 2022         | 2023         | 2024         | Total          | Merged        |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|----------------|---------------|
| Albany       | 13.1         | 15.2         | 21.2         | 29.8         | 20.7         | 13.0         | 12.9         | 126.0          | 1,848         |
| Broome       | 11.3         | 11.1         | 11.5         | 15.0         | 12.7         | 9.3          | 9.5          | 80.4           | 1,324         |
| Cayuga       | 3.7          | 3.8          | 4.7          | 5.3          | 4.6          | 3.4          | 3.5          | 29.0           | 501           |
| Chenango     | 1.0          | 0.9          | 0.9          | 1.3          | 1.0          | 0.8          | 0.8          | 6.7            | 134           |
| Columbia     | 3.2          | 3.1          | 4.2          | 4.9          | 4.0          | 2.4          | 2.3          | 24.1           | 354           |
| Cortland     | 2.1          | 2.0          | 2.3          | 3.1          | 2.5          | 1.8          | 1.8          | 15.6           | 278           |
| Delaware     | 2.1          | 2.0          | 2.7          | 3.4          | 2.5          | 1.9          | 2.0          | 16.6           | 258           |
| Dutchess     | 254.4        | 580.6        | 367.4        | 228.3        | 100.7        | 58.2         | 45.3         | 1,634.8        | 888           |
| Erie         | 48.9         | 48.7         | 55.3         | 67.3         | 49.6         | 39.2         | 38.0         | 347.1          | 1,019         |
| Essex        | 1.9          | 1.8          | 2.4          | 2.1          | 2.2          | 1.5          | 1.6          | 13.5           | 81            |
| Greene       | 2.2          | 2.2          | 2.7          | 3.6          | 3.2          | 2.0          | 1.7          | 17.6           | 447           |
| Herkimer     | 2.4          | 2.2          | 2.2          | 2.9          | 2.8          | 1.9          | 2.2          | 16.6           | 358           |
| Kings        | 156.7        | 89.1         | 105.7        | 137.4        | 110.5        | 57.0         | 65.0         | 721.4          | —             |
| Livingston   | —            | 1.5          | 1.7          | 2.1          | 1.7          | 1.4          | 1.3          | 9.8            | 131           |
| Monroe       | 14.4         | 14.1         | 15.1         | 18.7         | 16.4         | 13.7         | 6.7          | 99.1           | 1,705         |
| Nassau       | 138.7        | 57.8         | 47.6         | —            | —            | —            | —            | 244.1          | 2,741         |
| Oneida       | 8.0          | 8.7          | 9.6          | 11.4         | 10.4         | 7.5          | 8.2          | 63.7           | 1,529         |
| Onondaga     | 38.5         | 39.7         | 46.8         | 53.7         | 41.9         | 34.0         | 33.7         | 288.4          | 8,237         |
| Ontario      | 3.1          | 3.4          | 4.3          | 4.9          | 3.6          | 2.6          | 2.7          | 24.6           | 1,064         |
| Otsego       | 2.5          | 2.4          | 2.6          | 3.5          | 2.9          | 2.0          | 2.2          | 18.1           | 295           |
| Putnam       | 4.9          | 4.4          | 6.6          | 10.3         | 6.5          | 4.0          | 3.7          | 40.5           | 1,448         |
| Rensselaer   | 6.4          | 6.7          | 9.0          | 12.3         | 10.2         | 7.2          | 7.4          | 59.1           | 928           |
| Rockland     | 6.1          | 6.3          | 8.6          | 12.3         | 7.7          | 5.1          | 5.1          | 51.2           | 3,307         |
| Schoharie    | 1.3          | 1.3          | 1.4          | 2.0          | 1.8          | 0.8          | 1.1          | 9.6            | 187           |
| Steuben      | 11.0         | 11.2         | 11.8         | 13.4         | 11.5         | 8.4          | 8.6          | 76.0           | 713           |
| Tompkins     | 2.0          | 2.0          | 2.3          | 2.7          | 2.1          | 1.5          | 1.4          | 14.0           | 575           |
| Ulster       | 7.8          | 7.6          | 9.7          | 13.4         | 10.2         | 6.5          | 6.4          | 61.6           | 1,857         |
| <b>Total</b> | <b>747.7</b> | <b>929.8</b> | <b>760.5</b> | <b>664.9</b> | <b>444.0</b> | <b>287.1</b> | <b>275.2</b> | <b>4,109.3</b> | <b>36,958</b> |

Notes: Instruments scraped, in thousands, by county and recording year, read off the raw scrape filenames; the *Merged* column is the count matched to an originated HMDA loan and is in units, not thousands. “—” means no file for that county-year, which reflects portal coverage rather than an absence of recorded instruments. One merged file bundles the five New York City boroughs and is reported under *Nyc*. Counties scraped but not carried through cleaning, in most cases because the portal returned partial records: Chautauqua, Madison, Niagara, Orange, Oswego, Saratoga, Schenectady, Seneca, Warren, Westchester.

the surname distribution among deed-flagged loans that the HMDA age-gap rule independently confirms as intergenerational is statistically indistinguishable from the distribution among those it does not confirm ( $\chi^2_2 = 1.82$ ,  $p = 0.40$ ). Surname structure therefore carries no information about the parent-child question and is not reported as though it did. What remains is the overlap between the two rules, 9.4% of deed-flagged loans, and that is itself confounded: with three or more signatories HMDA reports two, and the age gap is computed on whichever two. The deed measure is accordingly used as a measure of multi-party borrowing that HMDA’s two-borrower cap conceals. Whether it carries intergenerational content of its own is a separate question, and the next paragraph tests it rather than settling it by argument.

**A positive test the deed rule can pass.** The surname check fails in one direction only: it cannot confirm the family reading, and it equally cannot refute it. A test that can confirm it uses the other instrument on the same loan. HMDA reports at most two borrowers and their age bands, so on deed-flagged loans that the age-gap rule does *not* flag, HMDA’s own second borrower is independent evidence about who is signing. If those loans are intergenerational they should report a second borrower more often, and an older one, than loans neither rule flags. The test is one-sided by construction: with three signatories and two HMDA slots the guarantor may sit outside the HMDA record entirely, so a skew is evidence for the family reading while its absence would not have been evidence against it.

Table OA12 runs it on the 1,108 deed-only loans against the solo group, excluding the 178 loans both rules flag so that nothing in the age rows can be mechanical. Deed-only loans report a second borrower on 65.9% of applications against 44.9% of solo ones, an adjusted difference of 18.71 points (2.35) once county and year fixed effects and the applicant controls are in place. That settles the first half: the rule is finding multi-party transactions rather than recording noise. On the age margin the reported second borrower is at least forty-five on 9.4% of deed-only loans against 5.2% of solo ones, an adjusted 3.74 points (1.72,  $p = 0.029$ ). The two sharper margins move the same way without reaching significance on 27 county clusters: at least fifty-five, 1.50 points ( $p = 0.130$ ), and a mean age gap 0.47 years wider ( $p = 0.143$ ).

The age-gap rule therefore still carries most of the intergenerational content of the augmented measure, but it no longer carries all of it. On the one margin HMDA permits us to observe, loans the age rule never touched are drawn from an older second-borrower distribution at the five percent level, which is the direct evidence the surname test could not supply.

**An external benchmark.** Both of the paper’s prevalence measures are constructed here, so it is worth asking whether an outside source puts cosigning in the same range. The National Survey of Mortgage Originations asks borrowers directly, and on the paper’s universe (conventional, first-time buyers, 8,249 respondents, survey-weighted) the answer depends entirely on which question is used. Asked whether they had to add a co-signer *or co-borrower* to qualify, 9.81% say yes — but that counts a spouse added to the note, which the intergenerational measures here deliberately exclude, so it is not a comparable figure and is reported only to forestall its being quoted as one. Restricting

TABLE OA12. What HMDA reports about the second borrower on deed-flagged loans

|                                                                                    | Solo      | Deed only | Deed (all) | HMDA age-gap |
|------------------------------------------------------------------------------------|-----------|-----------|------------|--------------|
| Loans                                                                              | 34,546    | 1,108     | 1,286      | 1,304        |
| Reports a co-applicant (%)                                                         | 44.9      | 65.9      | 70.6       | 100.0        |
| <i>Conditional on a reported co-applicant:</i>                                     |           |           |            |              |
| Co-applicant $\geq 45$ (%)                                                         | 5.2       | 9.4       | 27.2       | 100.0        |
| Co-applicant $\geq 55$ (%)                                                         | 0.7       | 2.3       | 19.2       | 87.7         |
| Mean age gap (years)                                                               | 0.57      | 1.28      | 7.52       | 33.22        |
| <i>Deed only minus solo, county and year fixed effects and applicant controls:</i> |           |           |            |              |
|                                                                                    | Estimate  | (cluster) | (HC1)      | <i>N</i>     |
| reports a co-applicant (pp)                                                        | +18.71*** | (2.35)    | [1.35]     | 35,652       |
| co-applicant $\geq 45$ (pp)                                                        | +3.74**   | (1.72)    | [1.11]     | 16,217       |
| co-applicant $\geq 55$ (pp)                                                        | +1.50     | (0.99)    | [0.59]     | 16,217       |
| age gap (years)                                                                    | +0.47     | (0.32)    | [0.27]     | 16,217       |

*Notes:* The 36,958 loans of the merged deed  $\times$  HMDA sample. *Deed only* is deed-flagged and *not* age-gap flagged, so its age columns cannot be mechanical; the 178 loans both rules flag sit in *Deed (all)* and in *HMDA age-gap*. Ages are HMDA bands evaluated at midpoints, the open bands at the midpoint of a ten-year interval on the closed side, which understates the top gap. Loans whose co-applicant age is coded not-provided count as reporting a co-applicant but drop out of the three age rows. The bottom block regresses each outcome on the deed flag over the age-gap-unflagged sample, with county and year fixed effects, log income, and applicant age band, race and sex. Standard errors clustered by county in parentheses, over 27 clusters, which is few enough that the heteroskedasticity-robust alternative is reported in brackets alongside rather than suppressed. Stars follow the clustered *p*-value. \*\*\*, \*\*, \*: 1, 5, 10%.

to respondents who also report that nobody else lives in the home, which isolates a cosigner who does not occupy the property, gives 4.20%.

That is the comparable number, and it falls between the two measures built here: above the HMDA age-gap rate on the matched sample and below the augmented rate. The ordering is what one would expect. The survey measure is stricter than mere presence, since it requires the cosigner to have been *needed* to qualify, but it catches non-occupant cosigners that HMDA’s two-borrower cap and twenty-year age-gap rule both miss. An independent source using a different instrument on a different sample therefore brackets the paper’s headline rather than contradicting it. It does not validate the deed rule specifically, and it inherits the usual limits of a self-reported survey measure.

**Who the deed records add.** The surname test above cannot say whether the deed rule finds parents. A different comparison can say something almost as useful: whether the people the deed rule finds *look like* the people the age-gap rule finds. Every descriptive fact in Section 4 is computed on the age-gap measure alone, so the roughly half of the augmented prevalence that the deed records contribute has never been described. Table OA13 describes it, profiling 1,108 loans flagged by the deed rule and *not* by the age-gap rule against 1,304 flagged by the age-gap rule and against solo borrowers on the matched sample.

Read the table in two blocks. On everything that does not depend on who is listed on the application — property value, loan amount, implied equity, the interest rate and combined loan-to-value — the deed-only group differs from solo in the same direction as the age-gap group and by

TABLE OA13. Who the deed records add: profile by cosign measure

|                            | Solo    | HMDA<br>age-gap | Deed<br>only | Union   |
|----------------------------|---------|-----------------|--------------|---------|
| Observations               | 34,546  | 1,304           | 1,108        | 2,412   |
| Income (\$000s)            | 130.00  | 210.31          | 147.86       | 181.63  |
| Property value (\$)        | 385,539 | 436,557         | 443,854      | 439,909 |
| Loan amount (\$)           | 315,069 | 343,972         | 355,217      | 349,138 |
| Implied equity (\$)        | 70,470  | 92,584          | 88,637       | 90,771  |
| Interest rate (pp)         | 4.61    | 4.81            | 4.88         | 4.84    |
| Combined LTV               | 84.97   | 81.46           | 82.51        | 81.94   |
| Share with DTI $\geq 43\%$ | 24.25   | 40.74           | 32.16        | 36.79   |

*Notes:* The merged deed×HMDA sample. *Deed only* are loans flagged by the  $> 2$ -signatory rule and not by the age-gap rule. A deed is recorded only when a loan closes, so the sample carries no rejected applications and rejection variables are omitted. Levels are local to New York and to a loan-amount-matched sample; the object of interest is the comparison across columns.

a similar amount, the ratio of the two gaps running between 0.70 and 1.39. On the two variables that *do* depend on the applicant list, the groups diverge sharply: the age-gap group reports income \$80k above solo where the deed-only group reports only \$18k, and the same pattern appears in the share above a 43% debt-to-income ratio.

That divergence is the point rather than a problem, and its direction is a check. HMDA reports the *combined* income of the parties on the note and computes debt-to-income from it. A cosigner caught by the age-gap rule is on the application, so the parent’s income sits inside the reported figure; a third signatory caught only by the deed rule is not on the application — which is exactly why HMDA misses them — so the reported income excludes it. Were the deed rule instead picking up couples, siblings or co-investors already visible to HMDA, the income gap would resemble the age-gap group’s. It does not. This is the one positive piece of evidence that the deed rule finds genuinely off-application parties, and it is weaker than a relationship test would be, but it is not nothing.

**Do the sample filters select on the outcome?** Two filters in the HMDA construction funnel are applied before the definitive-outcome filter and could in principle remove denials differentially, which would put every rejection result in the paper on a sample selected on its own outcome. Replaying the funnel over the raw register and tabulating outcomes at each of the two, the answer is not the one the ordering suggests. The property-value floor removes 3,517,970 records of which 61.1% are denials against 11.7% of those it keeps, an enormous imbalance: HMDA property value is frequently missing on denied applications, where no appraisal was obtained. The sex-reported filter, by contrast, removes 21,264,447 records whose denial share, 22.7%, is close to the 21.0% it retains.

Levels are not what matters here. Fact 2, the debt-to-income leniency evidence and the rejection difference-in-discontinuities are all cosigned-versus-solo *contrasts*, and a filter that removes

denials from both arms alike shifts the level without disturbing the contrast. Splitting each filter by cosign status reverses the ranking. The property-value floor removes denials at +47.1 points among cosigned applications against +49.5 among solo, a difference-in-differences of -2.4 points against a level effect near fifty: it is almost purely a level effect, and the contrasts survive it. The sex filter, innocuous in levels, carries a differential of +4.4 points. I report that as a caveat rather than a finding, because cosign status is itself constructed from the co-applicant age field, and records with sex unreported are records with weaker applicant reporting generally, so the flag is measured with more error in precisely the set being dropped.

**What the deed measure can and cannot be used for.** One further limitation governs how the deed records are used in the rest of the paper, and it concerns geography rather than levels. Across the 26 counties with more than a hundred matched loans, the deed-based cosign rate ranges from 0.38% to 12.85%, with a mass-weighted coefficient of variation of 0.85 against 0.51 for the HMDA age-gap rate — a variance ratio of 2.8. Thirty-fold variation in the same underlying behaviour invites the question of whether the deed rule is measuring cosigning or measuring how a county clerk fills out a form.

Section 4 supplies the benchmark for answering it. Fact 1 documents that cosigning rises where houses are expensive relative to local incomes, and the HMDA measure, built on a uniform federal reporting standard, carries the price gradient strongly: the county-level correlation between the HMDA cosign rate and the local house-price index is +0.90. The deed measure does not. Its correlation with the same index is +0.31, while its correlation with the mean number of parties recorded per instrument — computed over *all* instruments, including the one- and two-party majority the > 2 rule never flags, so a property of recording practice rather than of cosigning — is +0.52. The two measures correlate with each other at only +0.34 across counties.

The deed rate therefore tracks recording practice more closely than it tracks the economic driver of cosigning, and the reverse holds for the HMDA rate. I read this as a limit on scope rather than as grounds for discarding the deed data: the records remain the only source that sees the multi-party arrangements HMDA’s two-borrower cap conceals, which is what the augmented *level* is built from, but their cross-county variation is not interpretable as variation in family behaviour. Every geographic result in the paper — Fact 1, the county map, and the tract-income gradient — accordingly rests on the HMDA measure alone, and the extrapolation from New York to the nation inherits whatever recording heterogeneity the scraped counties carry. With 26 counties none of these correlations is precisely estimated; the exercise bounds the concern rather than settling it.

**Provenance of the intensive-margin estimates.** The estimates previously reported for this sample could not be retained. They were generated on 28 October 2025, against merged county files last rebuilt on 25 June 2026, and were drawn from the New York HMDA sample estimated *without* the deed records, with heteroskedasticity-robust rather than county-clustered standard errors. Table OA14 re-estimates the specification on the current merge, separately for the three cosign measures, since they are distinct objects.

TABLE OA14. Intensive-margin outcomes on the current deed-matched sample

|                             | Log property<br>value  | Log loan<br>amount     | Interest<br>rate (pp)  |
|-----------------------------|------------------------|------------------------|------------------------|
| <b>HMDA age-gap</b>         | −0.2133***<br>(0.0329) | −0.2456***<br>(0.0352) | +0.0902***<br>(0.0164) |
| <i>implied equity gap</i>   | +0.0323                |                        |                        |
| <b>deed (2 signatories)</b> | −0.0269<br>(0.0206)    | −0.0414<br>(0.0262)    | +0.1088***<br>(0.0243) |
| <i>implied equity gap</i>   | +0.0145                |                        |                        |
| <b>augmented (union)</b>    | −0.1207***<br>(0.0264) | −0.1446***<br>(0.0300) | +0.1061***<br>(0.0182) |
| <i>implied equity gap</i>   | +0.0239                |                        |                        |
| County FE                   | Yes                    | Yes                    | Yes                    |
| Year FE                     | Yes                    | Yes                    | Yes                    |
| Demographics                | Yes                    | Yes                    | Yes                    |
| Observations                | 36,956                 | 36,956                 | 36,889                 |

*Notes:* OLS on the merged deed×HMDA sample, re-estimated on the current merge (Appendix OA3.4). Demographics are applicant age bin, race, sex and log income; standard errors clustered by county. The *implied equity gap* is the property-value coefficient less the loan-amount coefficient, the share of the property-value difference not financed by a larger loan. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

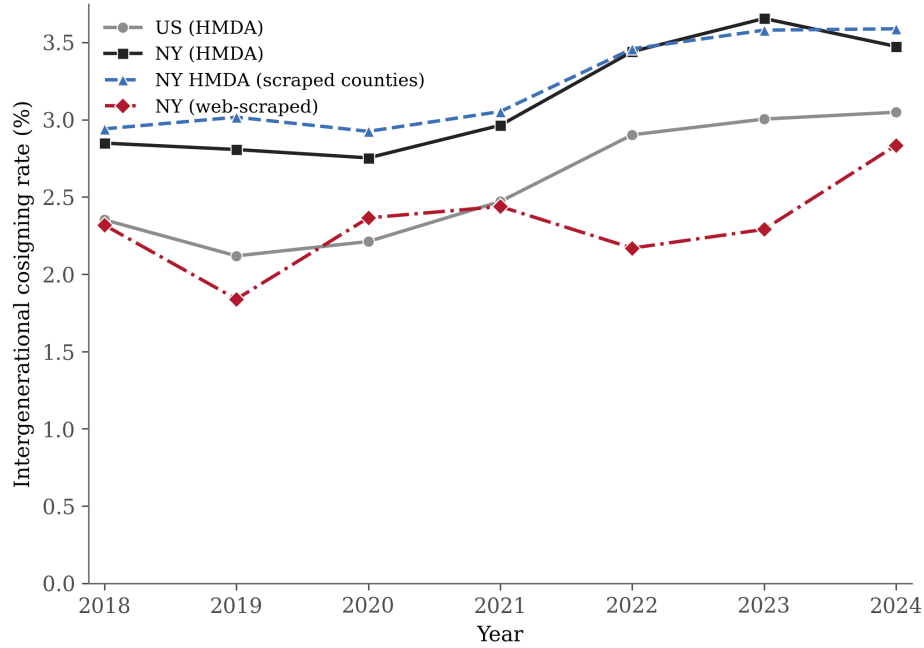
Two things follow. The conditional rate premium is unchanged in substance: the augmented measure gives +0.1061 percentage points with a county-clustered standard error of 0.0182, against the +0.1045 previously reported. The property-value and loan-amount coefficients, by contrast, change sign. That is a difference in sample rather than a reversal of the equity gap of Section 4.4: the deed-matched sample is matched *on loan amount*, which reweights the loan-size distribution mechanically, and it compares cosigned loans against all non-cosigned rather than against solo borrowers. What the body actually claims is about the difference between the two coefficients, since that difference is the part of the property-value gap not financed by a larger loan, and it is stable: +0.0239 log points here against +0.0212 in the earlier estimates. The guarantee still arrives with a down payment attached.

### OA3.5 Representativeness of the Web-Scraped Sample

Figure OA4 sets the scraped counties against the full New York HMDA series and that series against the national one. It is parallel movement rather than a common level that the argument needs, for the reason the note gives.

**How the augmented column is constructed.** The *Augmented* column of Table 3 adds a single national uplift of +3.00 percentage points to the HMDA age-gap baseline. I measure the uplift on the 36,958-loan New York deed-matched sample (Table OA10) as the share of loans flagged by *either*

FIGURE OA4. Intergenerational Cosigning Rates: US, New York, and Web-Scraped Sample



*Notes:* The level gap between the HMDA and deed series is definitional: HMDA identifies two-party cosigning by age difference, deeds capture additional signatories. Parallel trends, not levels, establish representativeness.

the deed rule or the age-gap rule, 6.53%, less the share flagged by the age-gap rule alone, 3.53%. It is applied as one national scalar rather than county by county because, as Appendix OA3.4 shows, the deed measure’s cross-county variation substantially reflects recording practice rather than family behaviour. The two columns of Table 3 therefore share no denominator — the baseline is a national count, the uplift a New York extrapolation — so the augmented column is an estimate, and no difference column is reported.

### OA3.6 Who Cosigns: Composition, Geography, and Checks on the Age-Gap Rule

Section 4.4 reports the three features of the cosigning population that Sections 5 and 6 use. This subsection collects the remainder: who the applicants are and how old the guarantor is, where cosigning happens, the evidence that the market’s thinness is not a supply-side phenomenon, and two checks on the age-gap rule that identifies intergenerational pairs.

**Thinness is not a supply-side phenomenon.** One candidate explanation for a market this thin is that few lenders write the product; they do. Of 3,883 lenders originating in the sample, the 2,348 with at least 100 originations are almost all in the business: only 2.9% never originate a cosigned loan, the median such lender’s cosigning rate is 2.37% and the 90th percentile is 4.38%. Cosigned originations are barely more concentrated than lending as a whole (Herfindahl 128 against 98; top-ten share 27.7% against 24.7%), the cosigning rate is flat across lender size deciles, and depositories and independent mortgage banks write the product at nearly the same rate (2.35%

and 2.86%). Nor do lenders keep the paper: cosigned loans are *more* likely than solo loans to be sold to Fannie Mae or Freddie Mac in the year of origination (58.7% against 53.1%) and less likely to be retained (14.3% against 17.5%, Table OA5), so cosigned lending sits further inside the loan-level price-adjustment grid than ordinary lending does, not outside it.

**Applicant composition: more women, fewer Black applicants.** Women are 49.8% of cosigned primary applicants against 38.6% of solo ones, a gap of +11.2 percentage points (s.e. 0.094). The racial pattern runs the other way and is larger in relative terms: Black applicants are 5.3% of cosigned applications against 8.0% of solo ones, so the family-guarantee channel is used least by the group with the widest homeownership gap and the thinnest parental balance sheets — the supply-side constraint of Section 8 appearing directly in the cross-section.

**Fathers cosign more often than mothers.** HMDA reports the sex of both parties, so the pair itself can be typed. Among the 280,649 intergenerational pairs in which both parties report male or female, the cosigner is male — a father, in the modal case — in 58.4% of pairs and female in 41.6%. The four cells are son–father 30.6%, daughter–father 27.8%, daughter–mother 22.2% and son–mother 19.4%. The two sides of the pair are tilted in opposite directions: sons and daughters receive a cosign in almost exactly equal numbers, which already makes daughters over-represented relative to their 38.6% share of solo applicants, while on the supply side fathers outnumber mothers by roughly three to two. 47.2% of pairs are of opposite sex, which is what makes retaining HMDA’s “Joint” category indispensable (Section 4.1). Because `co_applicant_sex` is not retained by the estimation pipeline, these shares are computed in a separate pass over the raw Loan Application Register (`empirics/utils/cosign_direction_and_sex.py`).

**Validating the age-gap rule.** The cosigning indicator infers a family relationship from an age difference, so it is worth asking whether the pairs it selects look like families. They do. Among applications where both parties report a race, 96.8% of intergenerational pairs share the primary applicant’s race, against 93.1% of same-generation joint applications. The age-gap rule selects pairs that are *more* racially concordant than married couples in the same data, which is what parent–child pairs should look like and not what unrelated co-investors would look like.

**What filing order hides.** HMDA’s “primary applicant” is a filing convention, not an economic role, and the estimation sample keeps only applications whose primary applicant is under 45. An intergenerational pair recorded with the *older* party first is therefore dropped outright. Replaying the sample construction without the age restriction turns up 94,061 such applications against the 294,333 the age-gap rule keeps, so 24.2% of two-party intergenerational pairs are filed parent-first. Counting both directions raises the two-party cosigning rate from 2.79% to 3.65%, a difference of 0.86 percentage points; Table OA7 gives the two measures year by year and age bin by age bin. The gap is widest exactly where cosigning matters most — 0.86 points on average but over three points among buyers under 25 — and it widens over the sample period, so the published series

understates both the level of cosigning and its growth. Read that as an upper bound on what filing order hides rather than as a correction: HMDA does not say which party is buying, so a parent-first pair may be a child's purchase recorded the other way round or a parent's own purchase with an adult child as co-borrower, and the two are indistinguishable in these fields. The sign is not in doubt in either case — the headline rate is a floor — which is a further reason to carry the deed records of Section 4.2, where every party to the instrument is named and filing order carries no information.

**Where cosigning happens.** Figure OA5 maps the rate across the 2,388 counties with at least 200 applications, together 98.8% of the sample. Dispersion is wide: the ninetieth-percentile county cosigns at 3.86% against 1.38% at the tenth, a ratio of 2.8 around a median of 2.29%. The geography is not the one a story about dense coastal metros would predict. Rates run highest through the mountain West and in amenity counties — Gunnison, Summit and Routt in Colorado, Gallatin in Montana, Mono in California, Dare in North Carolina, Monroe in Florida — and lowest across the interior Midwest and Plains. What those counties share is a high ratio of house prices to local earnings rather than a high price level as such, the county-level counterpart of the tract-level gradient in Table OA15 below. Counties with fewer than 200 applications are left grey rather than plotted, since a rate on a thin denominator is mostly noise and a choropleth would give it the same visual weight as a large county; 2,388 counties clear the threshold, covering 98.8% of the estimation sample.

Table OA15 reports that tract-level gradient itself, on a measure of neighbourhood income the parent's earnings cannot inflate: HMDA's applicant income cannot locate cosigning in the income distribution, because on a cosigned application that field already contains the parent's earnings. Measured instead by the tract's income position relative to its own metropolitan median, the rate is 3.14% in the lowest-income tracts, peaks at 3.50% in moderate-income ones, and falls across the remaining bins to 2.19% in the highest. Table OA16 asks the prior question, how many young adults have a parent able to help at all. The either-channel row is the headline because a cosign needs either earnings to add to the file or a balance sheet to stand behind the loan, and which channel binds differs across families. The second both-channel row proxies the parent's debt-to-income headroom by whether they own outright, a stock test standing in for a flow constraint that discards every parent carrying a mortgage they could comfortably service. Every share conditions on the parent being alive and interviewed, so young adults whose parents have died or left the survey, whose capacity is zero, sit outside the denominator: each figure is an upper bound on the population share with a capable parent.

### OA3.7 Descriptive Facts: Stability Across Stratification

Table OA17 reports the four descriptive facts of Section 4 across cell definitions: its first column is the property-value by loan-amount cells the body states, and the rest add year, state or county. The three facts separate. The interest-rate premium of Fact 3 is the most stable, positive at every level

TABLE OA15. Cosigning by the neighbourhood's income position

| Tract income relative to metro median | Applications | Cosigning rate (%) |                 | $\Delta$ (pp) |
|---------------------------------------|--------------|--------------------|-----------------|---------------|
|                                       |              | Child filed first  | Both directions |               |
| Low (<50% of metro median)            | 402,098      | 3.14               | 4.42            | 1.28          |
| Moderate (50–80)                      | 1,546,873    | 3.50               | 4.70            | 1.21          |
| Middle (80–100)                       | 2,136,092    | 3.06               | 4.06            | 0.99          |
| Middle-upper (100–120)                | 2,366,502    | 2.76               | 3.57            | 0.81          |
| Upper (120–150)                       | 2,339,769    | 2.51               | 3.22            | 0.71          |
| High ( $\geq 150$ )                   | 1,743,585    | 2.19               | 2.73            | 0.54          |

Notes: Applications grouped by the HMDA field `tract_to_msa_income_percentage`, the tract's median family income as a percentage of its metropolitan median, cut at the standard low/moderate/middle/upper points. Being a property of the tract, it is not inflated by the parent's earnings entering a cosigned file. Columns as in Table OA7.

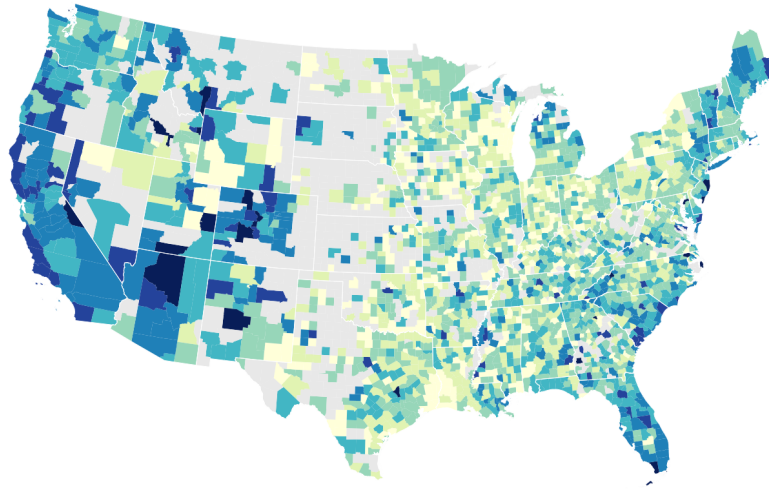
TABLE OA16. Who could cosign: parental capacity among young adults

| Parental capacity measure                                                                   | Share of young adults (%) |              |
|---------------------------------------------------------------------------------------------|---------------------------|--------------|
|                                                                                             | All aged 25–44            | Renters only |
| Parent owns their home                                                                      | 77.9                      | 70.5         |
| Parent owns outright (no mortgage)                                                          | 36.0                      | 31.0         |
| Parent housing equity $\geq$ \$50k                                                          | 68.6                      | 60.5         |
| Parent housing equity $\geq$ \$100k                                                         | 58.1                      | 50.9         |
| Parent net worth $\geq$ \$100k                                                              | 67.3                      | 58.8         |
| Parent net worth $\geq$ \$250k                                                              | 49.9                      | 42.8         |
| Parent income $\geq$ half the child's                                                       | 74.7                      | 81.6         |
| Parent income $\geq$ the child's                                                            | 51.1                      | 65.1         |
| <i>Either channel</i> : income $\geq$ half the child's <i>or</i> net worth $\geq$ \$100k    | 88.5                      | 88.4         |
| <i>Either channel, strict</i> : income $\geq$ the child's <i>or</i> net worth $\geq$ \$250k | 72.1                      | 74.8         |
| <i>Both channels</i> : equity $\geq$ \$50k and income $\geq$ half                           | 53.4                      | 51.7         |
| <i>Both, with DTI headroom</i> : outright owner, equity $\geq$ \$100k, income $\geq$ half   | 21.2                      | 19.6         |
| Person-years                                                                                | 9,382                     | 5,032        |

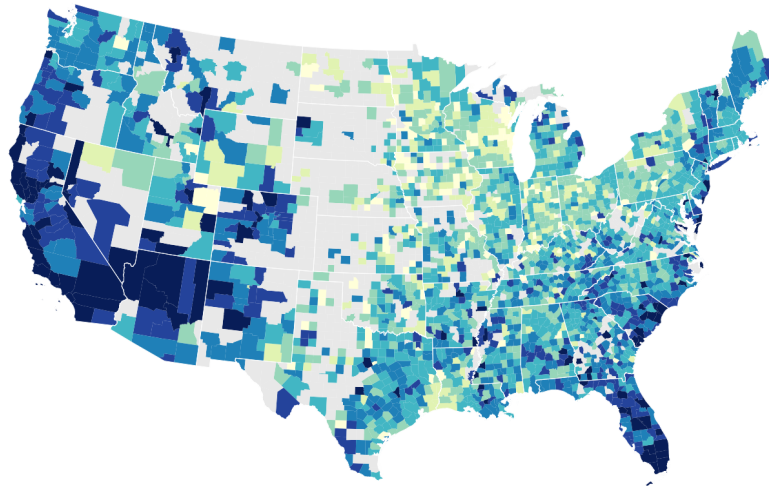
Notes: PSID matched parent–child file, 2019 and 2021 waves, weighted by the child's individual weight; young adults aged 25–44 linked to an observed parent. Housing equity is PSID imputed wealth including home equity less imputed wealth excluding it. The headline measure is the *either-channel* row; the two *both-channel* rows are reported for comparison only.

FIGURE OA5. Intergenerational cosigning by county

A. Published measure (child filed first)



B. Both filing directions



*Notes:* Cosigning rate as a percentage of applications, by county, pooled over 2018–2024. Panel A is the published child-filed-first measure, Panel B adds parent-filed-first applications (Table OA7); the two share one colour scale, so Panel B is uniformly darker and the geographic pattern should be read within each panel. Counties under 200 applications are grey; Alaska and Hawaii are in the data but off the map.

and largest under county cells. The property-value and loan-amount gaps of Section 4.4 attenuate once geography is absorbed, which is the same composition effect the fixed-effects estimates in Table OA14 pick up, though their difference, the equity gap, keeps its sign. The rejection difference of Fact 2 is the one that does not survive: it is  $-0.06$  points under year cells, so it should not be read as evidence that cosigned files are rationed more tightly, only that they are not rationed less.

TABLE OA17. Cosigning, credit access, and housing demand: stability across stratification

| Cosigned – solo         | Additional stratification |        |        |        |
|-------------------------|---------------------------|--------|--------|--------|
|                         | none                      | year   | state  | county |
| Rejection rate (pp)     | +1.26                     | -0.06  | +0.21  | +1.20  |
| Interest rate (pp)      | +0.169                    | +0.112 | +0.257 | +0.308 |
| Property value (\$000s) | +21.1                     | +16.2  | -0.7   | -1.4   |
| Loan amount (\$000s)    | +7.4                      | +3.6   | -8.8   | -5.2   |

*Notes:* The “none” column is the property-value by loan-amount cells reported in the text of Section 4; the rest add year, state or county to the cell definition. Cosigned minus solo-borrower means, excluding same-generation joint applications.

### OA3.8 Cosigning Mechanics: Additional Underwriting Detail

Section 2 summarizes how income pooling and weakest-link pricing move the admission and price margins in opposite directions. This subsection adds the underwriting mechanics behind that summary.

A primary borrower with income  $y$  and a cosigning parent with income  $y_{parent}$  qualifies on the basis of joint income  $Y_{co} = y + y_{parent}$ , which can clear the DTI threshold even when  $y$  alone cannot; for loans processed through Desktop Underwriter (DU), the automated underwriting track, the DTI is calculated on the combined income and liabilities of all borrowers. The representative credit score used for GSE pricing is determined by taking the middle bureau score for each borrower and then, for loans with multiple borrowers, selecting “the lowest applicable score from the group as the representative credit score for the loan.” That representative score is used “for pricing purposes (i.e., assessing LLPAs) on all loans” (Fannie Mae, 2026a). The *admission* side of this rule softened during the sample period while the pricing side did not: since September 2021, Desktop Underwriter applies the 620 eligibility floor to the *average median* credit score across borrowers, whereas loan-level pricing continues to use the lowest representative score (Fannie Mae, 2021). This mid-sample change widened the admission–pricing wedge and leaves the weakest-link pricing rule intact throughout 2018–2024.

For manually underwritten loans, the guide constrains the occupying borrower directly: “[u]sing only the income of the occupying borrower(s) to calculate the DTI ratio, the maximum allowable DTI ratio is 43%.” No occupant-only ceiling applies to loans underwritten through Desktop Underwriter. DU “analyzes the risk factors in the loan casefile for all borrowers” and evaluates the blended ratio within its overall risk assessment (Fannie Mae, 2020). A cosigner therefore operates

as a compensating factor in automated underwriting: debt loads that would trip the threshold for a solo applicant can be approved when a non-occupant co-borrower backs the loan.

The overlay is a ceiling that is slack for most cosigned loans rather than one that is irrelevant. Cosigned first-time buyers are materially less levered than solo ones — mean CLTV 83.7 against 86.9, and 5.4% above a 95% CLTV against 20.3% (Section 4) — and part of that gap is the overlay truncating the upper tail mechanically rather than selection on family equity. What is true is that the cap sits well above where the cosigned distribution is concentrated, so the estimated model sets the two limits equal. The approximation errs in a known direction: relative to the rulebook it gives cosigned borrowers *more* equity room than they in fact have, so the deposit relief priced in Section 8 is a conservative estimate of what the equity constraint costs a cosigning family.

### OA3.9 International Comparison of Family Guarantee Regimes

The sharp segmentation between access and pricing that characterizes the U.S. cosign market is not universal. Sorting major mortgage markets by the mechanism they use shows that take-up is substantially higher wherever a family guarantee that reaches the price or the deposit is available, and lowest in the two markets where none is.

**Private resolution: pooled pricing and collateral substitution.** Most private markets resolve the admission-pricing wedge that the U.S. leaves open. Norway offers two instruments and they divide the work between them. A *medlåntaker* (co-borrower) is assessed jointly with the buyer on income and debt-servicing capacity and adds nothing to the security, so it reaches the affordability test and stops there. A *realkausjon* is a third-party pledge of real property, typically the parents' home, and it is what satisfies the equity requirement: the pledge counts as *tilleggsikkerhet* inside the regulatory *belåningsgrad* calculation, so it relaxes the 85% loan-to-value cap directly (Smarte Penger, 2026; Finanstilsynet, 2024). A Norwegian family blocked by the deposit therefore has an instrument built for it; a U.S. family has none. Canada (17%) absorbs high-LTV risk into a national insurance pool run by the CMHC, with premiums largely invariant to credit score above an eligibility threshold, socializing the intensive-margin penalty for a risky young borrower (Government of Canada, 2024). Australia (10%) goes further: family guarantors pledge actual home equity as secondary collateral to avoid Lenders Mortgage Insurance, improving *both* margins through a direct collateral enhancement (Housing Australia, 2026; Aussie Home Loans, 2026).

The United Kingdom is the exception among the private regimes, and the instructive one. *Joint borrower sole proprietor* (JBSP) arrangements, on the order of 7% of borrowers, pool the parent's income for the affordability assessment and stop there. The co-borrower signs the mortgage but is not named on the title deeds and pledges no property of their own. The lender's security therefore remains the subject property alone, and the loan-to-value ratio, with it the deposit, is exactly what it would have been (Accord Mortgages, 2026). Nor does the parent's credit standing reach the rate. U.K. mortgages are priced off loan-to-value bands and product terms rather than a borrower-level risk grid, so a JBSP loan carries essentially the pricing of the equivalent standard

residential product at the same LTV (Duck, 2024). The parent’s income clears the door and reaches nothing else.

Two further details sharpen the parallel with the United States. Where the supporting parent still carries a mortgage of their own, several U.K. lenders cap the JBSP loan-to-value *below* the standard maximum. That is the same direction as the U.S. non-occupant overlay described above: on both sides of the Atlantic, bringing a parent onto the note can tighten the equity test rather than relax it. And the exception is beginning to appear. A small number of lenders now write JBSP above the ordinary ceiling, up to 100% of purchase price with family support (Mortgage Solutions, 2026). Such a product is close to the instrument Section 8 finds should exist, and its emergence at the edge of the market rather than at its centre is what the mechanism here would predict.

The United States and the United Kingdom therefore reach the same outcome by different institutional routes: in the U.S. the pricing rule explicitly discards the cosigner’s credit file, while in the U.K. there is no borrower-level risk price for it to enter. They are the two thinnest markets in the table.

**Public resolution: the state as cosigner.** Across much of Southern Europe and the Nordics the private guarantee is subsidized or replaced by a public one. State guarantees back roughly 24% of first-time-buyer loans in Portugal (The Portugal Post, 2026; Banco de Portugal, 2026, 2025; Banco Santander, 2026) and 20% in Italy (the Consap fund, for buyers under 36) (Finizio, 2026). Spain’s ICO *aval* guarantees up to 20% of the price for young buyers, though take-up remains low (about 15% of young buyers, barely 1% of originations) (López, 2025; Regidor, 2025), and Finland’s state-backed “ASP” system covers roughly 19% of first-time-buyer volume by pairing subsidized savings with sovereign guarantees (State Treasury of Finland, 2026). In the Iberian markets the public scheme also operates *in combination with* the family: the traditional instrument there is the parental guarantee (the Portuguese *fiadores* and the Spanish *aval*, typically provided by a parent), which long predates the state programs and continues alongside them. The quoted shares therefore describe parent-combination regimes in which the state supplements rather than replaces the family guarantor. In these bank-centric systems the state’s presence is a binary condition for access that circumvents the continuous risk-based pricing friction seen in the U.S.

**Comparability, sources and caveats.** The shares in Table 2 are orders of magnitude rather than comparable statistics. Definitions are not harmonized across countries, and the sources are of mixed type: official statistics for Canada, Norway and Finland, national press for Italy, Portugal and Spain, and industry or lender material for Australia and the United Kingdom. Three rows carry qualifications of their own, marked in the table. Norway’s two entries are both taken from the 2021 survey, the last vintage that publishes a level for each instrument on the same population and denominator: 18.5% of new amortising loans to first-time buyers had an older co-borrower outside the household, and 13.4% a guarantor or additional security in a dwelling owned by someone else (Finanstilsynet, 2021). The two overlap slightly and cannot be summed, since 1.6% of first-time buyers had both. Both series have moved since, and the co-borrowing one is volatile: it fell to 8%

by the 2024 survey before rising to 13% in 2025 (Finanstilsynet, 2025), while the guarantor series has continued to decline without a published level. Finland’s 19% is a share of first-time-buyer loan *volume* rather than of borrowers. Spain’s ICO *aval* dates only from 2024, so no period average exists: the 15% is take-up among eligible young buyers, and barely 1% of all originations. And no official U.K. series on JBSP take-up could be located, so the 7% is a trade-press estimate (Duck, 2024) and should be read as an order of magnitude; the *mechanism* entry for the United Kingdom is sourced to lender criteria (Accord Mortgages, 2026) and is not subject to that caveat. The U.S. row is the augmented HMDA-plus-deed measure of Section 4, computed on purchase applications to primary applicants under 45; HMDA carries no first-time-buyer flag, so age proxies the first-time margin, and the U.S. row is therefore not on exactly the population the peer rows describe.

Table 2, reported in Section 2, makes the point that a family guarantee which clears the access constraint and improves nothing else is close to an Anglo-American peculiarity. In seven of the nine markets some instrument that clears the access constraint, whether a creditworthy relative or a public guarantor, also reaches the price or the deposit. Norway supplies both kinds, and the contrast there is instructive: its co-borrowing reaches neither margin, exactly as U.S. cosigning does, but a family stopped by the deposit can pledge the parents’ home instead. Only in the United States and the United Kingdom does the family instrument reach neither margin with no alternative that does. The two regimes get there differently, and the distinction is worth keeping: the U.S. weakest-link rule is the only one that *explicitly* pools income for admission and then discards the cosigner’s credit file in the price. U.K. JBSP simply has no borrower-level risk price for a cosigner to enter. The mechanism is American; the outcome is shared. That wedge is the institutional foundation of what follows, and it is what makes the access value of cosigning separable from its pricing value.

A second regularity runs through the table, and it bears directly on the reform Section 8 evaluates: almost everywhere except the United States, a family or public guarantee is allowed to stand in for the *deposit*. In Norway the standard parental instrument is the *realkausjon*, a lien on the parents’ own property, used precisely when the child cannot meet the equity requirement of the lending regulation (Finanstilsynet, 2024). An Australian family pledge works the same way, securing the loan against the parents’ home so the effective loan-to-value ratio falls to the level at which lenders’ mortgage insurance no longer applies (Housing Australia, 2026; Aussie Home Loans, 2026). The public schemes are built on the same margin rather than on price: Italy’s *Consap* fund guarantees the tranche of the loan above the ordinary 80% loan-to-value ceiling, which is what makes a 100% mortgage possible for a buyer under 36 (Finizio, 2026). Spain’s ICO *aval* guarantees up to 20% of the purchase price (López, 2025), and Portugal’s scheme and Finland’s ASP pair a sovereign guarantee with the same objective (The Portugal Post, 2026; State Treasury of Finland, 2026). Even in the United Kingdom, where the guarantee instrument itself reaches neither margin, the deposit is addressed by a separate product rather than by JBSP. The Barclays Family Springboard has the helper post ten percent of the price into a linked account, so that the buyer needs no deposit at all (Duck, 2024).

The United States fails on both margins at once, and it is not quite alone, which is the useful part. A U.S. cosigner’s income clears the income test and is then discarded in the rate, and their balance sheet never touches the equity test, because the cosigner is liable for the debt rather than pledging collateral. A U.K. JBSP borrower ends up in the same position by a different route. The two regimes whose family instrument reaches neither the price nor the deposit are also the two thinnest markets in the table, while every market above ten percent lets a guarantee substitute for equity or absorbs the high-leverage risk publicly. Nine countries with very different housing and credit institutions identify nothing, so this is a pattern and not evidence. But it is the same pattern the estimated model produces from the inside in Section 8: pricing the guarantee is not what buys access, and letting it stand in for the deposit is.

## OA4 Natural Experiment: Further Robustness and Validity

### OA4.1 Mapping the Estimates to the Model

Each natural-experiment estimate is the empirical image of a structural object. The chain runs through the pricing condition (3): a common fee passes through as  $1/\mu_k$ , so the differential  $\beta_3^{co} - \beta_3^{so} = \Delta g^{reform}(1/\mu^{co} - 1/\mu^{so})$  is positive if and only if  $\mu^{co} < \mu^{so}$  (OA27) — and, because  $\mu^k$  is near one for any GSE-eligible pool, its magnitude is capped at 0.03–0.12 basis points (§5.1), below any detectable range. The reform’s rate estimate is therefore mapped to the model as a *bound* — no economically large repricing of the guarantee — rather than as a measurement of the pool-quality gap  $\bar{\mu}_{so} - \bar{\mu}_{co}$ , whose existence is established by the NSMO wedge and whose size the structural estimation identifies.

**Calibrating  $\mu^k$  from the NSMO.** Write expected repayment per dollar promised as  $\mu^k = 1 - p^k \cdot \text{LGD}$ , where  $p^k$  is the default rate of pool  $k$  and LGD is loss given default, the share of the balance a lender loses when a loan terminates in foreclosure. The NSMO wedge of Online Appendix Table OA27 gives  $p^{so} = 4.58\%$  and  $p^{co} = 4.58\% + 1.47 = 6.05\%$  on the 30/60-day rung. Two conversions bracket the object the pricing rule needs. Taking early delinquency at face value, with every spell terminating in default and  $\text{LGD} = 0.40$ , gives  $\mu^{so} = 0.982$  against  $\mu^{co} = 0.976$ ; allowing that most early delinquency cures, so that 30% of spells terminate in default, and taking  $\text{LGD} = 0.25$ , gives  $\mu^{so} = 0.997$  against  $\mu^{co} = 0.995$ .<sup>3</sup>

**The rule is the right order of magnitude on levels.** Fed these  $\mu^k$ , the pricing rule implies a level cosign premium  $r_{co} - r_{so} = (1 + \rho + g)(1/\mu^{co} - 1/\mu^{so})$  of 12 to 64 basis points. That is a real check and the rule passes it: the range brackets the compensating charge for cosigned default risk computed independently in Section 8 from the model’s own default rates (35.1–56.2 basis points) — tens of basis points either way — while sitting one order of magnitude *above* the 0.4 basis points

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<sup>3</sup>The loss-given-default band is the one Section 8 uses, from Qi and Yang (2009); the cure fraction is asserted rather than measured. Neither choice matters for the conclusion drawn here, since both conversions put the implied premium an order of magnitude above what lenders charge.

lenders actually charge. Both gaps point the same way: a rule that priced the measured risk would charge tens of basis points, and the market charges two. That mispricing is the subject of Section 8; what matters here is that the multiplicative rule, calibrated externally, produces a premium of the size the *risk* warrants.

TABLE OA18. Mapping the natural-experiment estimates to the model.

| Empirical object                                                      | Model counterpart                                                                          | Sign        |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------|
| Rate differential $\beta_3^{co} - \beta_3^{so}$ : null, bounded small | Prop. 4 via (OA27): mechanism-implied differential 0.03–0.12bp, below any detectable range | $\approx 0$ |
| Rejection differential $\beta_3^{co} - \beta_3^{so} < 0$              | Prop. 5: insulation via joint income $Y_{co} = y + y_p$ (C2)                               | –           |
| No economically large repricing of the guarantee on either margin     | The pricing rule reads the child’s file, not the guarantee                                 | bound       |
| NSMO default wedge $\hat{\gamma} > 0$                                 | Premise $\bar{\mu}_{co} < \bar{\mu}_{so}$ (trap term A)                                    | +           |
| Wedge concentrated in the GSE-priced channel                          | (3) binds via the risk-based g-fee/LLPA grid                                               | +           |

The reduced form thus returns what the model predicts at its resolution — no detectable repricing of the guarantee on either margin, with the insulating sign on rationing. The *magnitude* of the pool-quality gap  $\bar{\mu}_{so} - \bar{\mu}_{co}$  is what the structural estimation of Section 7 identifies. (That gap is the whole of (B.26); A names one component of it, the Region-2 trap loss, and the two should not be conflated.)

#### OA4.2 The Year-FE Primary: The Rate Margin’s Robustness Ladder

Table OA19 re-estimates the year-FE rate differential of Table C.1 with one design choice changed per row and everything else held fixed. The differential is a null in every row: across the variants and the four windows it spans  $-0.11$ – $+0.06$  pp, and no cell’s  $p$ -value falls below 0.66. The two choices that moved the retired estimate most do not move this one: the covariate convention, worth a third of a percentage point there (Appendix OA4.12), is worth  $-0.09$  against  $-0.06$  at  $h = \$20k$  here, and closing the post window at end-2022 — so the January-2023 LLPA re-grid never enters — returns  $-0.11$  pp, because each year carries its own intercept either way. The rejection margin’s full-sample counterpart, computed in the same pipeline as the denial variants (`rejection.variants.csv`), is  $-0.053$ – $-0.024$  across the wide grid, negative at 9 of 9 windows (smallest  $p = 0.26$ ): the insulating sign is not an artifact of the above-AMI restriction.

TABLE OA19. The Year-FE Primary Specification: Robustness Ladder (Rate Margin)

| Design variant                                    | Cosigned – solo differential, by bandwidth (\$000s) |              |              |              |
|---------------------------------------------------|-----------------------------------------------------|--------------|--------------|--------------|
|                                                   | 18                                                  | 20           | 22           | 24           |
| Year-FE primary (headline, Table C.1)             | −0.01 (0.23)                                        | −0.06 (0.23) | −0.06 (0.23) | −0.05 (0.23) |
| Donut: drop the \$5k bracket straddling the limit | −0.03 (0.25)                                        | −0.09 (0.25) | −0.06 (0.25) | −0.01 (0.25) |
| + quadratic in the running variable               | −0.10 (0.32)                                        | −0.06 (0.23) | −0.09 (0.23) | −0.07 (0.23) |
| Triangular kernel                                 | −0.09 (0.24)                                        | −0.07 (0.23) | −0.07 (0.23) | −0.07 (0.23) |
| Lender FE only (no covariates)                    | −0.06 (0.22)                                        | −0.09 (0.22) | −0.10 (0.22) | −0.09 (0.21) |
| Full sample (waiver-exposed)                      | +0.06 (0.22)                                        | +0.03 (0.22) | +0.03 (0.22) | +0.03 (0.22) |
| Post window closed end-2022                       | −0.07 (0.32)                                        | −0.11 (0.33) | −0.10 (0.32) | −0.10 (0.32) |
| Solo reweighted to cosigned LTV mix               | +0.02 (0.23)                                        | −0.03 (0.24) | −0.02 (0.23) | −0.02 (0.23) |

Notes: Each cell is the cosigned-minus-solo difference-in-discontinuities in the note rate under the year-FE primary (13) with only the named design choice changed, lender-clustered standard errors in parentheses; the headline row is the above-AMI estimate of Table C.1 ( $N$  at  $h = \$20k$ : 23,985 / 2,586). Each row is defined in the paragraph above. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

### OA4.3 The Four Pricing Cells and the Coefficient Mapping

The pricing rule of Section 3.2 — there  $1 + r_k = (1 + \rho + g)/\mu_k$  over menus  $k \in \{so, co\}$ , here  $1 + R^k = (1 + \rho_t + g_{s,t})/\mu^k$  over cosigning groups, the same index in both — gives an interest rate in each of the four cells defined by side of the conforming limit ( $s \in \{-, +\}$ , below or above) and reform timing ( $t \in \{Pre, Post\}$ ):

$$1 + R_{s,t}^k = \frac{1 + \rho_t + g_{s,t}}{\mu_{s,t}^k}, \quad \begin{aligned} \rho_t &= \rho + \Delta\rho \mathbf{1}\{t = Post\}, \\ g_{s,t} &= g_0 + \Delta g^{HB} \mathbf{1}\{s = +\} + \Delta g^{reform} \mathbf{1}\{s = +, t = Post\}, \end{aligned} \quad (OA24)$$

where  $\rho$  is the lender’s cost of funds,  $\Delta\rho$  a common macro funding shock between periods,  $g_0$  the baseline guarantee fee,  $\Delta g^{HB}$  the pre-existing high-balance fee premium for loans above the threshold,  $\Delta g^{reform}$  the 2022 fee increase, and  $\mu_{s,t}^k \in (0, 1]$  the average repayment quality of pool  $k$  in that cell. Keeping the multiplicative structure, rather than log-linearizing it away, is what allows the design to speak to heterogeneous pass-through.

Differencing the four cells recovers every coefficient of the Diff-in-Disc regression  $E[Y^k | D, Post] = \alpha^k + \beta_1^k D + \beta_2^k Post + \beta_3^k (D \times Post)$  in primitives. Under time-invariant composition,  $\mu_{s,Post}^k = \mu_{s,Pre}^k \equiv \mu_s^k$ , the above- and below-threshold changes are

$$R_{+,Post}^k - R_{+,Pre}^k = \frac{\Delta\rho + \Delta g^{reform}}{\mu_+^k}, \quad R_{-,Post}^k - R_{-,Pre}^k = \frac{\Delta\rho}{\mu_-^k}, \quad (OA25)$$

and under local continuity,  $\mu_+^k \approx \mu_-^k \equiv \mu^k$ , the common macro term cancels in their difference, leaving

$$\beta_3^k = \frac{\Delta g^{reform}}{\mu^k}. \quad (OA26)$$

The reform fee is scaled by the *inverse pool quality* of group  $k$ . Whenever cosigned pools carry lower average quality ( $\mu^{co} < \mu^{so}$ ), an ordering Fact 4 (§4.5) establishes for the borrowers’ own

repayment before any parental cover, the same reform fee passes through to a larger rate increase for cosigners:

$$\beta_3^{co} - \beta_3^{so} = \Delta g^{reform} \left( \frac{1}{\mu^{co}} - \frac{1}{\mu^{so}} \right) > 0. \quad (\text{OA27})$$

The differential is Sensitivity Decoupling on the intensive margin: heterogeneous pass-through of a *common* cost shock, generated by the level difference in pool quality rather than by any differential change in composition.

**Additivity.** The *additivity* condition of Picchetti et al. (2024) is mild here: the 2022 shock is a flat increment layered on the standing high-balance premium — cost components add in the numerator of (OA24) while pool quality scales within group — so nothing makes the reform scale the pre-existing discontinuity.

**Exposure.** High-balance status switches deterministically at the limit, so assignment is sharp. The fee is not. Fannie Mae and Freddie Mac charge it when they buy the loan, so a high-balance loan kept on balance sheet or sold privately never pays it. If the agencies buy a share  $s^k$  of pool  $k$ , (OA26) becomes  $\beta_3^k = s^k \Delta g^{reform} / \mu^k$ , and the  $\beta_3^k$  of Section 5 are intention-to-treat effects of high-balance status rather than effects of paying the fee. The cosigned pool is the more exposed one: the agencies buy 58.7% of cosigned originations in the year of origination against 53.1% of solo ones (Table OA5, all originations rather than the high-balance window alone). On the admission margin the gap runs against the result, since the more exposed pool is the one whose denials show no detectable response. On the rate margin it runs with (OA27), so a rate differential would have to be netted of the exposure gap before it could be read as pool quality; with no rate response detected in either pool, the point does not bind. The share  $s^k$  can itself respond — a lender facing the fee can keep the loan instead — and the estimates include that response.

Written out state by state, the four cells are

$$1 + R_{-,Pre}^k = \frac{1 + \rho + g_0}{\mu_{-,Pre}^k} \quad (\text{OA28})$$

$$1 + R_{-,Post}^k = \frac{1 + \rho + \Delta\rho + g_0}{\mu_{-,Post}^k} \quad (\text{OA29})$$

$$1 + R_{+,Pre}^k = \frac{1 + \rho + g_0 + \Delta g^{HB}}{\mu_{+,Pre}^k} \quad (\text{OA30})$$

$$1 + R_{+,Post}^k = \frac{1 + \rho + \Delta\rho + g_0 + \Delta g^{HB} + \Delta g^{reform}}{\mu_{+,Post}^k} \quad (\text{OA31})$$

and differencing them gives the three remaining regression coefficients in full:

$$\alpha^k = \frac{1 + \rho + g_0}{\mu_{-,Pre}^k} - 1, \quad (\text{OA32})$$

$$\beta_1^k = R_{+,Pre}^k - R_{-,Pre}^k = \frac{1 + \rho + g_0 + \Delta g^{HB}}{\mu_{+,Pre}^k} - \frac{1 + \rho + g_0}{\mu_{-,Pre}^k}, \quad (\text{OA33})$$

$$\beta_2^k = R_{-,Post}^k - R_{-,Pre}^k = \frac{1 + \rho + \Delta \rho + g_0}{\mu_{-,Post}^k} - \frac{1 + \rho + g_0}{\mu_{-,Pre}^k}. \quad (\text{OA34})$$

The intercept is the baseline (Below, Pre) rate;  $\beta_1^k$  is the pre-existing jump at the threshold, carrying both the high-balance fee and any cross-cutoff difference in pool quality;  $\beta_2^k$  carries the common macro funding shock together with any quality drift below the threshold.

**What each identifying assumption buys.** Local continuity does no work on  $\beta_1^k$ : the cross-cutoff composition in the pre-reform jump — the sorting that motivates a difference-in-discontinuities rather than a plain RD — is differenced out by *time-invariance*, not assumed to be zero. Smoothness is invoked only to cancel the macro residual  $\Delta \rho(1/\mu_+^k - 1/\mu_-^k)$  in the per-group  $\beta_3^k$ , so  $\beta_1^k$  keeps its composition term intact.

**The differential needs less than local continuity.** Written without it, the macro term in (OA27) enters only as  $\Delta \rho[(1/\mu_+^{co} - 1/\mu_-^{co}) - (1/\mu_+^{so} - 1/\mu_-^{so})]$ , which vanishes when the cross-cutoff sorting gap is *similar across cosign groups* — a weaker condition than full smoothness, and one the Difference-in-McCrory of Appendix C.3 supports, since bunching is stable and of similar magnitude in both pools.

**The denial margin.** The algebra above is for the interest rate, in which fees and the funding shock add. The headline outcome is a denial citing debt-to-income, a threshold outcome, and there time-invariance has to hold for more than pool composition. The pre-reform jump in denials compares the share of applicants past the cap just above and just below the limit, and each share depends on how many applicants sit near the cap. A rise in mortgage rates moves that mass, so the old jump — the high-balance fee’s effect and the sorting together — can change between periods with composition held fixed. What the denial estimates need is therefore that the pre-existing jump in denials would have stayed the same without the reform. The yearly event study of Figure 6 supports this over 2018–2021, when mortgage rates fell, but no pre-reform year saw a rise like the three hundred basis points of 2022–2024, so for that rise the condition is assumed rather than tested.

#### OA4.4 Selection on Observables: Balancing Tests

Even if the number of bunchers is constant, the *type* of borrowers sorting across the limit could have changed. I test for jumps in the composition of *pre-determined* borrower characteristics at the threshold using the same DiDisc specification, replacing the dependent variable with each characteristic. I restrict attention to characteristics not co-determined with the loan-amount choice — applicant income, age, gender, and whether the application is individual or joint — since loan size, property value, and LTV are jointly chosen with the running variable, so a discontinuity in

them cannot separate selection from a behavioral response. As shown in Table OA20, the cosigned-minus-solo DiDisc is statistically insignificant for every characteristic ( $p > 0.10$ ): the composition of borrowers at the cutoff did not shift differentially with the reform.

TABLE OA20. Composition Stability — Difference-in-Discontinuities on Pre-Determined Characteristics

| Characteristic       | Non-cosigned $\hat{\beta}_3$ | Cosigned $\hat{\beta}_3$ | DiDisc $\hat{\Delta}$ (cos – solo) |
|----------------------|------------------------------|--------------------------|------------------------------------|
| Log income           | −0.0533** (0.0243)           | +0.0163 (0.0760)         | +0.0851 (0.0773)                   |
| Applicant age (ord.) | −0.0163 (0.0239)             | −0.0204 (0.1055)         | −0.0228 (0.1118)                   |
| Female applicant     | +0.0208 (0.0175)             | +0.0921 (0.0877)         | +0.0514 (0.0893)                   |

*Notes:* DiDisc on pre-determined characteristics not co-determined with loan size. An insignificant cosigned-minus-solo differential ( $p > 0.10$ ) indicates no differential composition shift at the cutoff. SE clustered by lender.

Crucially, I test for *Sorting into Contracts*: if the reform caused safer borrowers to switch from solo to cosigned status to mitigate the fee hike, I would see a jump in the probability of cosigning at the threshold. My estimate ( $\beta_3^{Prob} = -0.000$ ) confirms that the share of cosigned mortgages remained unchanged at the discontinuity, ensuring that my results are not driven by endogenous subgroup selection. The same condition covers exits from the sample. A reform that moved applicants above the limit to an FHA loan, to a jumbo loan, or out of the market would thin the density just above the limit, the same signature as heavier bunching, and the Difference-in-McCrary of Appendix C.3 finds no change in that density’s jump in either pool. That test is more than a balance check. The model of Section 3 predicts that a fee increase drives Bad-type parents out of the cosign pool faster than Good-type parents, because the obligation the fee raises enters the cosigning propensity with weight  $(1 - \pi_\theta)$ . A composition response would therefore have surfaced precisely here, and its absence is a direct test of the time-invariant-composition condition on which (OA26) rests rather than a demonstration that the sample happens to be balanced.

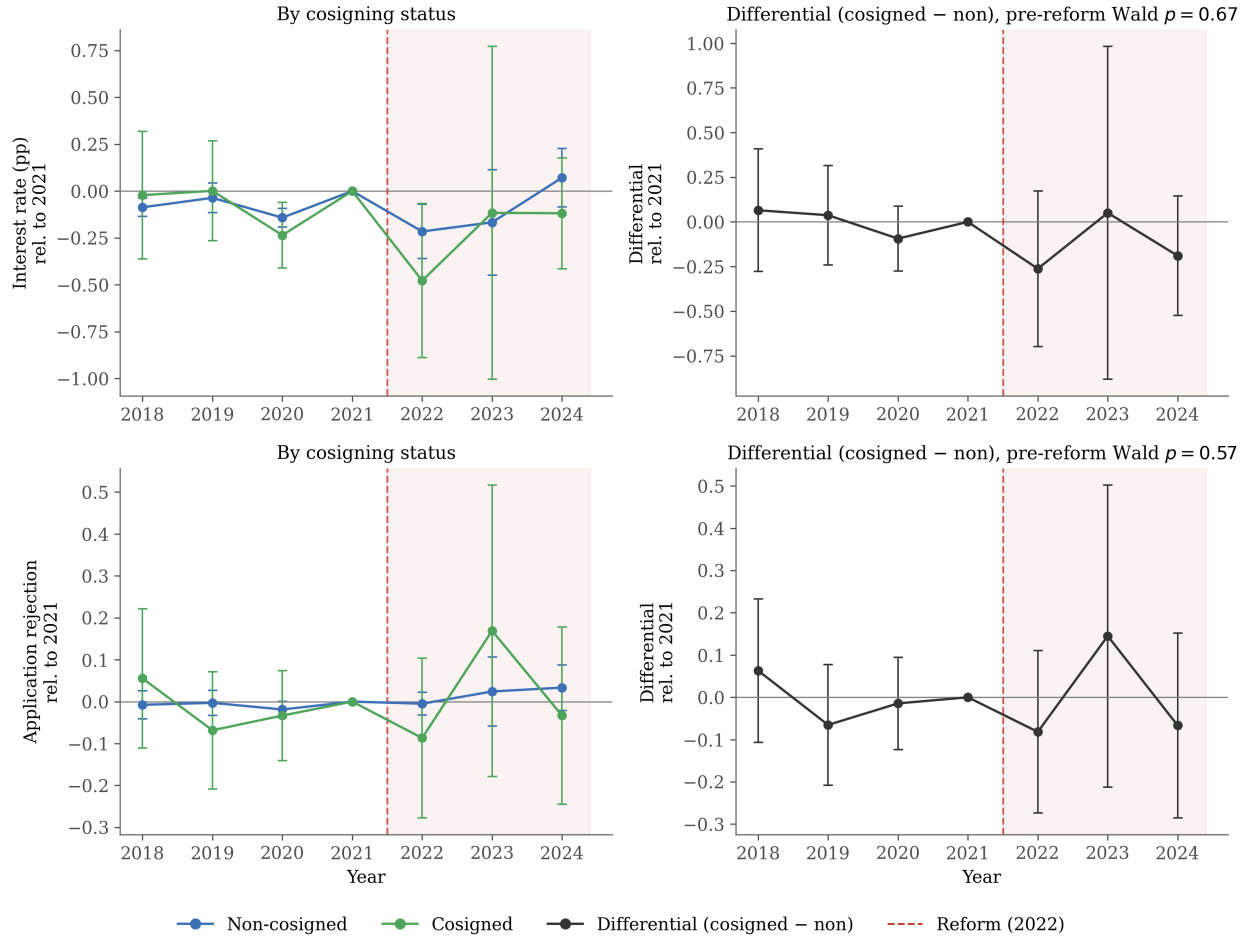
#### OA4.5 Parallel Pre-Trends: The Event Study

A second threat is that cosigned and solo borrowers might have been on diverging trajectories before the reform, so that the post-reform divergence reflects pre-existing dynamics rather than the cost shock. I assess this with an event study: I re-estimate the discontinuity at the limit separately for each year, normalized to the 2021 baseline, for each cosigning group.<sup>4</sup> Figure OA6 plots the resulting year-by-year series (left column) and the cosigned-minus-solo differential (right column). The pre-reform estimates are flat and statistically indistinguishable from zero for both

<sup>4</sup>One mid-sample institutional change touches the admission margin: since September 2021, Desktop Underwriter applies the 620 eligibility floor to the *average median* credit score across borrowers, while loan-level pricing continues to use the lowest representative score (Appendix OA3.8). The change is common to both sides of the conforming limit, so it differences out of the discontinuity; its only design implication is that 2021 is not a perfectly clean reference year on the admission margin. Re-referencing the event study to 2020 leaves the conclusion unchanged: the joint pre-reform Wald test fails to reject parallel trends on both margins ( $p = 0.46$  for the interest rate,  $p = 0.23$  for rejection).

groups, and the differential exhibits no pre-trend: joint Wald tests on the pre-reform years fail to reject equality of trends across groups for both outcomes (reported in the panel titles). These tests are precisely the validity check [Picchetti et al. \(2024\)](#) propose for the difference-in-discontinuities design, a stacked regression-discontinuity estimate across pre-treatment periods Wald-tested for a constant discontinuity, which formalizes the time-invariance of the confounding discontinuity that identification requires. The dynamic response emerges only from 2022 onward, consistent with a causal effect of the g-fee shock rather than a pre-existing divergence. The wide 2023 intervals reflect that year's thin above-limit cell and are not evidence of a reversal, and the underlying estimates, including the same event study at every headline bandwidth on the above-AMI sample, are in `event_study_grid.csv`.

FIGURE OA6. Event-study difference-in-discontinuities, by outcome



Each point is the estimated discontinuity in the outcome at the conforming limit in a given year, normalized to 2021; interest rate on top (percentage points), application rejection below (probability). Left column, the two groups separately; right column, their difference, with the joint pre-reform Wald  $p$ -value in the title. Bars are 95% intervals at the RD-optimal bandwidth; shading marks the post-reform period.

## OA4.6 Institutional Specificity: Placebo Cutoff Tests

A potential concern is that cosigners and non-cosigners may simply have different "local trends" in interest rate sensitivities or rejection probabilities that would have diverged regardless of the CLL reform. To rule out this possibility, I conduct a placebo experiment by shifting the institutional threshold to an arbitrary point in the distribution where no cost shock occurred.

Rather than rely on a single false threshold, I re-estimate my main specification across a grid of placebo cutoffs, set at fractions 0.70 through 1.30 of the actual Conforming Loan Limit (excluding the true cutoff). At each placebo the running variable is re-centered on the false threshold and observations within  $\pm\$5,000$  of the true CLL are removed, so the placebo cannot pick up the genuine discontinuity. Table OA21 reports the cosigned-minus-solo differential at every placebo. It is insignificant at the 5% level at all ten cutoffs for both outcomes (0/10); two rejection placebos (0.90 and 1.15 of the limit) clear 10%. Away from the institutional threshold, then, the two borrower pools respond to loan-size variation similarly enough that no placebo differential is resolved.

TABLE OA21. Placebo Cutoffs — Difference-in-Discontinuities at False Thresholds

| Placebo cutoff<br>( $\times$ CLL) | DiDisc $\hat{\Delta}$ (cosigned – solo) |                   |
|-----------------------------------|-----------------------------------------|-------------------|
|                                   | Interest rate (pp)                      | Rejection (prob.) |
| 0.70                              | +0.1916 (0.1544)                        | +0.0202 (0.0400)  |
| 0.75                              | +0.3106 (0.2323)                        | −0.0826 (0.0454)  |
| 0.80                              | +0.0044 (0.1812)                        | −0.0287 (0.0390)  |
| 0.85                              | −0.0380 (0.1756)                        | −0.0311 (0.0406)  |
| 0.90                              | +0.3064 (0.2287)                        | +0.0756 (0.0400)  |
| 1.10                              | +0.1455 (0.4399)                        | +0.0779 (0.0649)  |
| 1.15                              | +0.1063 (0.4836)                        | +0.1051 (0.0608)  |
| 1.20                              | +0.3955 (0.4310)                        | +0.0734 (0.0888)  |
| 1.25                              | +0.1496 (0.4144)                        | +0.0268 (0.0848)  |
| 1.30                              | +0.2062 (0.4083)                        | −0.0858 (0.1173)  |

*Notes:* Cosigned-minus-solo DiDisc at false cutoffs (fractions of the CLL), with cutoffs snapped to the \$5,000 reporting-heap grid and a per-placebo donut; observations within  $\pm\$5,000$  of the true limit removed. Differential insignificant at every placebo for both margins.

Under the retired pooled-intercept specification the true cutoff returned a positive rate differential and a negative rejection differential, whereas at every placebo threshold both differentials are indistinguishable from zero. The test is one-sided in what it can establish, however, and the placebo standard errors say how far: at the upper placebos they run 0.41–0.48, larger than the headline point estimate itself, so at those cutoffs the design could not have rejected a placebo differential equal to the true one. Insignificance at ten placebos is therefore consistent with, but not proof of, a zero placebo effect, and two of the lower placebos (0.75 and 0.90 of the limit) return point estimates near +0.31. Randomisation inference puts the headline differential at the  $p = 0.074$  percentile of the placebo distribution — suggestive, and short of conventional significance. That test runs over a finer grid of 26 false cutoffs than the ten tabulated here, since a randomisation distribution needs

more draws than a table needs rows. The placebo grid is best read as failing to find the generic loan-size divergence that would explain the headline away, rather than as ruling it out.

#### OA4.7 Sample: the Above-AMI Primary Specification against the Full Sample

The primary specification restricts to applications with income above 120% of the area median. The reason is a second fee change moving in the opposite direction: from late 2022 the first-time-buyer upfront-fee waiver zeroes the high-balance surcharge for qualifying borrowers, and the threshold is 100% of area median income in general but 120% *in a high-cost area*. This sample is high-cost by construction — a high-balance loan exists only where the conforming limit is raised — so the operative cut is the 120% one, and a restriction at the area median would leave waiver-eligible applicants in a specification that claims to be waiver-immune. HMDA income on a multi-borrower application is the blended figure, so a cosigned application is mechanically less likely to fall below any income threshold than an otherwise identical solo one, and the waiver therefore lands disproportionately on the *solo* pool: a full-sample differential compares a fee increase with a fee increase net of an exemption granted to one side. Above 120% of area median income no applicant in this sample is eligible and the shock is common by construction.

Under the retired pooled-intercept specification the same design on the full sample gives a rate differential of +0.62 to +0.71 pp at  $h \leq \$24k$  against +0.63–+0.74 on the above-AMI sample, and a rejection differential of –0.03 to –0.05 against –0.054––0.012: the two samples agree on sign and magnitude under that specification. They disagree about the composition of the rate differential. On the full sample the solo discontinuity is flat at every bandwidth (–0.19 to +0.06 pp), so the differential looks like a cosigned-only response; on the primary specification the solo jump at  $h = \$20k$  is +0.21 pp against a cosigned +0.88, so it is a gap between two positive responses. The waiver is the natural explanation — it suppresses the below-median part of the solo pool’s fee increase — and an earlier draft preferred the above-AMI decomposition on that ground. Under the year-FE primary specification the question dissolves: both samples return a null differential (full sample +0.03–+0.06 pp across  $h = \$18\text{--}24k$ , Table OA19), and the above-AMI restriction survives as the design’s waiver-immunity requirement rather than as a choice between decompositions.

#### OA4.8 A Second Falsification: the Untreated Jumbo Boundary

The placebo-cutoff grid of Appendix OA4.6 moves the threshold within the high-cost sample. A sharper test moves the *sample* instead, holding the threshold institutional. In non-high-cost counties the baseline conforming limit is a genuine boundary — loans above it are jumbo and GSE-ineligible — but the 2022 high-balance fee did not apply there. A differential at that cutoff cannot be pass-through of the reform, so a null is what the design requires.

Table OA22 does not deliver a clean null. The cosigned-minus-solo differential runs +0.32 to +0.75 pp across the grid, insignificant at every bandwidth but of the same sign and a larger magnitude than the +0.19 to +0.21 pp the treated cutoff returns when its post window is closed before the January-2023 LLPA re-grid.

TABLE OA22. Falsification: the Jumbo Boundary in Non-High-Cost Counties (Untreated)

| $h$ (\$000s) | Non-cosigned $\hat{\beta}_3$ | Cosigned $\hat{\beta}_3$ | DiDisc $\hat{\Delta}$ (cosigned – solo) | $N_{\text{non}}$ | $N_{\text{cos}}$ |
|--------------|------------------------------|--------------------------|-----------------------------------------|------------------|------------------|
| 18           | +0.9731*** (0.1161)          | +1.3006* (0.7770)        | +0.3274 (0.7700)                        | 66,780           | 2,380            |
| 20           | +0.9506*** (0.1189)          | +1.2664 (0.7867)         | +0.3158 (0.7800)                        | 67,727           | 2,396            |
| 22           | +0.7910*** (0.1200)          | +1.1999 (0.7839)         | +0.4089 (0.7746)                        | 72,786           | 2,555            |
| 24           | +0.7545*** (0.1201)          | +1.1570 (0.7819)         | +0.4025 (0.7719)                        | 73,046           | 2,564            |
| 26           | +0.7606*** (0.1036)          | +1.4386** (0.5990)       | +0.6780 (0.5982)                        | 80,493           | 2,849            |
| 28           | +0.7884*** (0.1030)          | +1.5408*** (0.5882)      | +0.7525 (0.5877)                        | 83,945           | 2,973            |

Notes: Identical design at the baseline conforming limit in NON-high-cost counties, where loans above the cutoff are jumbo (GSE-ineligible) and the 2022 high-balance fee did not apply. A null differential here rules out that the retired differential reflects generic cosigned-vs-solo divergence at loan-size thresholds. 2022-only post window. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

#### OA4.9 The DTI Margin: Stretch and Leniency

Two facts about the debt-to-income margin, documented directly from the underwriter-reported DTI in HMDA (computed on *joint* income for multi-borrower applications), pin down the mechanism behind the rejection insulation of Proposition 5. First, cosigned applications are *not* sparse near the underwriting threshold — they concentrate at and above the 43–45% range (Table OA23; Figure OA7): families stretch on joint income rather than using it to create slack, consistent with the leverage evidence from the NSMO. Second, the rejection hazard behaves asymmetrically through that range, and the asymmetry is on the passage *through* the cap rather than on the approach to it. From the 43–45 band to 46–49 the solo rejection hazard rises +0.46–+0.68 pp (largest  $p = 0.0077$ ) while the cosigned hazard does not move at all; the cosigned-minus-solo difference is –0.88–0.64 pp and is significant at 5% in all three specifications (largest  $p = 0.045$ ), including on pre-reform years alone, which is the right test of leniency as a standing institutional fact rather than a reform artefact (Panel B of Table OA23). Over the *approach* to the cap the differential runs the other way, cosigned applications closing on the threshold faster (+0.39 pp without controls), so the endpoint contrast from 36–42 to 46–49 nets the two segments against one another and returns –0.25 pp, indistinguishable from zero (Panel C). Both are reported, and the through-cap contrast is the one the insulation mechanism is about. The institutional basis is the two-track underwriting rule of Section 2: an occupant-only 43% ceiling applies on the manual track only, while the automated track evaluates the blended ratio of all borrowers within an overall risk assessment (Fannie Mae, 2020). Threshold *leniency*, not threshold sparsity, is what insulates the cosigned rejection margin — and it implies the fee shock’s predicted rejection differential is an order of magnitude below this design’s detectable range, so the near-zero rejection difference-in-discontinuities is read as a bounded null rather than a precise zero.

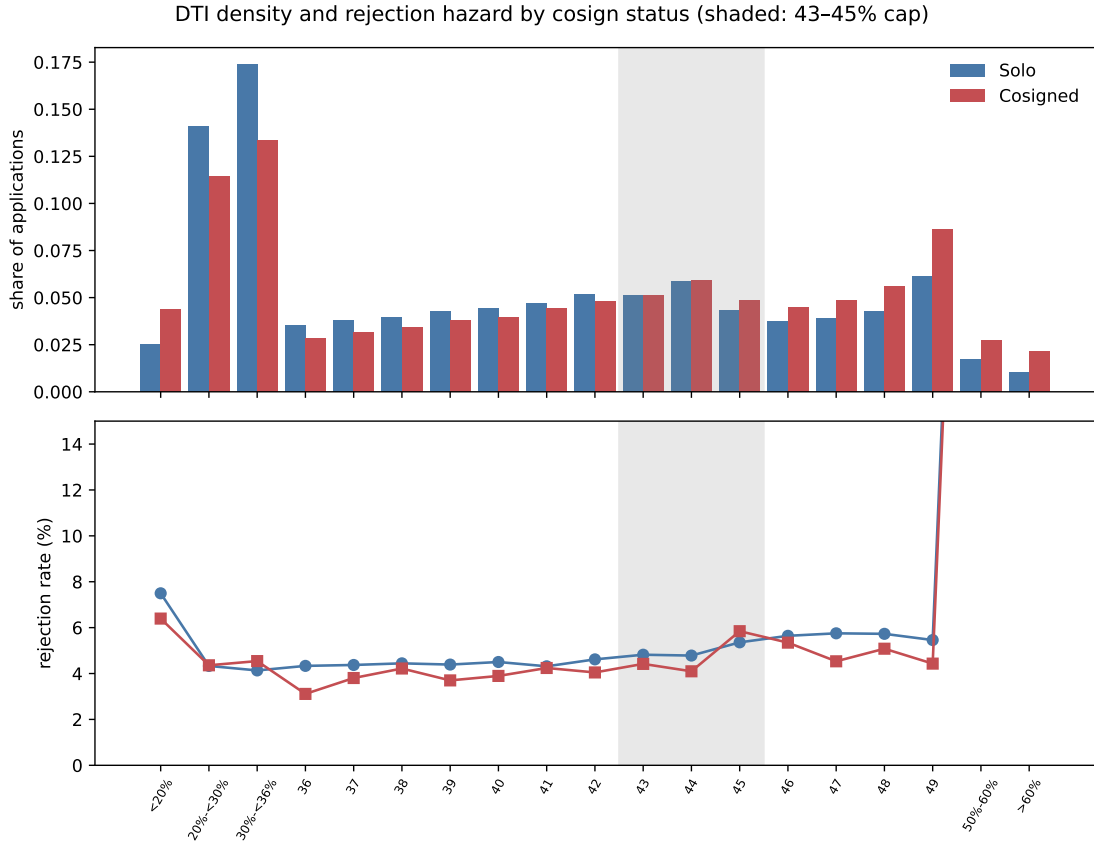
**The margin the leniency story predicts.** If the insulation works through the debt-to-income cap specifically, the reform’s rejection response should concentrate on denials that cite debt-to-income as a reason, rather than on rejection in general. Table OA26 runs the retired pooled-

TABLE OA23. DTI Density and Rejection by Cosign Status — Threshold Leniency, Not Sparsity

| Panel A: shares and raw rejection rates by DTI zone                  |                    |                |                   |                    |
|----------------------------------------------------------------------|--------------------|----------------|-------------------|--------------------|
| DTI zone                                                             | Solo share         | Cosigned share | Solo rejection    | Cosigned rejection |
| 36–42                                                                | 47.1%              | 40.0%          | 4.40%             | 3.82%              |
| 43–45 (cap)                                                          | 24.3%              | 24.1%          | 4.93%             | 4.73%              |
| 46–49                                                                | 28.6%              | 35.8%          | 5.59%             | 4.75%              |
| Panel B: passing through the cap — the step from 43–45 to 46–49 (pp) |                    |                |                   |                    |
|                                                                      | Solo step          | Cosigned step  | Cosigned – solo   |                    |
| No controls                                                          | +0.66***<br>(0.17) | +0.02          | −0.64**<br>(0.32) |                    |
| Year, county, lender FE + controls                                   | +0.46***<br>(0.17) | −0.21          | −0.67**<br>(0.32) |                    |
| Pre-reform years only (2018–21)                                      | +0.68***<br>(0.17) | −0.20          | −0.88**<br>(0.41) |                    |
| Panel C: the endpoint contrast — 36–42 to 46–49 (pp)                 |                    |                |                   |                    |
|                                                                      | Solo step          | Cosigned step  | Cosigned – solo   |                    |
| No controls                                                          | +1.19***<br>(0.17) | +0.94          | −0.25<br>(0.27)   |                    |
| Year, county, lender FE + controls                                   | +0.97***<br>(0.21) | +0.54          | −0.43<br>(0.28)   |                    |
| Pre-reform years only (2018–21)                                      | +1.26***<br>(0.22) | +1.00          | −0.26<br>(0.34)   |                    |

*Notes:* Conventional first-lien applications with underwriter-reported DTI in the integer brackets 36–49, spousal-joint co-applicants dropped;  $N = 544,622$  solo / 37,917 cosigned. HMDA's DTI is computed on *joint* income. Panels B and C are the *same* linear probability model of rejection on three DTI zones interacted with cosign status, lender fixed effects absorbed, errors clustered by lender; they differ only in the omitted zone, so the fit is identical and only the reported contrast changes. Panel B omits the cap zone and therefore reports the passage *through* the threshold, which is the margin the insulation claim is about; Panel C omits 36–42 and pools that passage with the approach to the cap, over which the differential runs the other way (cosigned applications approach the cap *faster*: +0.39 pp without controls), which is why the endpoint contrast is closer to zero. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

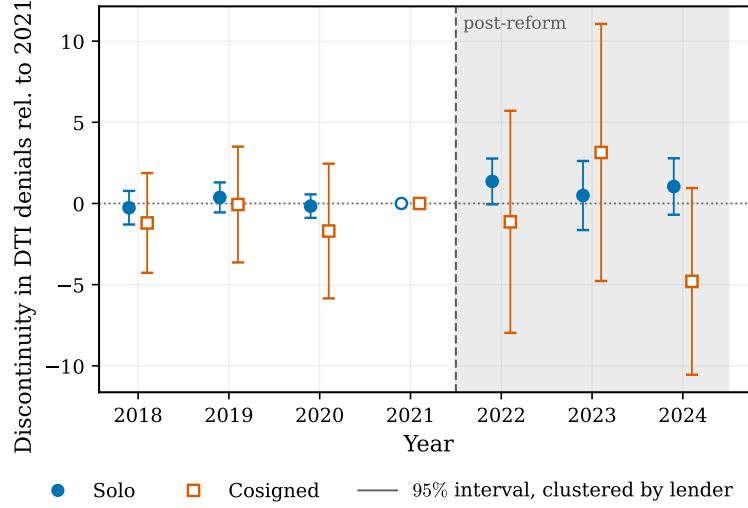
FIGURE OA7. DTI density and rejection rate, by cosign status



DTI density (top) and rejection rate (bottom) by cosign status across the underwriter-reported DTI distribution; the shaded band marks the 43–45% underwriting threshold.

intercept design on that outcome (2022-only window). The cosigned-minus-solo differential is negative at every bandwidth and shrinks monotonically as the window widens ( $-0.044$ ,  $-0.036$ ,  $-0.029$ ,  $-0.015$ ), the sign the leniency mechanism predicts, but it is insignificant throughout (standard errors  $0.022$ – $0.047$ ). Under the year-FE primary specification on the pooled window the solo leg carries conventional stars: Table OA24 puts the solo response at  $+0.96$ – $+1.95$  percentage points, significant at 5% at every window and confirmed by a score wild cluster bootstrap ( $p \leq 0.022$ ), with every cosigned estimate at or near zero. The placebo standard is applied in two batteries, and their disagreement is itself informative. The far battery — 20 cutoffs displaced  $\pm\$70$ – $160$ k, estimated at the wide windows — ranks the real cutoff's  $t$  second of twenty-one ( $p_{RI} = 0.095$ ), and its window-stability extension finds 2 of 12 eligible placebos reproducing the real estimate's full profile ( $p_{stab} = 0.23$ ): at distant thresholds this design does generate window-stable pseudo-responses. But distant is the operative word. The above-side far placebos sit in ceiling-selected counties against real area limits, and the far territory's composition differs from the cutoff's own. The near battery closes the gap: 12 placebo thresholds at  $\pm\$30$ – $80$ k — the estimate's own neighbourhood, in  $\$10$ k steps that preserve the reporting-heap geometry, each estimated at the same narrow windows as

FIGURE OA8. Year-by-year discontinuity in DTI-cited denials, by pool



The discontinuity in denials citing debt-to-income at the conforming limit, estimated year by year relative to 2021 for each pool on the above-AMI sample at  $h = \$67.1k$ , with year-specific slopes on each side, controls and lender effects; percentage points with 95% intervals clustered by lender. Hollow markers at 2021 are the normalisation; shading marks the post-reform years. The solo series is Panel A of Figure 6; no cosigned estimate is distinguishable from zero in any year.

the real cutoff, valid because no treatment discontinuity exists at any of them (below the limit nothing is treated, above it everything is) — produces a placebo distribution exactly as dispersed as the standard errors claim, and *no* placebo reaches the real estimate’s  $|t|$  at either bandwidth: the real cutoff ranks first of thirteen at both ( $p_{near} = 0.077$ ), where the local response is +2.39 pp ( $t = 2.50$ ), roughly double the wide-window estimate. The noise floor grows with distance from the cutoff; at the cutoff itself the response stands clear of it. The solo response is therefore reported as a detection, localised to the threshold’s neighbourhood, with the far battery disclosed as the bound on how far that claim travels. A *positive* cosigned response would have counted against the leniency reading, and none appears. Panel B of Figure 6 in the body draws the near battery; the year-by-year discontinuity for both pools is Figure OA8 below, where the cosigned series is flat before and after the reform (joint pre-reform Wald on the cosigned-minus-solo differential  $p = 0.90$ ; solo alone  $p = 0.76$ ). The year-by-year fit uses the headline window, \$67.1k, so that the pre-trend test runs on the sample the estimate is made on; that window is the MSE benchmark for the pooled jump, not for a single year’s, and since each year’s jump is estimated on a fraction of the observations its own optimum would be wider (the  $n^{-1/5}$  scaling puts it near \$90k) and its interval narrower. Re-estimating at  $h = \$90k$  changes nothing of substance: the solo pre-reform years stay flat (joint Wald  $p = 0.84$ ), and 2022 is +1.46 pp (s.e. 0.58) against +1.36 pp (s.e. 0.72) at the headline window. Underlying estimates: `rejection_variants.csv`, `denial_wcb_ri.csv`, `denial_ri_stability.csv`, `denial_ri_near.csv`.

The two grids behind Figure 5 are Table OA24 for the denial margin and Table OA25 for rejection, both below.

TABLE OA24. The Reform and DTI-Cited Denials

| $h$ (\$000s)      | Panel A: above-AMI (primary) |                          |                       | Panel B: full sample |                          |                       |
|-------------------|------------------------------|--------------------------|-----------------------|----------------------|--------------------------|-----------------------|
|                   | Solo $\hat{\beta}_3$         | Cosigned $\hat{\beta}_3$ | DiDisc $\hat{\Delta}$ | Solo $\hat{\beta}_3$ | Cosigned $\hat{\beta}_3$ | DiDisc $\hat{\Delta}$ |
| 40                | +1.95***<br>(0.59)           | −0.71<br>(3.31)          | −2.66                 | +0.91<br>(0.58)      | −0.77<br>(3.69)          | −1.68                 |
| 50                | +1.72***<br>(0.58)           | −1.35<br>(2.84)          | −3.07                 | +1.27**<br>(0.52)    | −0.95<br>(3.31)          | −2.22                 |
| 60                | +1.30**<br>(0.54)            | −1.62<br>(2.70)          | −2.92                 | +1.24***<br>(0.47)   | −2.61<br>(3.08)          | −3.85                 |
| 67.1 <sup>†</sup> | +1.11**<br>(0.47)            | −0.11<br>(2.58)          | −1.22                 | +0.84*<br>(0.48)     | −1.84<br>(3.08)          | −2.68                 |
| 70                | +1.08**<br>(0.47)            | +0.67<br>(2.53)          | −0.40                 | +0.79*<br>(0.47)     | −1.08<br>(3.03)          | −1.87                 |
| 76.6 <sup>†</sup> | +0.96**<br>(0.43)            | −0.14<br>(2.08)          | −1.11                 | +0.75<br>(0.46)      | −1.82<br>(2.33)          | −2.57                 |
| 80                | +0.96**<br>(0.42)            | −0.09<br>(2.00)          | −1.05                 | +0.70<br>(0.46)      | −1.86<br>(2.27)          | −2.55                 |
| 90                | +1.05***<br>(0.40)           | −0.16<br>(1.98)          | −1.21                 | +0.83*<br>(0.42)     | −1.48<br>(2.16)          | −2.31                 |

Notes: Outcome: application denied with debt-to-income cited as a reason (HMDA), in percentage points. Specification, controls and clustering as in the year-FE primary (13). Bootstrap and placebo batteries: Online Appendix OA4.9. <sup>†</sup>: this outcome's MSE-optimal bandwidths, solo then cosigned; the solo optimum is the estimate Section 5 leads with (on the heaped support, Online Appendix OA4.12). Panel A samples: 35,490–92,643 solo, 3,839–10,341 cosigned; Panel B: 76,206–213,082 and 5,147–14,433. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

TABLE OA25. Pass-Through to Application Rejection — Year-FE Difference-in-Discontinuities (Primary Specification)

| $h$ (\$000s)      | Solo $\hat{\beta}_3$ | Cosigned $\hat{\beta}_3$ | DiDisc $\hat{\Delta}$ | $p_{\Delta}^{WCB}$ | $N_{\text{non}}$ | $N_{\text{cos}}$ |
|-------------------|----------------------|--------------------------|-----------------------|--------------------|------------------|------------------|
| 30                | +0.0142 (0.0115)     | −0.0423 (0.0661)         | −0.0565 (0.0691)      | 0.47               | 35,490           | 3,839            |
| 40                | +0.0165 (0.0112)     | −0.0303 (0.0618)         | −0.0469 (0.0633)      | 0.50               | 45,527           | 4,928            |
| 50                | +0.0180* (0.0094)    | −0.0152 (0.0529)         | −0.0332 (0.0557)      | 0.59               | 55,779           | 6,028            |
| 60                | +0.0167* (0.0088)    | −0.0275 (0.0511)         | −0.0442 (0.0548)      | 0.45               | 65,700           | 7,142            |
| 70                | +0.0106 (0.0084)     | −0.0027 (0.0480)         | −0.0133 (0.0512)      | 0.81               | 75,257           | 8,264            |
| 77.2 <sup>†</sup> | +0.0117 (0.0078)     | −0.0275 (0.0464)         | −0.0393 (0.0485)      | 0.46               | 81,930           | 9,032            |
| 80                | +0.0119 (0.0077)     | −0.0284 (0.0453)         | −0.0404 (0.0472)      | 0.42               | 83,994           | 9,269            |
| 83.2 <sup>†</sup> | +0.0118 (0.0078)     | −0.0252 (0.0450)         | −0.0369 (0.0467)      | 0.46               | 86,390           | 9,555            |
| 90                | +0.0145** (0.0073)   | −0.0152 (0.0404)         | −0.0297 (0.0422)      | 0.52               | 92,643           | 10,341           |

Notes: Specification (13) on the above-AMI sample; all of calendar 2022 is coded post, HMDA reporting no origination month. Covariates and lender fixed effects as in Appendix OA4.15.1; standard errors clustered by lender,  $p_{\Delta}^{WCB}$  the differential's score wild cluster bootstrap  $p$ -value. Outcome is application rejection in probability units. <sup>†</sup> covariate-adjusted MSE-optimal bandwidths, benchmarks only. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

TABLE OA26. Differential Response of DTI-Reason Denials — Difference-in-Discontinuities

| $h$ (\$000s) | Non-cosigned $\hat{\beta}_3$ | Cosigned $\hat{\beta}_3$ | DiDisc $\hat{\Delta}$ (cosigned – solo) | $N_{\text{non}}$ | $N_{\text{cos}}$ |
|--------------|------------------------------|--------------------------|-----------------------------------------|------------------|------------------|
| 62           | +0.0123* (0.0067)            | −0.0317 (0.0440)         | −0.0440 (0.0473)                        | 137,848          | 9,030            |
| 82           | +0.0070 (0.0059)             | −0.0288 (0.0307)         | −0.0357 (0.0322)                        | 179,269          | 11,803           |
| 102          | +0.0066 (0.0049)             | −0.0227 (0.0219)         | −0.0293 (0.0230)                        | 221,094          | 14,507           |
| 123          | +0.0063 (0.0044)             | −0.0084 (0.0220)         | −0.0147 (0.0223)                        | 264,454          | 17,361           |

Notes: Outcome = application denied *with debt-to-income cited as a denial reason* (HMDA). Threshold leniency predicts the cosigned insulation concentrates on this margin: the DTI cap binds solo applicants but not cosigned ones (Selling Guide B2-2-04). 2022-only post window; controls and clustering as in the retired pooled-intercept tables. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

#### OA4.10 The Mechanical Bound on the DTI-Denial Response

Section 5 reads the solo DTI-denial response of +1.11 pp against the largest response the payment arithmetic could deliver. This subsection works that bound in full, on the estimation sample. Every input is measured on solo applications just above the limit within the  $h = \$67.1\text{k}$  window, above-AMI, in the post-reform years 2022–24, so the bound is sized where the estimate lives rather than on national distributions.

**From fee to payment.** The fee is tiered by loan-to-value ratio: 75 basis points of the loan amount above 75% and 50 basis points at 60–75% (the untreated tier at or below 60% is excluded from the sample, Online Appendix OA4.14). In the cell, 83% of loans sit in the upper tier and 17% in the lower, an average upfront fee of 71 basis points. The enterprises annualise upfront fees by a present-value multiplier of four (Federal Housing Finance Agency, 2025, fn. 7), so the fee is 0.18 pp of note rate, against the 0.19 pp of the top tier alone that the body uses as an upper bound. At the cell’s mean 2022–24 note rate of 5.73%, a 30-year payment per dollar borrowed is  $m/(1 - (1 + m)^{-360})$  with  $m$  the monthly rate; raising the rate by 0.18 pp raises that payment by 1.9%.

**Who crosses the cap.** If the mortgage is the borrower’s only debt, the debt-to-income ratio scales with the payment: a ratio  $r$  becomes  $r(1 + g)$  with  $g = 1.9\%$ , and it crosses the 50% cap when  $r \geq 50/(1 + g) = 49.05$ . HMDA reports ratios between 36 and 49 as whole numbers without stating whether the exact ratio is truncated or rounded into the bin. Taking ratios as uniform within a bin, the share of the reported-49 bin that crosses is  $50 - 49.05$ , that is 95%, under truncation (the bin is  $[49, 50)$ ), and  $49.5 - 49.05$ , that is 45%, under rounding (the bin is  $[48.5, 49.5)$ ). Nothing in the 48 bin crosses under truncation; under rounding only its top sliver does, which I ignore.

**The bound.** In 2022–24, 5.3% of solo applications just above the limit reported a ratio of 49, up from 2.5% before the reform as interest rates rose. That is the observed post-reform share, not a counterfactual: a 1.9% payment increase moves ratios of 49.05–49.99 out of the bin and ratios of 48.1–48.9 into it, so the fee leaves the bin’s size roughly unchanged and no imputation is needed.

Multiplying, the fee could have pushed between  $5.3 \times 45\% = 2.4$  pp and  $5.3 \times 95\% = 5.0$  pp of solo applications across the cap. The estimate of +1.11 pp sits inside that range: a share of between 0.22 and 0.47 of the applicants the fee pushed over the cap were denied for it. Other debts make the payment a smaller part of the ratio, so fewer cross; the bound is therefore an upper bound and the denied share a lower bound.

**The cosigned comparison.** The same arithmetic on cosigned applications just above the limit gives a reported-49 share of 8.9% in 2022–24 (6.2% before the reform), 1.69 times the solo share, and a mechanical bound of 4.0 to 8.4 pp. The fee’s push therefore reaches a larger part of the cosigned pool than of the solo pool, and the cosigned denial response is -0.11 pp (s.e. 2.58): no response the data can detect where the arithmetic predicts a larger one than for solo applicants. That is the leniency of Section 3, the hazard shifted out by the guarantor, though the cosigned standard error is too wide to size it.

#### OA4.11 Direct Evidence on the Mechanism: The Default Wedge

The body’s pass-through results bound the market’s repricing of the guarantee at economically small, so the premise doing the work in the mechanism is not a priced gap but a real one. I verify it directly — that cosigned pools *are* riskier,  $\bar{\mu}_{co} < \bar{\mu}_{so}$  — using external default data. This step is what distinguishes genuine adverse selection from a benign alternative in which the two pools share the same mean risk but differ only in variance: only the former predicts a positive conditional default differential.

I use the National Survey of Mortgage Originations (NSMO) Public Use File, which contains a direct question on whether the borrower *had to add a co-signer to qualify*, together with loan performance and underwriting variables. The primary outcome is whether the loan is *ever 30/60-day delinquent*. Terminal outcomes (foreclosure, charge-off) understate borrower risk because the cosigner’s role is precisely to cure missed payments, so the loan performs even when the underlying borrower is risky; early delinquency is the rung a later cure does not erase. I therefore read the early-delinquency wedge as the primary estimate and the 90+-day and terminal rungs as a severity ladder. I estimate  $\text{default}_i = \gamma \text{cosign}_i + X_i' \beta$  on pooled cohorts with survey weights, where  $\gamma > 0$  conditional on observables is the adverse-selection wedge.

**Measurement: the delinquency flags are administrative and pre-cure.** The performance outcomes in the NSMO Public Use File are not survey self-reports: they are servicer-reported credit-record variables appended from the National Mortgage Database — a quarterly panel of point-in-time severity codes (`perf_status`) running from “30 to 59 days past due” through foreclosure and charge-off.<sup>5</sup> The ever-30/60-day indicator scans this panel, so a delinquency spell is retained permanently even when the loan is subsequently brought current. This is precisely the pre-cure

<sup>5</sup>FHFA, *Appendix C: National Survey of Mortgage Originations Public Use File Codebook and Unweighted Tabulations*, NMDB Technical Report 2, v7.0 (May 2026), pp. C-146–C-148, <https://www.fhfa.gov/data/nsmo>.

object the mechanism requires: a cosigned borrower who misses payments and is then rescued by the family still appears delinquent at the early rungs, while the terminal codes record only the failures the cure did not absorb. Two measurement caveats both work against finding a wedge rather than for it. Because the archives are quarterly, a short 30–59-day spell that begins and cures between archive dates is missed entirely — and cosigned spells, cured faster by construction, are missed more often than solo spells — so the early-delinquency wedge is, if anything, attenuated (90+-day spells necessarily span an archive date and cannot be missed). And a family rescue that arrives before day 30 never registers at all, so the measured wedge understates the cosigned pool’s underlying missed-payment propensity. (Pandemic-era forbearance, 2020–21, additionally suppressed reported delinquency for both pools.)

Two estimators carry the result, and the difference between their headline numbers is attribution, not disagreement. The estimator of record, which calibrates the model mapping of Appendix OA4.1, is the Firth logit, the bias-reduced choice for a sample with 124 cosigned delinquency events: its within-cohort average marginal effect is +1.47 pp ( $p < 0.001$ ). Table OA27 reports the companion linear-probability ladder on the same sample, whose within-cohort wedge is +1.36 percentage points (significant at the 5% level), a 30% increase over the 4.58% baseline rate: under either estimator, cosigned borrowers are substantially more likely to fall behind. Conditioning on observable credit risk (FICO and CLTV) attenuates the wedge to +0.56pp, with a standard error of 0.56 — a point estimate of the magnitude the mechanism would predict, and one this sample cannot separate from zero.

**What this sample could have detected.** The decomposition rests on 124 cosigned delinquency events, and it is worth stating what that buys rather than leaving the reader to infer it from a standard error. At s.e. = 0.56pp, a two-sided 5% test has 80% power against a true residual wedge of about 1.6pp and essentially none against residuals below 1pp. The residual the mechanism plausibly implies is in the latter range. A null here is therefore close to uninformative: the honest statement is that the private-information component is not identified in these data, in either direction, and no conclusion — including the conclusion that it is absent — should be drawn from the insignificance. The estimate is robust in magnitude to a logit specification (average marginal effect +0.71pp) and to controlling for subordinate financing, but robustness of an imprecise estimate is not precision.

What the NSMO does establish, at conventional levels and on the outcome the mechanism cares about, is the raw wedge: cosigned pools reach early delinquency substantially more often than solo pools, so  $\bar{\mu}_{co} < \bar{\mu}_{so}$ . That is what the pricing argument requires. It also shows that most of the measured gap is *observable* risk — lower-FICO, higher-CLTV borrowers admitted through the joint-income door. This is sufficient for the paper’s mechanism and worth being explicit about: selection on observables through an admission rule that reads joint income, priced by a rule that reads the weakest single file, produces mispricing without any hidden information. A private-information component would reinforce that account; establishing one requires either a larger sample (a pooled vintage stack, or the restricted-use National Mortgage Database) or a design rather than a control

set, and I do not claim it here.

TABLE OA27. The Adverse-Selection Wedge in Default — Cosigners vs. Solo Borrowers (NSMO)

| Outcome                         | Baseline | Cosign coefficient $\hat{\gamma}$ (pp), controls ladder |                 |                 | Rel. (Cohort) |
|---------------------------------|----------|---------------------------------------------------------|-----------------|-----------------|---------------|
|                                 |          | +Cohort                                                 | +FICO/CLTV      | +Full           |               |
| Ever 30/60-day delinquent       | 4.58%    | +1.36**<br>(0.59)                                       | +0.56<br>(0.56) | +0.47<br>(0.56) | +30%          |
| Ever 90+ days delinquent        | 1.78%    | +0.51<br>(0.35)                                         | +0.22<br>(0.35) | +0.18<br>(0.35) | +29%          |
| Terminal (FC / charge-off / BK) | 0.28%    | +0.08<br>(0.11)                                         | +0.03<br>(0.11) | +0.02<br>(0.11) | +29%          |

*Notes:* Linear-probability estimates from the NSMO Public Use File, cohorts pooled with survey weights on a constant sample ( $N = 42,631$ ; 2,651 cosigned), cosigning being the survey item “had to add a co-signer to qualify”.  $\hat{\gamma}$  is in percentage points against column 1’s baseline; the last column is  $\hat{\gamma}_{+Cohort}$  over that baseline. Ladder: +**Cohort** wave and metro-LMI fixed effects, +**FICO/CLTV** observable credit risk, +**Full** DTI, PTI and income. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

#### OA4.12 Bias-Aware Inference on a Heaped Running Variable

The conforming loan limit is bunched, and lender originations heap on the \$5,000 HMDA reporting brackets, so the running variable is discrete and an MSE-optimal regression-discontinuity bandwidth in the sense of Calonico et al. (2014) is not well grounded. Even an inference-optimal (coverage-error-rate-minimizing) bandwidth (Calonico et al., 2020) presumes a continuous running variable; with a discrete, heaped support conventional local-polynomial confidence intervals undercover regardless of the window, and one response is bias-aware inference rather than a finer bandwidth (Kolesár and Rothe, 2018). I therefore report bias-aware (“honest”) inference in the sense of Armstrong and Kolesár (2018) alongside the conventional estimates: the conventional standard error is augmented with a worst-case bound on the local smoothing bias over the class of conditional mean functions whose second derivative is bounded by  $M$ , and the estimate and its honest interval are traced across a grid of bandwidths. Table OA28 reports the result. The interest-rate differential is +0.41 to +0.44 pp at the narrow windows the discrete support allows, attenuating toward +0.15 as the window widens, and the rejection differential is a tight near-zero. The honest intervals are wide and include zero — they deliberately price in worst-case curvature on a heaped support.

**Why this was an alternative and not the headline.** The bias-aware route imposes a second choice that the conventional one does not, and in the opposite direction from the usual reading. RDHonest accepts no covariates, so the honest pipeline residualises the *outcome* on the control set before the discontinuity is estimated and then runs the discontinuity on the raw running variable — it never touches the running variable or the treatment. Frisch–Waugh equates that to a covariate-adjusted local regression only if both sides are residualised, so the two are different estimands rather than one estimand with and without a bias correction, and they coincide only under local orthogonality of the controls, which fails here. The paper’s covariate convention is the standard

covariate-adjusted local regression (Calonico et al., 2019), which is what the estimates in Section 5 report. Bias-awareness itself is cheap: holding the estimator fixed it costs on the order of 0.01pp. The distance between +0.67 and +0.24 is almost entirely the covariate convention ( $\approx 0.33$  of it), then the kernel ( $\approx 0.09$ ), with the residualisation frame worth 0.01–0.03 and county-versus-state fixed effects and loan term nothing at all. Two further prices are worth naming. The honest pipeline pools the residualisation across cosigned and solo applications, because a cosigned post-reform cell of a few thousand observations cannot support a lender-effect vector estimated on both pools separately; a formal test rejects that pooling on the numeric controls alone. And the bounded-misspecification-error estimator of Kolesár and Rothe (2018), the natural discrete-support alternative, is computed on every call in this pipeline and used by none of them: its bias bound on this design is roughly  $\pm 12$  pp on the rate margin and larger than 100 pp on rejection, which is not an inference anyone can report.

TABLE OA28. Bias-Aware (Honest) Inference for the Difference-in-Discontinuities

| Bandwidth $h$ (\$000s)                        | DiDisc $\hat{\Delta}$ (cosigned – solo) | Honest 95% CI    |
|-----------------------------------------------|-----------------------------------------|------------------|
| <i>Panel A: Interest rate (pp)</i>            |                                         |                  |
| 20                                            | +0.413                                  | [−0.143, +0.968] |
| 25                                            | +0.436                                  | [−0.110, +0.982] |
| 30                                            | +0.176                                  | [−0.265, +0.616] |
| 40                                            | +0.137                                  | [−0.242, +0.516] |
| 50                                            | +0.172                                  | [−0.168, +0.512] |
| <i>Panel B: Application rejection (prob.)</i> |                                         |                  |
| 50                                            | −0.039                                  | [−0.147, +0.069] |
| 65                                            | −0.043                                  | [−0.141, +0.054] |
| 82                                            | −0.044                                  | [−0.135, +0.048] |
| 100                                           | −0.046                                  | [−0.129, +0.037] |

*Notes:* Bias-aware (honest) inference (Armstrong–Kolesár) on the heaped running variable: the differential with worst-case curvature bias bounded by  $M$ , traced across bandwidths. Honest intervals deliberately price in worst-case curvature and are wide.

**What the bandwidth grid in that table is not.** Each row fixes the worst-case bias at  $B_{\max}$  and sets the curvature bound accordingly,  $M = 2B_{\max}/h^2$ . The bound therefore falls with the square of the window, and each row assumes a *different* smoothness class rather than re-estimating one object under one class at several windows. Read the grid as a menu — what the differential looks like if one is willing to assume this much curvature over this much of the support — and not as a robustness check, which is what a constant- $M$  grid would be.

Two further things this table cannot show on its own. First, the bias-aware point estimate is insensitive to the curvature bound itself: varying  $B_{\max}$  widens the interval and leaves the point unchanged. What moves it is the sample and the data window. The +0.41 reported here is the full sample; the +0.28 from the primary-spec pass and the +0.24 from the common-window grid of Table OA35 are the above-AMI sample at successively wider windows. Only the step from +0.28 to

+0.24 is a window effect holding the sample fixed; the larger step is sample and window together, and I do not attribute it to either alone. Table OA29 sets every estimator of this single differential side by side, with the interval that belongs to each, and names the retired headline.

TABLE OA29. Reconciling Every Estimator of the Interest-Rate Differential at  $h = \$20k$

| Estimator                 | What it is                                                          | $\hat{\Delta}$ | 95%      |
|---------------------------|---------------------------------------------------------------------|----------------|----------|
| Conventional, above-AMI*  | Controls in the local regression (CCFT), donut, lender-clustered SE | +0.6669        | [+0.026, |
| Conventional, full sample | Same estimator, no above-AMI restriction                            | +0.6651        | [+0.052, |
| Bias-aware, rd_win=\$60k  | Residualise-first, Armstrong–Kolesár conservative bound             | +0.2760        | [−0.318, |
| Bias-aware, common window | Residualise-first, rd_win=\$125k, both margins on one grid          | +0.2408        | [−0.351, |
| Bias-aware, appendix grid | Residualise-first, separate pipeline, full sample                   | +0.4130        | [−0.143, |

*Notes:* Five estimators of the same cosigned-minus-solo rate differential at  $h = \$20k$ , differing in covariate convention, sample and data window row is the retired pooled-intercept headline (Appendix OA4.15.1; the year-FE primary is Table C.1). The 0.43pp gap between the conventional inference, which costs about 0.01pp; the decomposition is in the paragraph above.

**The admission margin.** The same machinery applied to DTI-cited denials, the outcome Section 5 leads with, adapts the single-cutoff procedure of Kolesár and Rothe (2018) to a difference of two jumps. Each pool–period cell is fitted separately at a common bandwidth and curvature bound; the response is the post-reform cell less the pre-reform cell; the standard error clusters the per-lender scores across the two cells; and the worst-case bias of the difference is bounded by the sum of the two cells’ bounds, which is conservative because it lets the two periods curve in opposite directions. The outcome is residualised on the control set and on lender and year effects, the closest the procedure comes to the year-effect specification (13), which it cannot reproduce: it takes no covariates, no year-specific slopes and a triangular kernel, so the honest point estimate at the solo MSE-optimal bandwidth, +1.56 pp (s.e. 0.50), is not the +1.11 pp of the body. Residualising on lender effects alone gives +1.43 pp, so the year effects are not what separates them.

The curvature bound has to be chosen, not estimated; no data-driven choice can be honest (Armstrong and Kolesár, 2018). Kolesár and Rothe (2018) recommend choosing it from what is plausible in the application, reporting a range, and reading the bound through the heuristic that it lets the conditional mean depart from a straight line by at most  $ML^2/8$  over any interval of length  $L$  on one side of the cutoff. The bound the body quotes,  $B_{\max} = 1$  pp, lets the denial rate stray 0.25 pp from a line over any \$67.1k interval, a quarter of its pre-reform level of 0.98%; the rate in fact moves by about 0.2 pp across the whole window before the reform. Table OA30 reports the honest intervals across the bandwidth grid at that bound and at half of it. The solo interval excludes zero at all 9 windows under both; no cosigned interval excludes zero under any bound, including the conventional one.

Tracing the bound at the optimal bandwidth gives the breakdown value. The solo interval excludes zero until the denial rate is allowed to stray from a straight line by 0.52 pp over a \$67.1k interval (an extrapolation error of 2.06 pp at the window’s edge), half its pre-reform level and more than twice the 0.2 pp by which the rate actually moves across the window before the reform.

TABLE OA30. Bias-Aware (Honest) Inference for the DTI-Denial Response

| $h$ (\$000s)      | Solo         |                |                      |                    | Cosigned     |                    |
|-------------------|--------------|----------------|----------------------|--------------------|--------------|--------------------|
|                   | Estimate     | Conventional   | Honest, Bmax = 0.5pp | Honest, Bmax = 1pp | Estimate     | Honest, Bmax = 1pp |
| 30                | +1.90 (0.71) | [+0.52, +3.29] | [+0.46, +3.35]       | [+0.30, +3.51]     | -0.92 (3.09) | [-7.03, +5.20]     |
| 40                | +1.91 (0.61) | [+0.71, +3.10] | [+0.66, +3.16]       | [+0.51, +3.31]     | -0.67 (2.77) | [-6.15, +4.81]     |
| 50                | +1.84 (0.56) | [+0.74, +2.94] | [+0.69, +3.00]       | [+0.55, +3.14]     | -0.73 (2.48) | [-5.64, +4.18]     |
| 60                | +1.68 (0.52) | [+0.65, +2.70] | [+0.60, +2.76]       | [+0.45, +2.90]     | -0.93 (2.26) | [-5.41, +3.55]     |
| 67.1 <sup>†</sup> | +1.56 (0.50) | [+0.57, +2.54] | [+0.52, +2.60]       | [+0.37, +2.75]     | -0.92 (2.15) | [-5.20, +3.36]     |
| 70                | +1.50 (0.49) | [+0.53, +2.47] | [+0.47, +2.52]       | [+0.33, +2.67]     | -0.83 (2.11) | [-5.03, +3.37]     |
| 76.6 <sup>†</sup> | +1.40 (0.48) | [+0.47, +2.34] | [+0.41, +2.40]       | [+0.27, +2.54]     | -0.59 (2.05) | [-4.67, +3.48]     |
| 80                | +1.35 (0.46) | [+0.44, +2.26] | [+0.38, +2.32]       | [+0.23, +2.46]     | -0.60 (1.96) | [-4.50, +3.31]     |
| 90                | +1.26 (0.44) | [+0.40, +2.11] | [+0.34, +2.18]       | [+0.19, +2.32]     | -0.52 (1.80) | [-4.13, +3.08]     |

*Notes:* Outcome: application denied with debt-to-income cited, percentage points; above-AMI sample. Each pool-period cell is an RDHonest local linear fit (triangular kernel, jump at the +\$2,500 donut cutoff) on the outcome residualised on the control set and lender and year effects; the response is the post-reform cell less the pre-reform cell, its standard error clusters per-lender scores across cells, and its bias bound is the sum of the two cells' worst-case biases. Intervals are 95%; Bmax is the largest extrapolation error allowed at distance  $h$ , equivalently a departure of Bmax/4 from a straight line over any interval of length  $h$  on one side of the cutoff (Kolesár and Rothe, 2018). Conventional: Bmax = 0. <sup>†</sup>: MSE-optimal bandwidth of Table OA24. Standard errors in parentheses.

RDHonest's rule-of-thumb bound, which reads the \$5,000 reporting brackets as curvature, allows 11 times that and yields an interval of  $[-7.4, +10.5]$  pp that excludes nothing; on a heaped support the rule of thumb violates its own premise, that local smoothness is no worse than smoothness at large scales.

Second, the per-group jumps are not the reassurance an earlier draft of this appendix took them for. On the common window at  $h = \$20k$  the cosigned jump is +0.81 pp  $[+0.21, +1.41]$  — but the *solo* jump is +0.57 pp  $[+0.28, +0.86]$ , also bounded away from zero, and bounded away from zero at three of the six windows against the cosigned jump's one. Both pools move under bias-aware estimation, and both move by three to four times full pass-through of the entire reform fee. That is the case for the difference-in-discontinuities, since whatever inflates both jumps is common and differences out. It is not a case for reading either level as pass-through, and it rules out the account in which cosigned borrowers responded to the reform and solo borrowers did not.

#### OA4.13 A Temporal Placebo: The False Reform Year

The event-study estimates (Figure OA6) and the spatial placebo-cutoff grid (Appendix OA4.6) both speak to parallel trends; I add a direct temporal placebo. Restricting the sample to the pre-reform years and assigning a counterfactual reform in 2020 — two years before the genuine g-fee increase — I re-estimate the design. As Table OA31 shows, the per-group level coefficients capture ordinary pre-period dynamics — the non-cosigned level is +0.05 and statistically significant, reflecting rate movements common to the pool around the false date — but the object the design identifies, the cosigned-minus-solo *differential*, is small and insignificant under both constructions ( $-0.01$ ,  $p = 0.93$  preserve;  $+0.03$ ,  $p = 0.79$  pooled). No differential discontinuity arises at a date with no cost shock, reinforcing that whatever the retired specification measured was tied to the

2022 window rather than to a pre-existing divergence between the two pools.

TABLE OA31. Placebo Reform Year — Difference-in-Discontinuities at a False (Pre-Reform) Treatment Date

| Estimate          | $\hat{\beta}_3$ (above $\times$ false-post) | SE     | p-value |
|-------------------|---------------------------------------------|--------|---------|
| Non-cosigned      | +0.0502                                     | 0.0211 | 0.018   |
| Cosigned          | +0.0405                                     | 0.1138 | 0.722   |
| DiDisc (preserve) | −0.0097                                     | 0.1151 | 0.933   |
| DiDisc (pooled)   | +0.0270                                     | 0.1005 | 0.788   |

*Notes:* Pre-reform sample with a counterfactual reform in 2020. An insignificant cosigned-minus-solo differential confirms no differential discontinuity at a date with no cost shock.

#### OA4.14 Fee-Intensity Balance and LTV Reweighting

Because the fee is LTV-tiered, a difference in the LTV composition of cosigned and solo loans could mechanically generate a differential shock. Table OA32 rules this out. Cosigned high-balance loans sit at slightly *lower* LTV than solo (mean 83.2 vs. 85.0) — an imbalance the non-occupant co-borrower overlay produces in part, since it caps LTV/CLTV at 95 (90 on the manual track) against 97 for a solo transaction (Fannie Mae, 2020) — so under the tiered fee they received a marginally *smaller* fee shock (−3.23 bps): the imbalance runs against, not toward, the result. The retired specification’s positive differential was nonetheless present *within* LTV bins where the reform fee is equalized, and inverse-propensity reweighting of solo borrowers to the cosigned LTV mix leaves both margins essentially unchanged (rate +0.350  $\rightarrow$  +0.367; rejection −0.051  $\rightarrow$  −0.049). Whatever the retired specification measured was therefore not an LTV-dose artifact, and the rejection insulation — which survives the year controls — is not an artifact of cosigned borrowers holding safer collateral. Under the year-FE primary the same reweighting is a ladder row and changes nothing either (−0.03–+0.02 pp; Table OA19).

#### OA4.15 The Retired Pooled-Intercept Specification: Diagnosis and Archive

Everything below describes the pooled-intercept specification an earlier draft used as its headline. The diagnosis that retired it comes first, and the common-window comparison it turned on follows. No positive rate differential quoted here describes the year-FE primary, whose own ladder is Appendix OA4.2.

##### OA4.15.1 Year Composition: The Diagnosis

An earlier draft of Section 5 estimated the difference-in-discontinuities with a single *Post* intercept per pool and reported a cosigned-minus-solo rate differential of +0.67 pp ( $p = 0.041$ ) at  $h = \$20k$ . This subsection documents why that specification is retired and that table with it. The setting stacks three features no single one of which is fatal but whose combination is: the conforming limit

TABLE OA32. Fee-Intensity Balance — Within-LTV-Tier and IPW-Reweighted Difference-in-Discontinuities

| <i>Panel A: LTV mismatch and the implied reform fee dose (pre-reform, above-CLL, in-window)</i> |                   |                     |                                                           |
|-------------------------------------------------------------------------------------------------|-------------------|---------------------|-----------------------------------------------------------|
|                                                                                                 | <b>Solo</b>       | <b>Cosigned</b>     | <b>Difference</b>                                         |
| Mean LTV                                                                                        | 85.01             | 83.20               | −1.82                                                     |
| Implied $E[\Delta g]$ (bps)                                                                     | 62.17             | 58.94               | −3.23                                                     |
| <i>Within-LTV-bin interest-rate DiDisc (pooled)</i>                                             |                   |                     |                                                           |
| LTV $\leq 80$                                                                                   |                   | −0.0278 (0.2861)    |                                                           |
| LTV 80–90                                                                                       |                   | +0.3066 (0.4719)    |                                                           |
| LTV $> 90$                                                                                      |                   | +1.0731 (0.5904)    |                                                           |
| <i>Panel B: IPW-reweighted DiDisc (solo reweighted to the cosigned LTV mix)</i>                 |                   |                     |                                                           |
|                                                                                                 | <b>Unweighted</b> | <b>IPW-weighted</b> | <b><math>p</math> (unw. <math>\rightarrow</math> IPW)</b> |
| Interest rate (pp), preserve                                                                    | +0.350            | +0.367              | 0.155 $\rightarrow$ 0.140                                 |
| Application rejection, preserve                                                                 | −0.051            | −0.049              | 0.263 $\rightarrow$ 0.276                                 |

*Notes:* Panel A compares the LTV distributions behind the tiered 2022 fee (Kolmogorov–Smirnov  $p = 2e-08$ ); Panel B reweights solo borrowers by inverse propensity to the cosigned LTV mix. Standard errors clustered by lender. The reading of both panels is in the paragraph above.

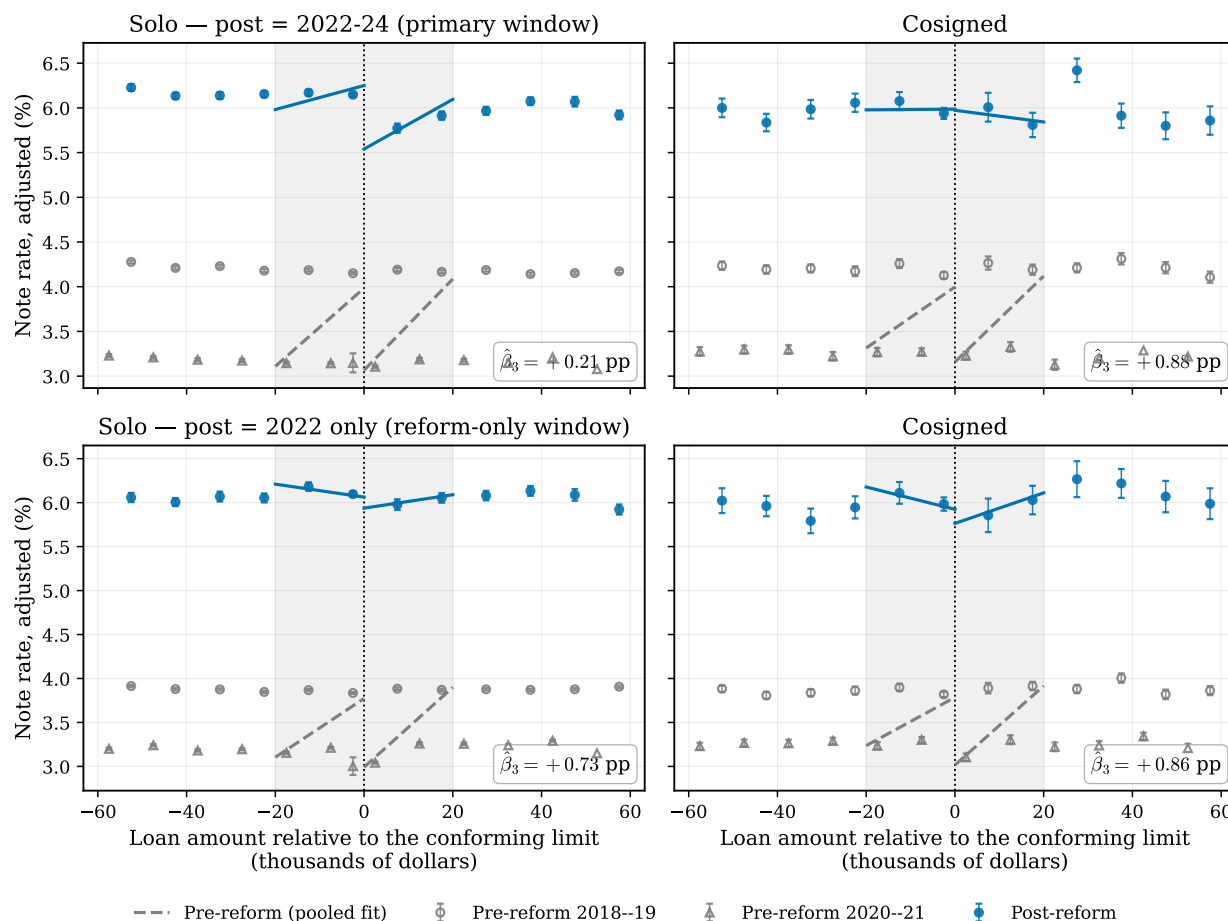
is reset annually, so the pooled design is a multi-cutoff RD with one cutoff per year (Cattaneo et al., 2016); mortgage rates moved three hundred basis points across the sample, so outcome *levels* differ across cutoffs by two orders of magnitude more than the treatment; and the running variable heaps on \$5,000 reporting brackets whose offset from the cutoff changes with each reset, so the composition of the estimation window is year-dependent — the above-cutoff share ranges from 17% (2024) to 52% (2020). A pooled intercept then books year-level differences into the fitted jumps wherever the year mix differs across sides of the cutoff.

Table OA33 collects the diagnostics. Panel A holds the estimator fixed and adds year controls: year fixed effects alone flip the differential’s sign and tighten its standard error, and year-specific slopes change little more, converging on the primary estimates of Table C.1. Panel B diagnoses the retired specification directly. The placebo row runs it on an outcome that cannot respond to the fee — each loan’s origination-year mean below-cutoff rate, constant within year by construction — and recovers most of the retired differential with conventional significance: pure composition leakage, measured under the estimator’s own weights. The reweighting row equalises the year distribution above and below the cutoff within each pool and removes the share-level component. Two further facts complete the picture. Estimated one pre-year against one post-year — the retired estimator itself, with nothing to pool — the twelve possible comparisons span −0.42 to +0.23 pp (median −0.03; 9 of 12 negative), so the retired coefficient lies outside the range of every year-consistent comparison and is therefore not a weighted average of them. And a maximally year-saturated event study (Figure OA6) puts the differential at −0.12, −0.07 and −0.13 pp in 2022, 2023 and 2024 against a pre-period mean of −0.00 — flat, and on the opposite side of zero from the retired estimate.

For completeness, the retired specification's own estimates are reproduced below, both margins in one table (Table OA34). The rejection differential, unlike the rate differential, is insensitive to the year controls — the same sign and order of magnitude under both specifications — which is why the body reports its sign with some confidence and its magnitude with none.

Figure OA9 shows the discontinuity itself, pool by pool and before against after, which is the picture the retired specification was reading a single intercept off. The pre-reform period is drawn in its two rate regimes, 2018–19 near 4.2% and 2020–21 near 3.2%, but fitted pooled, because pooled is what the retired equation does: its only time control is the *Post* dummy, where the primary specification (13) gives each year its own intercept and slopes. Neither regime shows a visible step at the cutoff on its own and the pooled pre-reform fit runs between them, which is the year-composition problem in a picture. The drawn jumps reproduce the reported coefficients of Table OA34 (+0.21 against +0.21 solo, +0.83 against +0.88 cosigned, top row).

FIGURE OA9. The rate discontinuity at the conforming limit, before and after the reform



Binned means of the note rate against the loan amount centred on the conforming limit, in \$5,000 bins (the HMDA reporting bracket), for solo applications (left) and cosigned (right), on the primary above-AMI sample. The top row codes 2022–2024 as post-reform, the bottom row 2022 alone. Lines are local-linear fits estimated either side of the cutoff over the shaded  $h = \$20k$  window, on rates adjusted for the control set of Table OA34.

TABLE OA33. Retiring the Pooled-Intercept Specification: Year Composition and the Old Differential

| Specification / test                                                           | Cosigned – solo differential, by bandwidth (\$000s) |                |                |                 |
|--------------------------------------------------------------------------------|-----------------------------------------------------|----------------|----------------|-----------------|
|                                                                                | 18                                                  | 20             | 22             | 24              |
| <i>Panel A: the differential under progressively year-honest time controls</i> |                                                     |                |                |                 |
| Pooled intercepts (retired headline)                                           | +0.63* (0.32)                                       | +0.67** (0.33) | +0.70** (0.34) | +0.74** (0.32)  |
| Pooled triple-difference, no year controls                                     | +0.43 (0.29)                                        | +0.45 (0.29)   | +0.48 (0.30)   | +0.55 (0.29)    |
| + year fixed effects                                                           | −0.13 (0.20)                                        | −0.13 (0.20)   | −0.09 (0.20)   | −0.09 (0.19)    |
| + year-specific slopes                                                         | −0.15 (0.20)                                        | −0.15 (0.20)   | −0.15 (0.20)   | −0.15 (0.20)    |
| Year-FE primary (per-group, cov.-corrected)                                    | −0.01 (0.23)                                        | −0.06 (0.23)   | −0.06 (0.23)   | −0.05 (0.23)    |
| <i>Panel B: diagnostics of the pooled-intercept specification</i>              |                                                     |                |                |                 |
| Placebo outcome: year-level mean rate                                          | +0.56** (0.24)                                      | +0.61** (0.24) | +0.62** (0.24) | +0.66*** (0.24) |
| Year-reweighted (IPW, as Table OA32)                                           | +0.44 (0.32)                                        | +0.48 (0.32)   | +0.53 (0.33)   | +0.52 (0.32)    |

*Notes:* All rows: above-AMI sample, interest-rate margin, lender-clustered standard errors. Panel A adds year controls to a fixed estimator; Panel B's placebo outcome is the origination year's mean in-window below-cutoff rate, constant within year and so unable to respond to the fee. Each row is read in the paragraph above.

TABLE OA34. Differential Pass-Through on Both Margins — Retired Pooled-Intercept Specification (Above AMI)

| $h$ (\$000s)                                              | Solo $\hat{\beta}_3$ | Cosigned $\hat{\beta}_3$ | DiDisc $\hat{\Delta}$ | $N_{\text{non}}$ | $N_{\text{cos}}$ |
|-----------------------------------------------------------|----------------------|--------------------------|-----------------------|------------------|------------------|
| <i>Panel A. Interest rate (percentage points)</i>         |                      |                          |                       |                  |                  |
| 18                                                        | +0.0986 (0.1764)     | +0.7264** (0.3501)       | +0.6278* (0.3243)     | 22,453           | 2,433            |
| 20                                                        | +0.2137 (0.1796)     | +0.8805** (0.3505)       | +0.6669** (0.3270)    | 23,985           | 2,586            |
| 22                                                        | −0.0280 (0.1807)     | +0.6762* (0.3534)        | +0.7041** (0.3353)    | 25,682           | 2,768            |
| 24                                                        | −0.0190 (0.1767)     | +0.7181** (0.3393)       | +0.7371** (0.3241)    | 26,882           | 2,880            |
| 26                                                        | +0.0875 (0.1416)     | +0.4430* (0.2580)        | +0.3556 (0.2572)      | 30,751           | 3,328            |
| 28                                                        | −0.0141 (0.1373)     | +0.2573 (0.2479)         | +0.2714 (0.2558)      | 32,037           | 3,476            |
| <i>Panel B. Application rejection (probability units)</i> |                      |                          |                       |                  |                  |
| 62                                                        | +0.0146* (0.0081)    | −0.0394 (0.0477)         | −0.0540 (0.0512)      | 67,413           | 7,350            |
| 82                                                        | +0.0100 (0.0074)     | −0.0345 (0.0430)         | −0.0445 (0.0445)      | 85,640           | 9,503            |
| 102                                                       | +0.0119 (0.0073)     | −0.0153 (0.0338)         | −0.0272 (0.0358)      | 102,721          | 11,598           |
| 123                                                       | +0.0107* (0.0063)    | −0.0016 (0.0272)         | −0.0123 (0.0287)      | 119,499          | 13,772           |

*Notes:* The retired pooled-intercept specification (Appendix OA4.15.1; year-FE primaries: Tables C.1 and OA25) on applications above 120% of area median income, post period 2022 onwards (HMDA reports no origination month). Covariates enter the local regression (Calonico et al., 2019): county fixed effects, house-price index, log income, applicant and co-applicant demographics, lender fixed effects absorbed; standard errors clustered by lender, the DiDisc column cross-cluster-corrected. Panel B's grid is wider because the cosigned rejection cell is thin at narrow windows, and its minimum detectable effect at 80% power is 8–14 pp: uninformative intervals, not precise zeros. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

#### OA4.15.2 Both Margins on a Common Window

The contrast an earlier draft drew from the retired specification — the price moves and the rationing does not — needed one more thing before it can be read that way. The rate differential is estimated at  $h \leq \$28k$  and the rejection differential at  $h \geq \$62k$ , because each margin’s own bias–variance trade-off points there. Read literally, the contrast therefore sets a local estimate against a global one. Table OA35 re-estimates both margins at the same six bandwidths inside the same \$125k window, under both covariate conventions, and reports the number of lender clusters behind each fit — 959 on the rate margin at  $h = \$20k$ , rising with the window. Under the convention of record the rate differential runs +0.25–+0.67 pp, positive at all 6 windows and separated from zero at 4; under the residualise-first convention it runs +0.07–+0.24 pp, positive at all six and separated from zero at none. The rejection differential is negative at all six under both (–0.088–0.025 conventional, that is two to nine percentage points) and separated from zero under neither. What survives across conventions is the ordering rather than either magnitude: price positive, rationing negative, at every window. The rejection differential also attenuates as the window widens, which is consistent with dilution of a local effect. I do not press that reading. The six windows are nested, so the estimates are strongly correlated and a near-monotone attenuation path is close to mechanical; the rate differential’s own path across the same windows is not monotone. The two margins also remain estimated on nested but different populations, originated loans against all applications, and no choice of window can change that.

TABLE OA35. Both Margins on a Common Bandwidth Grid and a Common Data Window, under Both Covariate Conventions

| $h$ (\$000s)                                        | <i>Conventional (CCFT)</i> |                  | <i>Bias-aware (AK)</i> |                  | MDE   | $G$   |
|-----------------------------------------------------|----------------------------|------------------|------------------------|------------------|-------|-------|
|                                                     | $\hat{\Delta}$             | 95% CI           | $\hat{\Delta}$         | 95% CI           |       |       |
| <i>Panel A. Interest rate (percentage points)</i>   |                            |                  |                        |                  |       |       |
| 20                                                  | +0.667** (0.327)           | [+0.026, +1.308] | +0.241                 | [−0.351, +0.833] | 0.916 | 959   |
| 30                                                  | +0.287 (0.254)             | [−0.210, +0.784] | +0.074                 | [−0.399, +0.547] | 0.710 | 1,036 |
| 40                                                  | +0.487*** (0.182)          | [+0.130, +0.843] | +0.114                 | [−0.299, +0.527] | 0.510 | 1,091 |
| 50                                                  | +0.463*** (0.167)          | [+0.135, +0.791] | +0.195                 | [−0.182, +0.572] | 0.469 | 1,123 |
| 62                                                  | +0.369** (0.177)           | [+0.022, +0.716] | +0.211                 | [−0.137, +0.560] | 0.496 | 1,169 |
| 75                                                  | +0.248* (0.148)            | [−0.042, +0.538] | +0.198                 | [−0.122, +0.518] | 0.415 | 1,209 |
| <i>Panel B. Application rejection (probability)</i> |                            |                  |                        |                  |       |       |
| 20                                                  | −0.088 (0.079)             | [−0.242, +0.067] | −0.097                 | [−0.308, +0.114] | 0.221 | 965   |
| 30                                                  | −0.078 (0.064)             | [−0.205, +0.048] | −0.089                 | [−0.268, +0.091] | 0.180 | 1,047 |
| 40                                                  | −0.063 (0.058)             | [−0.178, +0.051] | −0.076                 | [−0.240, +0.089] | 0.163 | 1,103 |
| 50                                                  | −0.049 (0.051)             | [−0.149, +0.051] | −0.064                 | [−0.218, +0.090] | 0.143 | 1,134 |
| 62                                                  | −0.054 (0.051)             | [−0.154, +0.046] | −0.058                 | [−0.204, +0.088] | 0.143 | 1,183 |
| 75                                                  | −0.025 (0.047)             | [−0.116, +0.066] | −0.047                 | [−0.189, +0.095] | 0.131 | 1,223 |

Notes: Both margins at the same bandwidth  $h$  inside the same \$125k window on the above-AMI sample, each estimator beside *its own* interval. *Conventional (CCFT)* puts covariates inside the local regression (Calonico et al., 2019); *bias-aware (AK)* residualises the outcome first, so the two estimate different objects (Appendix OA4.12). MDE is the minimum detectable effect at 80% power,  $G$  the lender clusters.

Within the retired specification, then, the price margin returned an apparent estimate and

the rationing margin a sign, and the common grid showed the contrast was not an artifact of the windows. The year-composition diagnostics of Appendix OA4.15.1 subsequently retired the price-margin estimate; the rationing sign carries over to the year-FE primary unchanged.

## OA5 Quantitative Model: Computational Detail

### OA5.1 Mapping the Stylized Game into the OLG Economy

Table OA36 states the object-by-object correspondence between the asymmetric-information game of Section 3 and the quantitative economy of Section 6.

TABLE OA36. Mapping the asymmetric-information game (Section 3) into the OLG economy.

| Section 3 object                                                                                | Quantitative counterpart                                                                                                                                             | Parameters / equations                                                                                                    |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Two types $\theta \in \{G, B\}$ , repayment $\pi_G > \pi_B$                                     | Same types; $\pi_\theta$ prices the pool, but the default <i>decision</i> is endogenized through a type-specific stigma and limited commitment                       | $\pi_B, \Delta\pi; \bar{\xi}, \Delta\xi$ with $\xi_\theta = \bar{\xi} \pm \Delta\xi/2; V_j^{PD}$                          |
| Income-ordered prior $\lambda(y), \lambda'(y) > 0$                                              | Prior on the child's type, increasing in own productivity                                                                                                            | $\lambda(z_c), \partial\lambda/\partial z_c > 0$                                                                          |
| Guarantee $o \in \{0, 1\}$ with offer propensity $q_\theta(y, y_p)$                             | Two-sided: discrete help choice by the parent at $j = 43$ times the child's take-up at $j = 18$ . Both are equilibrium objects under rational expectations           | $q_\theta \leftrightarrow \pi_\kappa(z_p) \cdot \Pr(\text{accept} \mid \theta); \kappa \in \{1, 2, 3, 4\}; \sigma_{menu}$ |
| Taste shock $\eta_p \sim F_\eta$ (full support)                                                 | Non-financial help cost plus Type-I extreme-value taste shocks: <i>location</i> of $F_\eta \rightarrow$ mean cost, <i>dispersion</i> $\rightarrow$ logit temperature | $\eta_\kappa \leq 0$ (location); $\sigma_{help}$ (dispersion)                                                             |
| Marginal Cost Dominance (Assumption 1) giving $q_G > q_B$                                       | Not assumed: parents observe $\theta$ and internalize a <i>type-specific</i> expected liability, larger for Bad children, so $q_G > q_B$ arises endogenously         | $L_\theta \approx (1 - \pi_\theta)m$ , eq. (OA36)                                                                         |
| Bayesian posterior $\Lambda_1(y)$ ; pool quality $\mu_{co}$                                     | Realised Good-type share of each $(h, z_c)$ pool, read off the equilibrium measure of buyers; quality linear in that share                                           | $q_G^k(h, z_c); \mu_k = \tilde{\pi}_B + (\tilde{\pi}_G - \tilde{\pi}_B)q_G^k$                                             |
| Rule-constrained pricing $1 + r_k = (1 + \rho + g)/\mu_k$ , computed on the occupant's own file | Rate <i>level</i> from the observable schedule; the cosign-solo <i>differential</i> is set cell by cell by the institutional pricing rule                            | $r_{base}(z_c); \frac{\Delta r(h, z_c)}{R(z_c)[\mu_{co}^{-1} - \mu_{so}^{-1}]} =$                                         |
| Guarantee fee $g$ (the FHFA shock)                                                              | Above-conforming-limit rate shock — the lever of the natural experiment                                                                                              | rate shock on $h \geq 2$                                                                                                  |
| Wedge: price off $y$ , accept off $Y_{co} = y + y_p$ , DTI cap $\kappa$                         | Base rate from the child's own-income risk bucket; eligibility on <i>joint</i> income                                                                                | $\bar{\omega}_{solo} = 0.40, \bar{\omega}_{joint} = 0.42; \Omega_{cos}$                                                   |
| Three-region cutoffs $y_F < y_G^{pool} < y_B^{sep}$ (Prop. 3)                                   | Cross-section of $(h, z_c)$ cells; the trap is the set with a positive premium                                                                                       | $\Delta r(h, z_c) > 0$                                                                                                    |

**Does the estimated economy reproduce the three-region structure?** The last row of Table OA36 asserts a correspondence, and it can be checked rather than asserted. Proposition 3 predicts a specific signature at the good types' exit cutoff  $y_G^{pool}$ : the cosigned pool's Good-type share *collapses* as the good types leave, and the cosigned rate *jumps* to the separating level. On

an eleven-state income grid a discontinuity cannot be observed as such, but the signature can: it should appear as a large upward step in the cosign–solo gap at the same state where  $q_G^{co}$  falls sharpest.

It does. Read off the estimated equilibrium, both fire at  $z_c = 6$ : the gap steps +4.48 bp — the largest step of either sign anywhere on the grid — while  $q_G^{co}$  moves  $-0.281$ , the largest fall anywhere on the grid. Above that state the gap stays positive over the range carrying 76.8% of cosigned buyers, and the cosigned pool is adversely selected on type at every income state where it is thick, with  $q_G^{co} - q_G^{so}$  reaching  $-0.139$ . The trap of Section 3 therefore exists in the estimated economy, sits where the theory puts it, and holds most of the cosigned population.

What does not carry over is magnitude. The whole schedule spans 7.76 bp and peaks at +2.45 bp, against a separating benchmark an order of magnitude larger, so the “jump” is a jump in the ordering rather than in the price a borrower would notice. Region 3 never forms either: cosigned take-up is flat across the upper half of the income grid, because a premium of a few basis points is too small to price anyone out of the menu. Both are the same fact the premium decomposition reports from the price side — the structure of the stylized model survives quantification; its magnitudes do not.

## OA5.2 When the Fixed Point is Declared Solved

The joint fixed point of Appendix A.8 is reached iteratively, and the stopping rule is not a single tolerance. It is worth stating in full, because the map is not a contraction and a naive rule certifies the wrong thing.

Convergence requires the max-norm of the change across every block — default policy, premium, help probabilities, cash cost and pool composition — to fall below a tolerance. Three qualifications are layered on that. First, a cell carrying no mass can flip on a knife edge forever without the equilibrium being unsettled, so the criterion is also satisfied by an *inhabited-cell* norm together with a mass-weighted default change an order of magnitude tighter; which of the two certified a given solve is recorded rather than assumed. Second, the criterion must hold on *consecutive* iterations, because the pool-composition block cycles when undamped and a criterion met once is as consistent with a cycle passing through the tolerance band as with a fixed point. Third, the state residuals settle well before the reported moments do — the cosigned share can still be drifting when the premium has been inside tolerance for a dozen iterations — so certification is withheld while any reported moment is still moving relative to its own tolerance. Against these, a plateau rule aborts a solve whose inhabited norm stops improving, reporting failure rather than exhausting the iteration budget. The inner inheritance loop carries its own separate tolerance, and the composition block a separate one again. Every certified solve reports the residual  $\|BR(q_G) - q_G\|$  at the halt, so the tightness of a fixed point is a number in the log rather than a claim.

**What the reported estimate meets.** The equilibrium the estimates and counterfactuals are computed at meets that criterion. It certifies on the inhabited-cell norm at outer iteration 18 of

a budget of 50, on 2 consecutive readings of 0.0077 and 0.0098 against a tolerance of 0.01. Since the criterion layers three requirements on top of a tolerance, this is a stricter statement than the single-tolerance rule the literature usually reports rather than a weaker one.

Three facts distinguish a settled point from a stalled one, and all three hold here. First, every block that carries mass falls by two to three orders of magnitude along the path: the default policy from 0.0736 to 0.00079, the help probabilities from 0.226 to  $8.0 \times 10^{-5}$ , the cash cost from 0.0798 to 0.00057, and the premium from 0.0221 to  $3.8 \times 10^{-5}$ . The inhabited norm falls from 0.226 with 2 back-steps, the larger reaching 0.177 at iteration 11; both are isolated, and neither resembles the alternating  $q_G$  cycle that the damping was introduced to remove. Second, the object that does *not* settle is the strict max-norm, which halts at 0.138 on a cell carrying 0.045 percent of the mass, against a mass-weighted default change of  $1.9 \times 10^{-5}$ : the binding cell is a norm artefact, and the diagnostic trail says so cell by cell. Third, and independently of any criterion, a second solve at the same vector three days later reproduces the four headline moments to  $6.7 \times 10^{-5}$ , the largest disagreement falling on the Cosign share.

One qualification survives. The selection pool is a fixed point on its inhabited cells only: the  $q_G$  residual is 0.049 there, inside the separate tolerance of 0.05, but 0.69 on the strict norm. Cells the pool never visits are therefore unsettled, and a quantity read off one of them would not be reliable. Whether the residual that remains is small enough to license the reported precision is a separate question from whether the solve converged, and Online Appendix OA6.4 answers it by propagating that residual through to the standard errors.

**Housing supply and the construction sector.** In the baseline stationary equilibrium the house price  $p_{own}$  is calibrated to match the homeownership rate, which is equivalent to a perfectly elastic supply of owned housing at that price. To evaluate the general-equilibrium incidence of the counterfactual pricing reform of Section 8, I close the model with an explicit construction sector that delivers an upward-sloping supply curve. A representative competitive construction firm produces new housing by combining a fixed factor (developable land) with a variable input through a Cobb-Douglas technology; profit maximisation yields an iso-elastic supply schedule

$$H_s(p) = C p^\varepsilon, \quad \varepsilon = \frac{\alpha_c}{1 - \alpha_c}, \quad (\text{OA35})$$

where  $\alpha_c$  is the variable-input share in construction and  $\varepsilon$  is the price elasticity of housing supply. This is the construction technology of [Castellanos et al. \(2024\)](#), who use it to study the Irish loan-to-value and loan-to-income reform of 2015 in a model with endogenous house prices and rents. The scale coefficient is calibrated to pass the supply curve through the baseline equilibrium  $(p^*, H^*)$ ,  $C = H^*/(p^*)^\varepsilon$ , so the baseline equilibrium is left exactly unchanged and  $\varepsilon$  governs only the counterfactual price response. The owned- and rental-housing markets each carry their own supply curve,  $H_s^{own}(p_{own})$  and  $H_s^{rent}(p_{rent})$ , calibrated to their respective baseline prices and quantities.

I set  $\varepsilon = 2.2$ . Under Cobb-Douglas with competitively priced factors the exponent is  $(1 - s)/s$ , where  $s$  is land's share of the value of housing, so the parameter can be read off a land-share series.

The usual source is the national accounts: [Davis and Heathcote \(2007\)](#) find that residential land accounts for 36 percent of the value of the aggregate housing stock on average over 1975–2006, rising to 46 percent by 2006, implying elasticities of 1.78 and 1.17. But the aggregate stock is the wrong population for this model. Every buyer here is a first-time buyer, first-time buyers purchase cheaper houses than the average home, and cheaper houses are less land-intensive, so the aggregate land share overstates the fixed factor in the technology that actually constrains them.

That distinction can be measured rather than assumed. [Davis et al. \(2021\)](#) estimate residential land prices and land shares for 54,515 census tracts, applying the assessed-to-market correction and covering the large majority of the U.S. single-family stock. I weight those tracts by the 831,000 young-buyer originations in the 2021 HMDA data of [Section 4](#) and evaluate them at the price point the model’s buyers occupy, a price-to-income ratio of 3.56 that the model reproduces at its house-size floor. That gives a land share of 0.306, and hence  $\varepsilon = 2.27$ . The figure is stable at 0.303–0.310 ( $\varepsilon = 2.22$ – $2.30$ ) across the age definitions and price-to-income targets that could reasonably stand in for the model’s buyer. I use  $\varepsilon = 2.2$ , the round value at the least elastic end of that range, so the calibration understates rather than overstates the ownership gain that survives in general equilibrium.

The gap from the aggregate is entirely a composition effect, and a large one. Sorting tracts by value, the land share climbs monotonically from 0.183 in the decile of tracts where the cheapest tenth of young buyers purchase to 0.577 in the most expensive, a range far wider than the disagreement in the literature, and the reason the population matters more here than the data source. Weighting the same tracts by purchase volume without conditioning on price gives 0.399 ( $\varepsilon = 1.51$ ), close to the national accounts figure and to the 0.356 that [Davis et al. \(2021\)](#) report for all single-family parcels, because mortgage-financed young buyers are not in fact concentrated in cheap markets: the median young-buyer purchase sits at the 67th percentile of the national tract-value distribution. What separates 0.306 from 0.399 is not where these buyers live but which house they buy once there.

The result is corroborated from a second and unrelated direction. [Saiz \(2010\)](#) imposes a common slope across metropolitan areas and estimates  $\varepsilon = 1.54$  (his Table III, column 1), placing it inside the range of one to three that the literature proposes at the national level; his population-weighted metropolitan average is 1.75. The value used here sits above both but well inside that range, which is what one expects given that Saiz estimates an average across all housing rather than the starter segment. For reference, [Castellanos et al. \(2024\)](#) set a land share of 0.33 for Ireland, close to the figure obtained here for U.S. first-time buyers. I report robustness to  $\varepsilon \in \{0, 1.5, 4.0\}$  — the fixed-supply limit, the aggregate-stock calibration this appendix previously used, and an elastic end beyond the national range — each a separate general-equilibrium solve rather than an interpolation ([Table 7](#)). Note that  $\varepsilon = 0$  is the *most* price-responsive case, not the least: with the housing stock fixed the entire demand shift falls on the price. The partial-equilibrium column is the opposite endpoint,  $\varepsilon \rightarrow \infty$ , at which supply absorbs the shift and the price does not move — which is why the baseline calibration of  $p_{own}$  to the homeownership rate is equivalent to perfectly

elastic supply. The four solves therefore span the full range from complete capitalisation to none.

Three caveats bound what this parameter can be asked to do. First, the empirical literature does not estimate the object in (OA35). [Saiz \(2010\)](#) derives his elasticities from a monocentric-city model in which construction costs are constant and land rents rise as the city expands outward, with no substitution between land and structures; the elasticity he estimates therefore falls as a city builds out, and treating it as the constant exponent here is a local approximation — defensible in this application because the counterfactual moves the house price by less than one percent. Other frequently cited figures are further still from the present object: [Aastveit et al. \(2023\)](#) estimate the response of building *permits*, a construction flow rather than the stock governed by  $\varepsilon$ , and report it falling from 2.63 over 1996–2006 to 1.75 over 2012–2017. I take from that literature the direction of travel — supply responsiveness has declined — rather than a level. For the same reason I use land’s share of the value of the housing *stock* rather than the roughly 11 percent that raw land represents in the cost of a newly built house ([Davis and Heathcote, 2007](#)): the fixed factor in (OA35) stands for the scarcity of desirable locations, which is capitalised into the stock and largely absent from acreage at the margin of new construction.

Second, elasticities are considerably lower in the land-constrained metropolitan areas from which the natural experiment of Section 5 is identified: [Saiz \(2010\)](#) puts them below one in coastal California, Miami, New York, Boston and Chicago. Since a lower  $\varepsilon$  implies a larger price response and hence less surviving ownership gain, the national calibration makes the reported homeownership effect an upper bound for precisely those markets. Third, the counterfactual has been solved separately at four elasticities, spanning the calibrated value, the aggregate-stock figure it replaces, an elastic end below anything the U.S. data supports, and the fixed-supply limit. The surviving homeownership effect, as a share of its partial-equilibrium counterpart, is  $-1$ , 34, 43 and 57 percent at  $\varepsilon = 0, 1.5, 2.2$  and 4.0. The cosign share is nearly untouched across the same range — 93 to 95 percent — with the income gradient likewise flat between  $-0.45$  and  $-0.48$ .

The fixed-supply endpoint makes the mechanism unambiguous. With the housing stock unable to grow, aggregate ownership cannot grow either: the entire demand shift becomes price, which rises 1.23 percent, and the reform delivers no additional homeowners at all. Yet the cosign share still rises 4.55 pp, the cosigned rate still falls 22 basis points, and the income gradient still moves from  $-0.012$  to  $-0.448$ . The access margin therefore requires an elastic supply to exist; the substitution and incidence margins do not depend on the supply side at all. That is the sense in which the qualitative conclusion of Section 8 holds at every value considered, while the magnitude of the access effect is a supply-elasticity question rather than a property of the reform.

**The liability the parent *decides* on is not the one their budget pays.** Equation (A.16) is the transfer the parent’s budget constraint carries, and it is type-blind: once the guarantee is invoked the parent pays, whatever the child turned out to be. The object entering the *help decision* at  $j = 43$  is different, and the difference is what generates Marginal Cost Dominance. A parent

observes the child's type  $\theta$ , so the liability they price ex ante is

$$L_\theta(z_p, z_c) = \sum_h \pi_{cos,\theta}^h(z_p, z_c, h) \varphi(1 - \pi_\theta) p_{cover}(h, z_p, z_c) m(h, z_p, z_c), \quad (\text{OA36})$$

where  $\pi_{cos,\theta}^h$  is the equilibrium distribution of house sizes among cosigned children of type  $\theta$ . Two features are deliberate. The default probability is the *structural* rate  $1 - \pi_\theta$  rather than the behavioural one: a parent with private type information expects type-consistent repayment, not the inflated rate that includes the guarantee's own effect on the child's default incentive, so the moral-hazard channel is absent from the ex-ante expectation while remaining in the realized transfer. And the house-size weights are type-specific, because the two types buy differently.

Because  $\pi_G > \pi_B$ , equation (OA36) gives  $L_G < L_B$ : guaranteeing a Bad child is more expensive to the parent than guaranteeing a Good one. That is exactly Assumption 1, which the stylized model of Section 3 imposes by fiat, and here it is an implication of the parent's problem rather than a restriction on it. A reader who took (A.16) alone as the parent's decision object would find the liability identical across types and  $q_G > q_B$  unobtainable; the two equations are different objects and both are needed. The robustness switch that replaces  $1 - \pi_\theta$  with the behavioural typed default rate is reported in Appendix OA6.1.

The aggregate adding-up condition,

$$\int_j \tau_{j,p}(\mathbf{s}_{j,p}) d\mathcal{M}_j(\mathbf{s}_{j,p}) = \int_j d_{j,c}(\mathbf{s}_{j,c}) m_{j,c}(\mathbf{s}_{j,c}) d\mathcal{M}_j(\mathbf{s}_{j,c}) \quad (\text{OA37})$$

— aggregate parental transfers  $\tau_{j,p}$  equal the aggregate payments made on behalf of defaulting children,  $d_{j,c} \in \{0, 1\}$  being the child's default indicator — then holds by construction rather than as a separate market-clearing requirement, since  $\bar{d}$  is itself the equilibrium default policy. What the certainty-equivalent treatment gives up is not the level of the transfer but its dispersion across parents, and hence any precautionary response to having cosigned.

## OA6 Structural Estimation: Further Identification and Validation

### OA6.1 Parameters Set Outside the SMM

Tables OA37 and OA38 list every parameter fixed before estimation, grouped by how it was fixed: Panel A external evidence, Panel B contract and policy facts, Panel C internal calibration, Panel D structural taste dispersion, Panel E numerical. Section 7 discusses the two that carry the argument, the involuntary-repayment gap  $\Delta\pi$  and the common loan-to-value limit.

### OA6.2 External Validation of the Type Gap $\Delta\pi$

Four strands of evidence bound the involuntary type gap. Gupta and Hansman (2022) attribute 30–60% of the cross-sectional leverage–default relationship to unobserved-type selection; Gerardi et al. (2018) put strategic, able-to-pay default under 1%, so the type gap is almost entirely invol-

TABLE OA37. Parameters set outside the SMM: external evidence and institutional design

| Parameter                                                                                      | Symbol                       | Value | How set                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------|------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Panel A: set outside the model (external evidence, or fixed with literature motivation)</i> |                              |       |                                                                                                                                                                                                                                                                                                                                                                                           |
| Involuntary repayment gap                                                                      | $\Delta_\pi = \pi_G - \pi_B$ | 0.05  | fixed. Attributes 37% of the log typed default gap to ability-to-pay and 63% to the strategic margin, against the 30–40% / 60–70% selection–moral-hazard split of <a href="#">Gupta and Hansman (2022)</a> ; see also <a href="#">Gerardi et al. (2018)</a> ; <a href="#">Karlan and Zinman (2009)</a> . Robustness band and the hazard arithmetic in Appendix OA6.2                      |
| Altruism elasticity to parent income                                                           | $\gamma$                     | 1.00  | unit elasticity (the homothetic benchmark), fixed in the SMM. <a href="#">De Nardi (2004)</a> ; <a href="#">Lockwood (2012)</a> motivate income-dependent altruism; their luxury finding implies $\gamma > 1$ , so $\gamma = 1$ is the conservative choice                                                                                                                                |
| Discount factor                                                                                | $\beta$                      | 0.960 | annual: 0.96 in <a href="#">Cocco (2005)</a> , 0.9606 estimated by <a href="#">Nakajima and Telyukova (2020)</a>                                                                                                                                                                                                                                                                          |
| Return on savings                                                                              | $r$                          | 0.030 | annual, riskless in the model and read as the return on the household’s portfolio: the top of the 1–3% peacetime range of real safe returns in <a href="#">Jordà et al. (2019)</a> , below their returns on risky wealth; <a href="#">Li et al. (2016)</a> use 2.7%. Mortgage rates sit 1.3–1.8 points above $r$ (data mortgage–T-bill spread: 2.9), so debt costs more than savings earn |
| Risk aversion                                                                                  | $\sigma$                     | 1.50  | exactly the 1.5 of <a href="#">Bacher (2022)</a> ; the modal value in life-cycle housing models is 2 ( <a href="#">Kiyotaki et al., 2011</a> ; <a href="#">Garriga and Hedlund, 2020</a> ; <a href="#">Kaplan et al., 2020</a> ), with portfolio-choice settings higher. Low end of the range                                                                                             |
| Housing expenditure share                                                                      | $\theta$                     | 0.25  | <a href="#">Davis and Ortalo-Magné (2011)</a> set $\theta = 0.24$ ; their MSA estimates span 0.20–0.28 and are near-constant across decades                                                                                                                                                                                                                                               |
| Housing depreciation                                                                           | $\delta_h$                   | 0.015 | <a href="#">Harding et al. (2007)</a> : $\approx 2\%$ net of maintenance (2.5% gross); the model has no maintenance choice, so the net figure is the comparison. Set conservatively below it                                                                                                                                                                                              |
| Income persistence                                                                             | $\rho$                       | 0.96  | <a href="#">Heathcote et al. (2010)</a> , PSID: $\rho \approx 0.97$                                                                                                                                                                                                                                                                                                                       |
| Income innovation s.d.                                                                         | $\sigma_\varepsilon$         | 0.110 | <a href="#">Heathcote et al. (2010)</a> ; at the lower edge of the 0.11–0.21 range in the PSID literature                                                                                                                                                                                                                                                                                 |
| Age profile of income                                                                          | $P_j$                        | PSID  | head labour income by 3-year age group net of year means, PSID family files 2003–2021, heads reporting no retirement; model age $j = \text{age} - 18$ . Relative to the purchase age (36): 0.24 at entry, 1.07 at 53, 0.92 at 66 ( <a href="#">Appendix A.5</a> )                                                                                                                         |
| <i>Panel B: institutional design</i>                                                           |                              |       |                                                                                                                                                                                                                                                                                                                                                                                           |
| Loan-to-value limit, solo                                                                      | $\lambda_m$                  | 0.90  | US conforming: 10% down, as in <a href="#">Benetton et al. (2022)</a> . NSMO first-time buyers: mean CLTV 88.1 (median 95)                                                                                                                                                                                                                                                                |
| Loan-to-value limit, cosigned                                                                  | $\lambda_m^{\text{co}}$      | 0.90  | <b>equal to the solo limit</b> : the NSMO shows no cosigned–solo divergence (median 95 both; means 87.5 vs 88.2; share $> 90$ 0.605 vs 0.612)                                                                                                                                                                                                                                             |
| Recourse fraction                                                                              | $\varphi$                    | 1.00  | the US non-occupant cosigner is jointly and severally liable from origination rather than a contingently liable guarantor: full recourse ( <a href="#">Section 2</a> )                                                                                                                                                                                                                    |
| Mortgage term (years)                                                                          | $M$                          | 30    | 30-year fixed                                                                                                                                                                                                                                                                                                                                                                             |
| Conforming-limit house tier                                                                    | $h^{CLL}$                    | 3     | tier at which the conforming limit binds                                                                                                                                                                                                                                                                                                                                                  |

*Notes:* Panels are ordered by how each value was fixed, not by what it does; Panels C–E are in Table OA38. **A** is external evidence, each entry checked against the cited paper, with  $\Delta_\pi$  and  $\gamma$  motivated rather than point-estimated there and two entries at or below their anchor range as the row states. **B** records contract and policy facts.

untary rather than strategic; and the identified-selection magnitudes in consumer credit (Karlan and Zinman, 2009; Adams et al., 2009; Einav et al., 2012) bound it at a few percentage points. A five-point gap is consistent with all four.

Because  $\Delta\pi$  is calibrated rather than estimated, I also check ex post that  $\Delta\pi = 0.05$  is consistent with that evidence by reproducing Gupta and Hansman’s decomposition *inside* the estimated model. Gupta–Hansman attribute 30–60% of the cross-sectional leverage–default relationship to unobserved-type selection, the remainder being the causal effect of leverage. In the model the higher-leverage margin is cosigning (the income-constrained stretch), and with hidden type  $\theta$  and repayment probabilities  $\pi_G = \pi_B + \Delta\pi$  a pool’s pre-cure default rate is linear in its Good-type share,  $d(\text{pool}) = (1 - \pi_B) - q_G \Delta\pi$ . The cosign-solo default gradient therefore decomposes exactly into a *selection* component  $\Delta\pi (q_G^{\text{solo}} - q_G^{\text{cosign}})$  — the pools differ in type composition — and a *causal* component (leverage raising default within type). The computation is implemented in `validate_delta_pi_selection.jl` and reported both unconditionally and within income–house cells.

One feature of the current estimate bears on this check directly. The selection component is signed by the ordering of the two pools’ Good-type shares, and at this estimate the solo pool is the cleaner one ( $q_G^{\text{solo}} = 0.73$  against  $q_G^{\text{cosign}} = 0.72$ ), so the component is positive — selection and leverage push the gradient the same way, as in Gupta–Hansman. Under the previous specification the ordering ran the other way and the selection component was negative, which sat awkwardly against the evidence being matched. The share itself is re-computed at this estimate rather than carried over.

**What the gap is, and is not, disciplined by.** Since  $\pi_\theta$  is a repayment probability over the life of the loan, the object comparable to a delinquency statistic is the annual involuntary hazard  $1 - \pi_\theta^{1/M}$ . Table OA39 reports it across the robustness band. Two things are visible. First, the *level* of the involuntary floor is governed by  $\pi_B$  and not by  $\Delta\pi$ : the Bad-type hazard is 0.351% per year at every value in the band, because it depends on  $\pi_B$  alone, and Bad types are the ones who default. The gap moves only the Good-type hazard, so the mass-weighted floor slides from 0.296% to 0.226% across the band — a range that brackets the 0.23% observed 90-day hazard without selecting a point inside it. I therefore report  $\Delta\pi$  as fixed with a band rather than as calibrated to that moment. That the floor already sits at the observed 90-day level is also why  $M_2$  targets the 30-day rung instead. Second, what the band *does* discipline is the split. Because the typed involuntary ratio rises from  $1.45\times$  to  $3.45\times$  while the total typed ratio targeted by  $M_5$  is  $6.93\times$ , the share of the log typed default gap attributable to ability-to-pay runs from 19% to 64%, and equals 37% at  $\Delta\pi = 0.05$ . That is the quantity comparable to Gupta–Hansman, who attribute 30–40% of the leverage–default correlation to selection and the remaining 60–70% to the causal channel, with 36% in their Option ARM sample. The correspondence between a 37/63 split here and a 36/64 split there is close, and it is the reason the mid-band value is retained; it is a consistency check on two different decompositions rather than a calibration, so it is not treated as a target.

One clarification prevents a misreading: the residual carried by  $\Delta\xi$  is dispersion in the *cost* of defaulting, which governs how far into distress a household goes before it stops paying. It is not ruthless, able-to-pay default, which the estimated economy’s own default-composition diagnostics put at 1–3% of defaults, in line with Gerardi et al. (2018). A typed gap that runs mostly through default costs is therefore compatible with a default population that is overwhelmingly distressed.

### OA6.3 Untargeted Moments

Six objects are computed at the estimate but held out of the SMM objective; Table 6 in Section 7 reports them, and this subsection gives the construction of each target. Each is a prediction the fitted model should reproduce without having been fit to it, and together they discipline the parts of the mechanism the six targeted moments leave free: how the cosign–solo default gap survives conditioning on income and house size, how much of it the parental cure absorbs, what survives to terminal loss, what the rate charges for it, and how cosigning varies with child and with parent income.

$U_1$  and  $U_5$  are the conditional and terminal relatives of the one wedge the objective does target ( $M_6$ , the *pooled pre-cure* wedge): the estimator is asked to match the raw cosign–solo delinquency gap and nothing more, so how that gap survives conditioning ( $U_1$ ) and how much of it the parental cure removes ( $U_4$ ,  $U_5$ ) remain genuine out-of-sample predictions.  $U_1$  is the within- $(z, h)$  wedge measured before the cure, benchmarked to the +0.56 pp the NSMO within-cohort wedge shows once FICO and CLTV are added; the raw +1.36 pp version, itself a lower bound, is the targeted  $M_6$  (Appendix OA4.11). The model’s wedge is right-signed and about half the conditioned benchmark, so the estimated economy understates the conditional wedge rather than manufacturing it.

$U_2$ , the cosign rate premium, is economically negligible in the data whichever way it is measured, though the two measures differ sharply in precision and should not be blurred. Conditioning within the NSMO on FICO and CLTV gives  $\approx +0.01$  pp, indistinguishable from zero; the HMDA APR-minus-APOR gap is +0.055 pp with a standard error of 0.002, estimated on millions of loans and so distinguishable from zero many times over while remaining a twentieth of a percentage point. What matters for the mechanism is the magnitude and not the  $t$ -statistic. This is a different object from the unconditional cosign rate gap of Section 4 (Fact 3), which is an order of magnitude larger because it conditions on far less. A near-zero  $U_2$  beside a positive  $U_1$  is the *decoupling*: the cosigned pool is worse, and barely priced.

$U_3$ , the cosign income gradient, is measured on the paper universe (conventional  $\times$  first-time buyers) at  $-0.051$ , 95% CI  $[-0.242, +0.141]$  (NSMO,  $n = 8,249$ ): flat and imprecise. The model’s gradient is  $-0.01$ , inside that interval and close to its centre, where the previous specification fell outside it with the wrong sign. It is also inside the interval of the non-occupant refinement ( $+0.317$ , CI  $[-0.021, +0.655]$ ), which restricts cosigners to the non-resident ones the model describes and is the closer conceptual counterpart. Both flags are reported because neither is distinguishable from zero and they carry opposite signs; the all-originations value  $-0.148$  is driven by repeat buyers outside the paper universe. Within first-time buyers, cosigning is a cohort-wide instrument, not an

TABLE OA38. Parameters set outside the SMM: internal calibration, taste, and numerical settings

| Parameter                                                  | Symbol             | Value     | How set                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------|--------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Panel C: calibrated to an internal target</i>           |                    |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Bad-type repayment prob.                                   | $\pi_B$            | 0.90      | <i>internal target</i> : aggregate default level                                                                                                                                                                                                                                                                                                                                                                                                      |
| Rent-to-price ratio                                        | $p_{rent}/p_{own}$ | 0.090     | gross rental yield on a <i>constant-quality</i> unit, so the price–rent ratio of 11.1 is not the quoted 15–20: <a href="#">Urban Institute (2025)</a> find owner-occupied homes $\approx 37\%$ more valuable than rentals after quality adjustment, giving a comparable ratio of 15.2 and a 6.6% yield against a US average of 6.7%                                                                                                                   |
| Owner utility premium                                      | $\nu$              | 1.100     | <i>internal target</i> : the homeownership rate. <a href="#">Kaplan et al. (2020)</a> also calibrate this parameter internally, to 1.015; the present value is higher, which the rental-grid cap below partly stands in for                                                                                                                                                                                                                           |
| Cash-transfer size                                         | $\psi_{cash}$      | 0.100     | share of parent income at $j = 43$ , scaled by house size: 6.0–10.8% of parent income, or 0.3–2.9% of house value (0.9% at the median parent). Sits between the median and mean of the PSID parental-transfer distribution in <a href="#">Brandsaas (2025)</a> (\$1,000 / \$3,960 / \$10,000 at the median / mean / p95, 2016 USD). The <i>incidence</i> is a separate targeted moment (26.2% against a target of 24.95%). Re-solved at $\pm 25$ –50% |
| Rental-grid cap                                            | $h_{rent}^{max}$   | 0.40      | <i>internal target</i> : makes $HO(z)$ monotone increasing. <b>Cost</b> : owner/renter size ratio $3.4\times$ vs $1.5\times$ in data (untargeted)                                                                                                                                                                                                                                                                                                     |
| Type–income gradient scale                                 | $\lambda_\alpha$   | 2.545     | normalisation: $\sqrt{1 - \rho^2}/\sigma_\varepsilon$                                                                                                                                                                                                                                                                                                                                                                                                 |
| <i>Panel D: structural taste dispersion</i>                |                    |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Parental-help taste dispersion                             | $\sigma_{help}$    | 0.50      | <i>structural</i> : dispersion in the parent’s willingness to help. Deliberately an order of magnitude above the numerical smoothers in Panel E — at 0.1 cosigning vanishes altogether, so the parameter is bounded below by the existence of the margin                                                                                                                                                                                              |
| <i>Panel E: numerical (solution method, not economics)</i> |                    |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Default-choice smoothing                                   | $\sigma_{def}$     | 0.100     | logit smoothing; not calibrated                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Rent/buy smoothing                                         | $\sigma_{rb}$      | 0.100     | logit smoothing; not calibrated                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Menu smoothing                                             | $\sigma_{menu}$    | 0.100     | logit smoothing; not calibrated                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Asset / income / house grid points                         | $n_a, n_z, n_h$    | 30, 11, 6 | discretisation                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Lifespan (periods)                                         | $J$                | 70        | ages 1–70                                                                                                                                                                                                                                                                                                                                                                                                                                             |

*Notes*: Continues Table OA37. **C** is internal calibration, chosen so the model matches a stated target and therefore not independent evidence for what the model then delivers; **D** is a preference parameter and **E** affects only the numerical solution. Estimated parameters: Table D.2.

TABLE OA39. Involuntary default hazard across the  $\Delta\pi$  robustness band

|                            | $\Delta\pi = 0.03$ | $\Delta\pi = 0.05$ | $\Delta\pi = 0.07$ |
|----------------------------|--------------------|--------------------|--------------------|
| Good-type hazard (%/yr)    | 0.242              | 0.171              | 0.101              |
| Bad-type hazard (%/yr)     | 0.351              | 0.351              | 0.351              |
| Mass-weighted floor (%/yr) | 0.296              | 0.261              | 0.226              |
| Typed involuntary ratio    | $1.45\times$       | $2.05\times$       | $3.45\times$       |
| Share of log typed gap     | 19%                | 37%                | 64%                |

*Notes*: Annual involuntary hazard  $1 - \pi_\theta^{1/M}$  at  $M = 30$ ,  $\pi_B = 0.90$ , the floor mass-weighted at the stationary type mix ( $\Pr(\text{Good}) = 0.500$ ). Last row:  $\log(\text{typed involuntary ratio})/\log(6.93)$ , the denominator being the typed default ratio  $M_5$  targets.

income-graded one. It is reachable only through  $p_{own}$ , opposite to homeownership, and so is held out.

$U_4$  is the recovery-side complement to  $U_1$ : the share of the cosign pool's excess early-stage risk that never reaches terminal loss. In the data it is one minus the terminal-to-pre-cure ratio of the within-cohort wedge,  $1 - 0.08/1.36 = 0.94$  raw and  $1 - 0.08/0.56 = 0.86$  with FICO/CLTV controls; both ingredients are attenuated by the administrative retention of delinquency flags, so the range is indicative rather than sharp. In the model it is the mass-weighted coverage probability  $p_{cover}$  over delinquent cosigned borrowers.  $U_5$  completes the ladder with the wedge that survives the cure, cosigned delinquency  $\times (1 - p_{cover})$  less solo default; it is also the fair premium of Section 8 up to loss given default, since netting the composition charge against the recovery credit cancels the counterfactual hazard. This is the model's clearest miss. At  $-0.69$  pp against the NSMO's  $+0.08$  pp the estimated economy resolves too many cosigned delinquencies in the parent's favour, and the attenuation in the data pushes the target up rather than down, so the gap is not a measurement artifact. Together  $U_1$ ,  $U_4$  and  $U_5$  identify what the level premium cannot: the pool is worse, the guarantee absorbs most of the excess and too much of it, and the rate prices none of it.

#### OA6.4 Standard Errors: Construction and What They License

Standard errors on  $\hat{\theta}$  are delta-method. The moment map is differentiated numerically at  $\hat{\theta}$  by central differences at a 5% relative step — one baseline and 12 perturbed equilibrium solves — and combined with the  $6 \times 6$  sampling covariance  $\Sigma$  of the targeted data moments as  $V = \Lambda \Sigma \Lambda'$ .  $\Sigma$  is assembled from the moments' own sources rather than from the tolerances entering  $\mathbf{W}$ , which are chosen weights and not sampling variances: five of the six diagonal entries are measured (CPS ASEC, SHED, and a joint NSMO bootstrap) and the cosign share carries a construction spread. Cross-source blocks are zero by independent samples, so the only non-zero off-diagonals lie inside the NSMO block {default, typed ratio, wedge}. Because the system is just-identified by construction — six moments, six parameters — the weighting matrix cancels and  $\Lambda = -G^{-1}$ ; this holds numerically to  $3 \times 10^{-12}$ , and  $\text{cond}(G) = 3.5 \times 10^3$ .

**Does the Jacobian survive the solver's own residual?** Each of the 12 perturbed solves is an equilibrium computed to the tolerance of Online Appendix OA5.2, not to machine precision, and  $G$  is then inverted. How much of the reported precision that leaves can be measured from the solves themselves. Every parameter contributes three evaluations of every moment, and the central difference uses only the antisymmetric part  $|m_+ - m_-|$ ; the symmetric part  $|m_+ + m_- - 2m_0|$  is  $O(h^2)$  for a smooth map, so what remains in it is the solver's own noise. Their ratio measures how far each entry of  $G$  stands above that floor, and Panel A of Table OA40 reports all 36 of them.

26 of the 36 entries clear the floor, at a median of 2.5 to one. The 10 that do not are, with 2 exceptions, entries  $G$  itself places near zero: they sit at a median 0.2% of the largest entry in their own row, against 24% for the rest, so they carry almost no weight in  $G^{-1}$ . Both exceptions lie in the Cosign row, which is the one place where the noise falls on an entry large enough to matter.

TABLE OA40. The moment map’s numerical resolution, and what it does to the standard errors

| <i>Panel A. Resolution of each Jacobian entry, <math> m_+ - m_-  /  m_+ + m_- - 2m_0 </math></i> |             |             |             |             |              |             |
|--------------------------------------------------------------------------------------------------|-------------|-------------|-------------|-------------|--------------|-------------|
|                                                                                                  | $\bar{\xi}$ | $p_{own}$   | $\eta_{co}$ | $\eta_{ca}$ | $\Delta_\xi$ | div         |
| HO                                                                                               | 4.8         | 5.1         | 10.5        | <i>0.14</i> | 1.9          | <i>0.68</i> |
| Default                                                                                          | 11.2        | 1.1         | <i>0.29</i> | 1.5         | 164          | <i>0.09</i> |
| Cosign                                                                                           | 1.4         | <i>0.58</i> | 15.6        | <i>0.03</i> | <i>0.89</i>  | <i>0.32</i> |
| Cash                                                                                             | 13.8        | 1.3         | 1.2         | 75.4        | 52.8         | 1.8         |
| TypeRatio                                                                                        | 8.3         | 1.2         | 10.8        | 3.2         | 6.9          | 4.4         |
| Wedge                                                                                            | 6.1         | 4.0         | <i>0.29</i> | <i>0.09</i> | 11.3         | 57.1        |

| <i>Panel B. Standard errors under the measured numerical noise in <math>G</math></i> |          |          |        |           |             |
|--------------------------------------------------------------------------------------|----------|----------|--------|-----------|-------------|
|                                                                                      | Reported | 5th pct. | Median | 95th pct. | 95th / rep. |
| $\bar{\xi}$                                                                          | 0.8298   | 0.7080   | 0.8340 | 1.0190    | 1.23        |
| $p_{own}$                                                                            | 0.0888   | 0.0685   | 0.0906 | 0.1267    | 1.43        |
| $\eta_{co}$                                                                          | 0.1069   | 0.1053   | 0.1208 | 0.1572    | 1.47        |
| $\eta_{ca}$                                                                          | 0.0704   | 0.0694   | 0.0705 | 0.0716    | 1.02        |
| $\Delta_\xi$                                                                         | 1.5123   | 1.1957   | 1.5131 | 2.0583    | 1.36        |
| div                                                                                  | 0.3782   | 0.3420   | 0.3826 | 0.4413    | 1.17        |
| cond( $G$ )                                                                          | 3486     |          | 3513   | 4521      | 1.30        |

*Notes:* Built from the one baseline and 12 perturbed equilibrium solves behind the delta-method standard errors, at a 5% central-difference step. Panel A divides the antisymmetric difference the derivative uses by the symmetric residual it discards; the latter is  $O(h^2)$  for a smooth map, so it measures the solver’s own noise plus curvature and *overstates* the noise. Entries below one are set in italics: they sit at a median 0.2% of the largest entry in their own row, against 24% for the rest, so they are entries  $G$  itself places near zero. Panel B redraws  $G$  20,000 times with each entry perturbed by that residual and recomputes  $\sqrt{\text{diag}(G^{-1}\Sigma G^{-\prime})}$ .

Panel B turns that into a statement about the standard errors. Perturbing every entry of  $G$  by its own measured residual and redrawing 20,000 times, the median standard error stays within 13% of the reported value for all six parameters, and the 95th percentile runs from 1.02 to 1.47 times it, the widest being  $\eta_{co}$ ; cond( $G$ ) moves from 3486 to a median of 3513 and a 95th percentile of 4521. Two readings follow. The standard errors are good to a factor of about one and a half rather than to the digits Table D.2 prints, so the  $t$ -statistics should be read as orders of magnitude. But they are not uninformative, and the ranking of precision across the six parameters is unchanged in every draw. Because the symmetric residual contains genuine curvature as well as noise, it overstates the noise, and this bound is therefore conservative.

The variance decomposition explains the uneven precision of Table D.2. Splitting each parameter’s variance by moment block,  $\eta_{cash}$  draws 99.8% of its variance from the cash-gift share alone and  $\eta_{cosign}$  draws 95% from the cosign share: both are locally identified by the moment they were chosen to match. The three default-technology parameters —  $\bar{\xi}$ ,  $\Delta_\xi$ , and the divisor — draw essentially all of theirs from the typed default ratio and the pre-cure wedge, so their imprecision is inherited from the sampling variance of those two NSMO targets rather than generated by the estimation.

One further check a reader might expect is unavailable for a structural reason: setting the model’s response to the 2022 fee reform beside the estimates of Section 5. The model has six house

sizes and therefore no running variable or threshold, and one might hope to compare *estimands* instead — the differential pass-through of a reform-sized fee shock. The obstacle is support, not resolution. At the estimated parameters 95.0% of buyers sit in the smallest house tier, only 1.9% sit above the model’s conforming limit at all, and the cosigned mass above that limit is *zero*, in the origination flow and the owner stock alike, against 4.9% below it; the cell the reform design identifies from is therefore empty in the model, and a differential computed there would price a schedule no household faces. Nor is placement the issue: the largest above-threshold share attainable on the house grid is 5.0%, against the 17.8% share of the reform’s estimation sample. The two exercises accordingly discipline the same mechanism on different populations, and neither validates nor refutes the other.

## OA6.5 Where the Coverage Defect Sits

Two questions about the coverage probability are answered here: why matching the NSMO’s post-cure gap appears to demand an implausibly large parental obligation, and where the specification genuinely strains.

Take the first. Reproducing that gap through coverage alone requires an obligation near six years of scheduled debt service — the parent servicing the loan outright, a balance-scale liability rather than the payment-scale one joint-and-several obligation creates. The reason the moment asks for it is a measurement artifact rather than an economic fact. The NSMO records a loan only once it is thirty days late, so a cosigning parent, who receives the same notices as the child, and who meets a missed payment inside that window takes the loan out of the delinquent pool altogether. In the data that episode is indistinguishable from a loan that never fell behind; in the model it is coverage. Matching the post-cure gap therefore asks the coverage channel to produce the whole observed gap after day thirty when part of it was resolved before, and identifies coverage only among the families whose parent did *not* act in the first month. Read that way the exercise bounds coverage from below and corroborates the one-year normalization rather than competing with it. What would settle it is a payment-level record that observes cures inside the thirty-day window and identifies who made them, which neither the NSMO nor the servicing data used here provides.

Now the second. Section 7 names  $p_{cover}$  as the likely source of the terminal-wedge miss ( $U_5$ ), and the diagnosis a reader reaches for first is the wrong one. It is tempting to say the obligation is too *small* — that a cosigner is in practice liable for the accelerated balance rather than for one payment — but that fails here on two counts.

First,  $m$  is a full year of debt service, while a cure in practice runs one to four monthly payments. This economy already charges the guarantor several times what reinstatement actually costs, so moving the obligation toward a realistic arrears path pushes  $p_{cover}$  further *toward* one, not away from it.

Second, there is no acceleration node for a balance-scale obligation to attach to. A cosigned child who invokes keeps the house and re-faces the same choice next period, so no terminal event

exists at which a balance falls due; adding one is a change to the contract, not a rescaling of this one.

Where the specification does strain is that  $p_{cover}$  is asked to do two jobs at once. It measures the parent’s capacity, and it also multiplies the expected liability the parent prices at  $j = 43$ , so the parent is charged in precisely the states where they can pay and not at all in the states where they cannot. A shortfall therefore reduces the guarantee’s cost to the family at the same time as it reduces its value to the child, which is the source of the scale asymmetry Section 8 reads off the pricing null. Separating the two — a capacity measure built on the parent’s resources rather than gross labour income, and a liability that survives a shortfall as a deficiency rather than vanishing with it — is the repair the estimate needs. It requires structure the present model does not carry and re-opens the whole parameter vector, which is why it is flagged as the leading open question at this estimate rather than folded into the present vintage.

## OA6.6 Decomposing the Cosign Premium

The operative cosign premium is the net of two forces, and separating them, as Table OA41 does, shows that its near-zero level is a cancellation rather than an absence. At origination ( $j = 19$ , all cells) I evaluate a no-signal counterfactual in which the bank prices the cosigned pool on the income prior  $\lambda_z$  alone — as if cosigning conveyed nothing about borrower type. The gap between that counterfactual and the equilibrium rate is the value of the signal. At the current estimate, pricing on the prior alone would give cosigners a *discount* of  $-0.013$  pp, because cosigners are drawn disproportionately from income cells whose prior Good-type share is high; updating on the cosigning decision itself then *raises* the rate by  $+0.021$  pp, because the realised Good-type share of the cosigned pool falls short of what its income composition predicts. The two leave a net premium of  $+0.008$  pp. That is the same object as the  $+0.4$  basis points charged in Table 8, evaluated on the baseline diagnostics solve rather than on the estimate’s.

The signs on the price side are the reverse of those on the quantity side: the type-channel split of Section 7 attributes  $+1.48$  pp to need-based adverse selection against a screening correction of  $-1.20$  pp. The two decompositions are not the same object — one is mass-weighted over overlap cells on the default-hazard basis, the other spans all cells on the rate basis at  $j = 19$ , and the pools are weighted at different ages — so a sign difference is possible without either being wrong. I have not established which weighting drives it, and I record the disagreement rather than assert a reconciliation. What both agree on is the structure that matters for the paper: two offsetting forces leaving a premium under one basis point.

## OA7 Counterfactual: Supporting Diagnostics

### OA7.1 Two Nulls: Hidden-Type Sorting and House Size

Two results follow from the design of §8.3 rather than qualifying it, and both are nulls. The reform prices and lends against *observables*, so it should not re-sort borrowers on hidden type, and it

TABLE OA41. Cosign Premium Decomposition ( $j = 19$  origination, all cells)

| Component                                                     | Premium (pp)   |
|---------------------------------------------------------------|----------------|
| Gross adverse selection ( $\lambda_z$ prior, no screen)       | −0.0130        |
| Screening correction ( $q_G^{\text{imp}} - \lambda_z$ effect) | +0.0206        |
| Net premium (prior + screen) — model                          | +0.0076        |
| Net premium — data (NSMO / HMDA range)                        | +0.01 to +0.06 |

*Notes:* Computed at the SMM estimate from the baseline equilibrium diagnostics (`baseline_premium_decomposition.csv`); the components sum to the net exactly. The first row is which income cells cosigners come from, the second how their realised type mix departs from what those cells predict.

does not: the solo pool’s Good-type share does not move at all, 0.733 against 0.732, while the cosigned pool’s falls 5.9 points as deposit relief admits marginal entrants (Panel C of Table E.1). The estimated economy stays in the adverse-selection configuration across the states carrying 93 percent of buyers, and the configuration in which the parental signal dominates appears where Section 3 places it but on too little mass to interpret (Figure OA12). What changes is who takes the contract, not how the two pools are ranked.

House size behaves the same way, and Panel D of the same table rules out the reading that movements merely cancel: cosigned buyers trade slightly *down*, buyers who do not cosign do not move at all, and what rises is the count rather than the size. The deposit relief brings new households into ownership at the prevailing house size instead of moving incumbents up the ladder, so the intensive margin is not where this reform acts.

## OA7.2 Supporting Figures for the Main Results

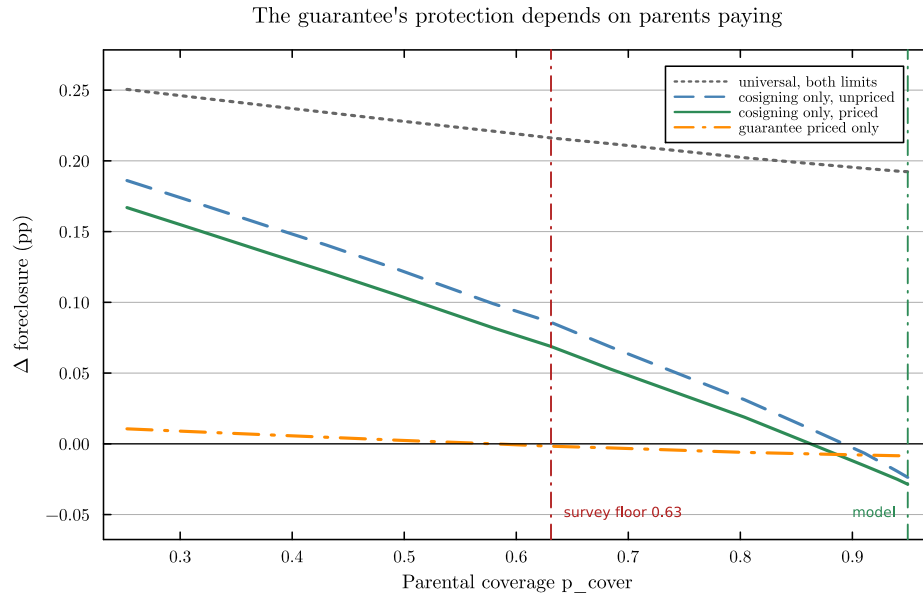
The five exhibits below support results stated in Section 8 and are reported here because the main text reads their numbers off a table rather than off the graphic.

Figure OA10 traces the two coverage columns of Table 9 across the full range of the parental obligation and locates the crossing point at which the targeted designs stop reducing foreclosure — the point the model’s 0.950 and the survey’s floor of 0.631 sit either side of. The cosigned designs cross zero at  $p_{\text{cover}} \approx 0.87$ , which leaves the model’s own coverage just above the sign change and the survey’s floor of 0.631 below it. Universal relief stays above zero throughout, drifting from +0.19 to +0.25 pp rather than holding flat, since it has a guarantee to weaken only through the cosigning it induces.

Figure OA11 gives the by-state profile behind the intensive-margin null of §OA7.1, whose aggregates are Panel D of Table E.1.

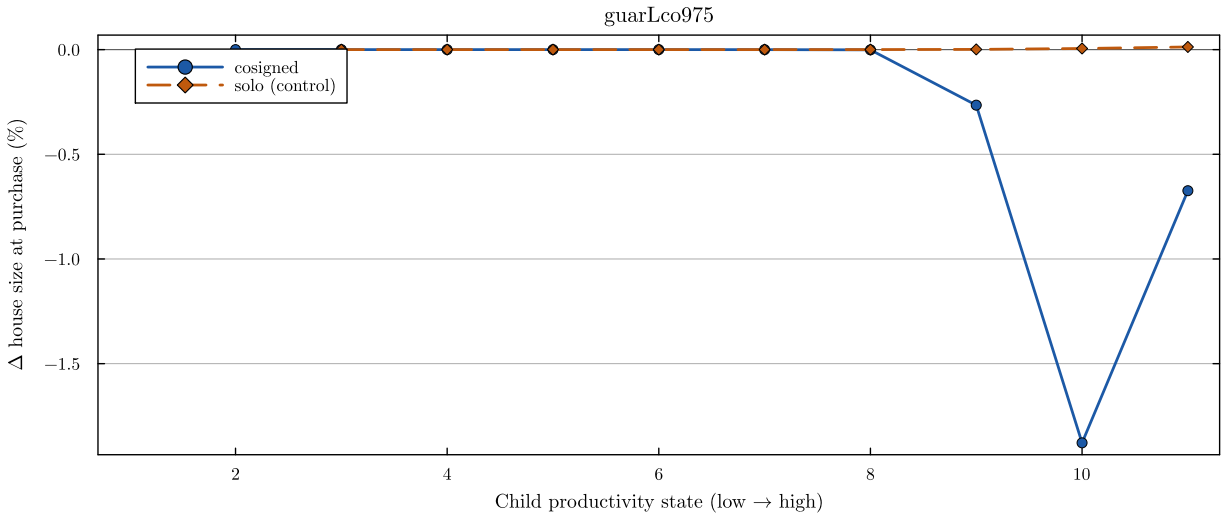
Figure OA12 asks whether the estimated economy reproduces the three-region structure of Section 3, and reports the mass behind every cell so the answer can be weighted. It does, with the qualification that the regions are of very unequal size. Panel A gives the cosign–solo rate gap: positive across  $z_c$  6–9, the adverse-selection configuration of Region 2, and negative at  $z_c$  4–5, where

FIGURE OA10. The guarantee's protection depends on parents paying



Change in the aggregate foreclosure rate as parental coverage varies, for the four reforms of Table 9.

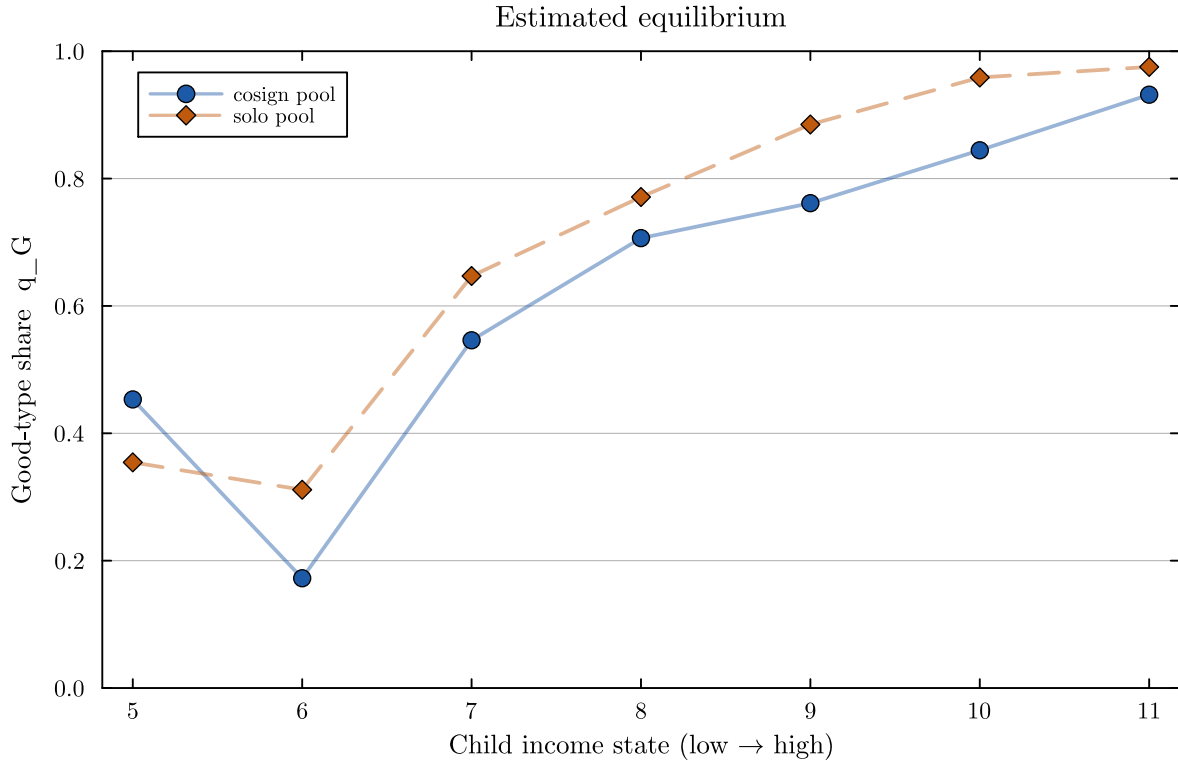
FIGURE OA11. Intensive margin by child productivity



Change in house size at purchase, cosigned against solo. *Caveat:* baseline house size sits at the bottom rung of the grid for  $z_c \leq 7$ , so the flat region there is partly the grid floor rather than an absence of response.

the parental signal dominates as in Region 1. Panel B gives the pool composition that gap is priced from. Panel C is the reliability check the other two must be read against, and it is what settles the weighting: the cells carrying Region 1 sit at or below the  $10^{-4}$  mass floor, which is why their markers are faded in all three panels.

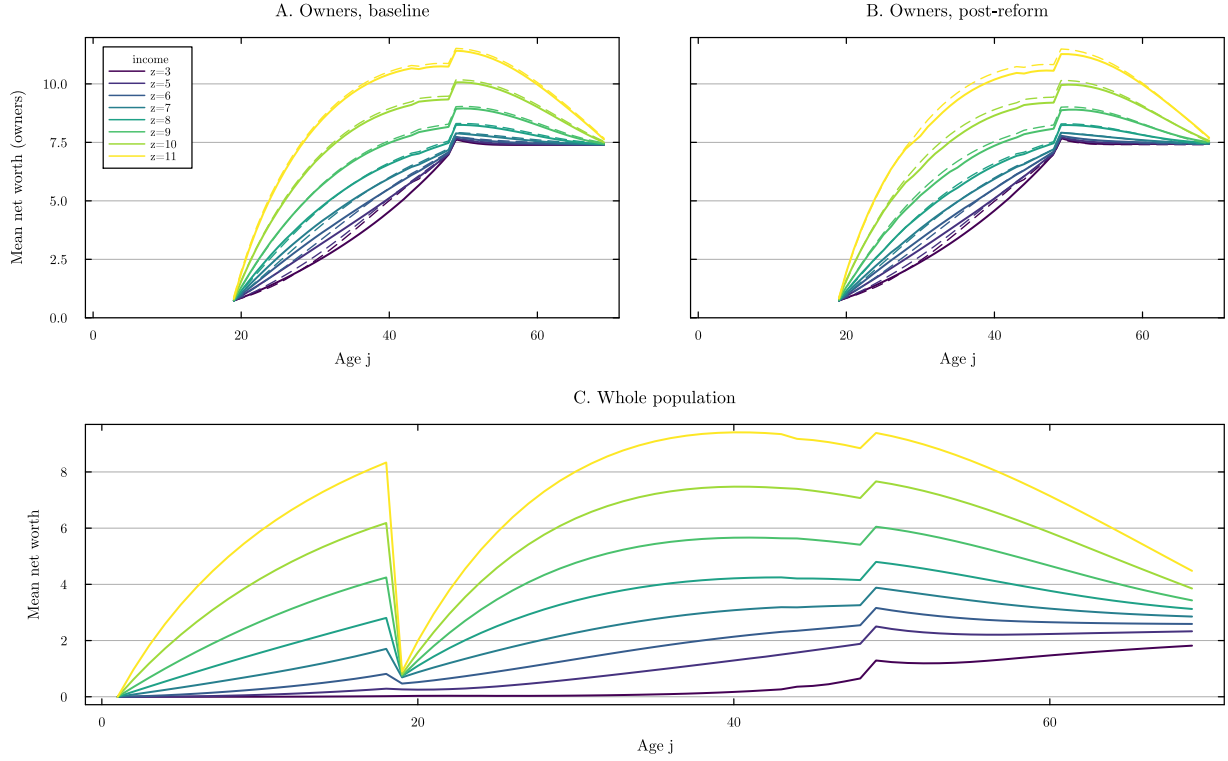
FIGURE OA12. Does the estimated economy reproduce the three-region equilibrium?



**A.** Cosign minus solo rate gap in basis points. **B.** Good-type share of each pool. **C.** Buyer mass supporting each cell, on the decision ( $j = 18$ ) and origination ( $j = 19$ ) bases, log scale against a  $10^{-4}$  floor; faded markers everywhere are cells at or below that floor. A single-economy diagnostic at the estimate, not a pre/post comparison.

Figure OA13 follows the cosigned–solo net worth comparison of §8.4 from origination to death. All owners converge to the same net worth at  $j = 19$  regardless of income, since the deposit has just been paid and nobody holds anything else. Adding renters lowers the levels and exposes the purchase cliff: an eight-fold spread in wealth at  $j = 18$  becomes under two-fold at  $j = 19$ . Across the owners-only panels the cosigned line sits below its solo counterpart at every age and income in the baseline; post-reform the mid-range states converge onto and cross it, while the top state takes a mid-life leverage penalty before recovering. The bequest jump at  $j = 48$  is proportionally largest at the bottom:  $z_c = 3$  roughly doubles,  $z_c = 11$  gains about 7%.

FIGURE OA13. Net worth over the life cycle, cosigned versus solo



Solid lines are cosigned owners, dashed lines solo owners at the same income state; colour runs from low income (dark) to high (light). **A** and **B** are owners only and share a vertical scale, so the reform can be read across them. **C** adds renters, so its levels sit lower; the jump at  $j = 48$  is the bequest. Income state is measured at age  $j$ , not at purchase.

### OA7.3 Welfare: CEV by Child Income State

Welfare in Section 8 is reported as the consumption-equivalent variation

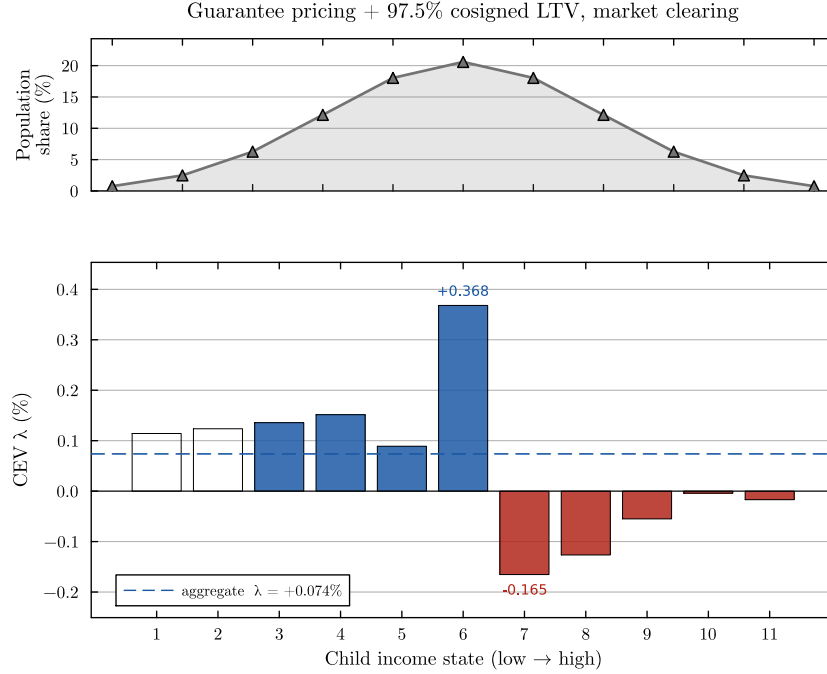
$$\lambda = \left( \frac{V^{CF} + gB}{V^{BL} + gB} \right)^{1/\alpha} - 1, \quad (\text{OA38})$$

with  $g = 1/(1-\sigma)$  and  $\alpha = (1-\theta)(1-\sigma)$  — at  $\sigma = 1.5$  and  $\theta = 0.25$ ,  $g = -2$  and  $\alpha = -0.375$  — and  $B$  the survival-discounted horizon. The aggregate is taken over the stationary income distribution across states with positive buyer mass.  $V$  is the lifetime value of an agent entering at  $j = 18$ , so both stages of the life cycle sit inside it: the same agent reaches the parental stage at  $j = 43$  and bears the offer cost and the contingent liability there.

Figure OA14 plots the state-level gains against the income distribution that weights them, and Table OA42 gives the same values as levels. Both are read under market clearing at  $\varepsilon = 2.2$ ; the fixed-price counterparts are the  $\Delta\text{EV}(\text{PE})$  column of Table 7. The peak of  $+0.368\%$  at  $z_c = 6$  coincides with the mode of the income distribution, which is why a gain concentrated on one state still moves the aggregate. The profile is redistributive: states  $z_c \leq 6$  gain,  $z_c \geq 7$  lose up to  $-0.165\%$ , and the aggregate is  $+0.074\%$ . At fixed prices every state gains instead and the aggregate

is +0.237%, so the sign reversal at the top is a general-equilibrium object: house prices rise 0.68% while rents fall, which transfers from incumbent owners to renters (Table 7).

FIGURE OA14. Welfare gains and where the population is



Consumption-equivalent variation by child income state *under market clearing* ( $\varepsilon = 2.2$ ), with the stationary income distribution above; the two panels must be read together. Hollow bars carry no buyer mass.

TABLE OA42. Welfare: CEV by Child Income State — Guarantee Pricing + 97.5% Cosigned LTV, Market Clearing

| $z_c$         | 3      | 4      | 5      | 6      | 7      | 8      | 9      | 10     | 11     |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| $\lambda$ (%) | +0.136 | +0.151 | +0.089 | +0.368 | -0.165 | -0.127 | -0.055 | -0.004 | -0.017 |

Notes: CEV from (OA38) for guarantee pricing with a 97.5% cosigned LTV limit, under market clearing ( $\varepsilon = 2.2$ ), by child income state, over states with positive buyer mass; the aggregate, weighted by the stationary income distribution over these states, is +0.074%. Values of  $|\lambda| \leq 0.03\%$  are within solution tolerance (2 of 9 states here).

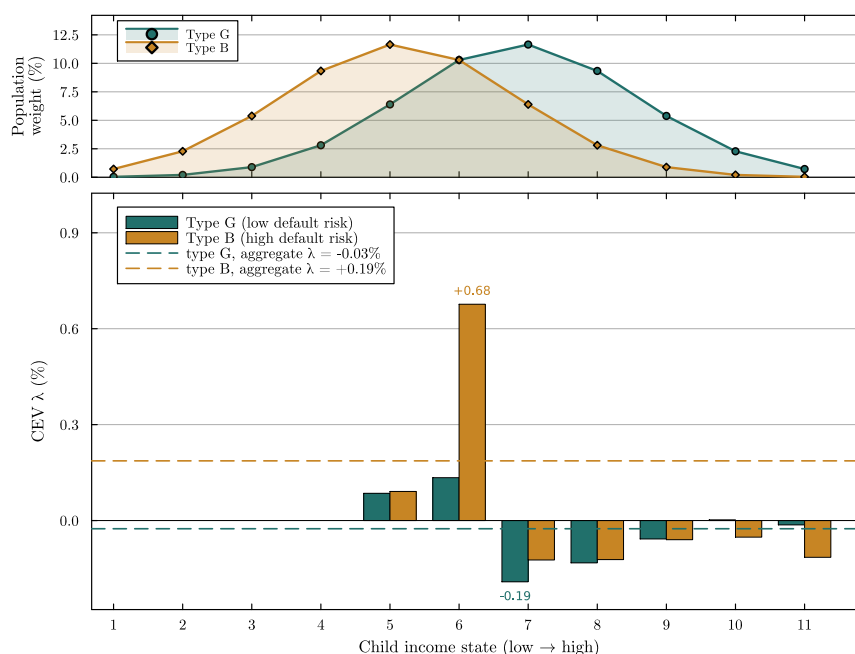
A second split asks who gains by hidden borrower type rather than by help type. Figure OA15 reports it: the reform is worth +0.19% of lifetime consumption to high-risk borrowers and -0.03% to low-risk ones. Part of that difference is composition, since the probability of being low-risk rises with income and the profile turns negative at the top of the distribution, but most of it survives conditioning on income. At  $z_c = 6$ , where the two types are equally likely and therefore carry identical population weight, the gain is +0.68% for the high-risk type against +0.13% for the low-risk one.

The mechanism is the value of the contract rather than the number who take it. At that state

the reform raises the cosigned-versus-solo value gap roughly five times as much for the high-risk type as for the low-risk one, while cosigning take-up rises by 18 and 13 percentage points respectively. Constraint relief is worth most to the borrower who would otherwise be rationed, which is also why the difference closes above  $z_c = 7$ : once the deposit constraint stops binding, the two types value the contract almost identically.

Two things about the aggregation. The probability of being the low-risk type rises from 0.05 at  $z_c = 1$  to 0.95 at  $z_c = 11$ , so the low-risk weight peaks at  $z_c = 7$  and the high-risk weight at  $z_c = 5$ ; about two-fifths of the +0.21 pp gap between the two aggregates is that difference in composition and about three-fifths the state-level gains themselves. And each economy is weighted by its *own* help-type shares, so the bars carry each type's financing response to the reform. Holding those shares at their baseline values instead leaves  $-0.09\%$  and  $+0.04\%$ , which is the sense in which the result runs through what switching into the guarantee is worth rather than through the terms of a contract a household already holds.

FIGURE OA15. Welfare gains by hidden type



Consumption-equivalent variation by child income state, split by the hidden borrower type, under market clearing ( $\varepsilon = 2.2$ ). The panel above plots the aggregation weights, the population share of each type at each income state. States below  $z_c = 5$  carry no homebuyers of either type and are omitted.

## OA7.4 Which Friction Binds, by Child Income

Table OA43 gives the state-by-state evidence behind the claim of Section 8.2 that the income test is slack everywhere once a parent is on the application, and that what excludes the bottom of the distribution is the deposit.

TABLE OA43. Which friction binds, by child income

| $z_c$ | own DTI | joint DTI | can fund deposit | buyers | binding constraint |
|-------|---------|-----------|------------------|--------|--------------------|
| 1     | 1.249   | 0.288     | 0.0%             | 0.0%   | own capacity       |
| 2     | 0.986   | 0.271     | 0.1%             | 0.0%   | deposit            |
| 3     | 0.779   | 0.253     | 0.5%             | 0.0%   | deposit            |
| 4     | 0.616   | 0.233     | 2.5%             | 0.0%   | deposit            |
| 5     | 0.486   | 0.211     | 10.4%            | 4.8%   | deposit            |
| 6     | 0.371   | 0.183     | 32.7%            | 35.2%  | deposit            |
| 7     | 0.289   | 0.159     | 70.0%            | 88.5%  | —                  |
| 8     | 0.227   | 0.139     | 92.9%            | 99.9%  | —                  |

Notes: At the smallest house. Own DTI is the payment over the child's own income (cap 0.40); joint DTI adds the median parent's (cap 0.42).

### OA7.5 The Asset Threshold for Purchase, by Help Status

Section 8.2 argues that cosigning relaxes the income test and never the equity test. Table OA44 is the direct evidence: it reports the own assets at which a child's probability of buying at  $j = 18$  first reaches one half, separately for cosigned and solo contracts. In the baseline the two columns coincide wherever a solo buyer reaches the threshold at all, because the two contracts face the same loan-to-value limit,  $\lambda_m^{co} = \lambda_m$ . What cosigning buys at low income is not a cheaper deposit but admission: a solo buyer at  $z_c \leq 5$  never reaches the threshold at any asset holding, while a cosigned one does. Raising  $\lambda_m^{co}$  to 0.975 is what separates them, and it is what turns the guarantee into a deposit substitute. Required assets fall by 28–48% at  $z_c \leq 5$  and to zero at  $z_c = 6$ , while every solo column is unchanged and  $z_c \geq 7$  is unaffected because those children already needed almost no assets. The extensive-margin response is therefore concentrated at  $z_c = 6$ , the one state where the constraint both binds and is cleared.

### OA7.6 The Fair Premium Across the Parental Obligation

Write  $d_{solo}$  and  $d_{cos}$  for the realized annual default hazards of the solo and cosigned pools, and  $d_{cos}^{CF}$  for the hazard the cosigned pool would face with the guarantee withdrawn and its composition held fixed. A lender should charge for the cosigned pool being the riskier one and credit itself with what the guarantee is worth — the defaults it deters, plus the invocations parents meet — so that, as annual spreads per dollar of balance at loss given default  $\ell$ ,

$$\begin{aligned}
\text{composition charge} &= (d_{cos}^{CF} - d_{solo}) \ell, \\
\text{recovery credit} &= (d_{cos}^{CF} - d_{cos} (1 - \bar{p})) \ell, \\
\text{fair premium} &= \text{composition} - \text{recovery} = [d_{cos} (1 - \bar{p}) - d_{solo}] \ell.
\end{aligned} \tag{OA39}$$

TABLE OA44. Own assets required to buy at  $j = 18$ , by help status

| $z_c$ | Baseline |          | Guarantee pricing + 97.5% LTV |          |
|-------|----------|----------|-------------------------------|----------|
|       | solo     | cosigned | solo                          | cosigned |
| 1     | never    | never    | never                         | never    |
| 2     | never    | 17.18    | never                         | 8.98     |
| 3     | never    | 12.42    | never                         | 8.98     |
| 4     | never    | 8.98     | never                         | 6.50     |
| 5     | never    | 6.50     | never                         | 4.70     |
| 6     | 1.77     | 1.77     | 1.77                          | 0.00     |
| 7     | 0.00     | 0.00     | 0.00                          | 0.00     |
| 8     | 0.00     | 0.00     | 0.00                          | 0.00     |
| 9     | 0.02     | 0.02     | 0.02                          | 0.02     |
| 10    | 0.17     | 0.17     | 0.17                          | 0.17     |
| 11    | 0.34     | 0.34     | 0.34                          | 0.34     |

*Notes:* Own assets at which the probability of buying at  $j = 18$  first reaches 0.5, averaging the policy function over parent income and parental housing with the equilibrium eligible-renter mass. “never” means the buy probability does not reach that level at any asset holding.

The counterfactual hazard enters both components and cancels exactly, not to a first order, which is why the fair premium carries no separate credit for deterrence: deterred defaults are already netted out of  $d_{cos}$ , and what is left is the terminal post-cure wedge priced at  $\ell$ .

Section 6 sets the parent’s obligation at one year of scheduled debt service as a normalization. The sign of (OA39) is therefore a function of the coverage  $\bar{p}$  that obligation implies and of nothing else. Table OA45 reports it across the scale, together with the value at which the sign turns. That value is not a separate object: writing  $\mu_j = 1 - d_j$ , the turning point  $1 - d_{solo}/d_{cos}$  is exactly the coverage threshold  $\bar{p}^*$  of Proposition 2(ii), so the table evaluates the stylized model’s condition at the estimated hazards. The result reported in the main text requires the parent’s effective obligation to stay below roughly 4.0 years of debt service. What the NSMO terminal rung supplies is a *floor* on coverage, 0.631, rather than an estimate of it: the survey observes a loan only once it is thirty days late, so a parent who meets a missed payment inside that window takes the loan out of the delinquent pool altogether and the coverage identified from what remains is a lower bound. Evaluating the premium there accordingly bounds it from *above* rather than measuring it. At that floor the fair premium is +2.0 to +3.2 basis points against the +0.4 the market charges, so the sign does turn at the bottom of the range, but by an amount an order of magnitude below the headline transfer; across the admissible bracket  $[0.631, 1]$  the premium is negative over 91% of its width.

The obligation scale of 4.0 that the last row implies is the point of the table. Section 6 argues that it is large against the one-year normalization precisely because the early cures are missing from the moment being matched, so the row bounds coverage from below rather than rivalling the model’s value.

TABLE OA45. The actuarially fair cosign premium as a function of the parental obligation

| Obligation scale                  | Parental coverage $\bar{p}$ | Fair premium (bp) |
|-----------------------------------|-----------------------------|-------------------|
| 1.0                               | 0.949                       | -27.3 to -17.1    |
| 1.5                               | 0.940                       | -26.4 to -16.5    |
| 2.0                               | 0.911                       | -23.7 to -14.8    |
| 2.5                               | 0.863                       | -19.0 to -11.9    |
| 3.0                               | 0.804                       | -13.4 to -8.4     |
| 4.0                               | 0.684                       | -2.0 to -1.2      |
| 4.5 <sup>†</sup>                  | 0.631                       | +2.0 to +3.2      |
| 5.0 <sup>†</sup>                  | 0.579                       | +5.1 to +8.1      |
| 6.0 <sup>†</sup>                  | 0.495                       | +10.1 to +16.1    |
| 7.0 <sup>†</sup>                  | 0.430                       | +14.0 to +22.4    |
| 8.0 <sup>†</sup>                  | 0.378                       | +17.1 to +27.4    |
| 10.0 <sup>†</sup>                 | 0.303                       | +21.6 to +34.6    |
| 12.0 <sup>†</sup>                 | 0.252                       | +24.6 to +39.4    |
| <i>Sign turns at</i>              |                             | $\bar{p} = 0.66$  |
| <i>Premium the market charges</i> |                             | +0.4              |

*Notes:* The fair premium nets the composition charge against the recovery credit; the counterfactual hazard cancels, leaving  $[d_{cos}(1 - \bar{p}) - d_{solo}] \ell$  — the terminal post-cure wedge priced at loss given default, the same object as  $U_5$ . Its sign therefore depends on  $\bar{p}$  alone and turns at 0.66: rows marked <sup>†</sup> have coverage below that value, where cosigned loans should be priced *above* solo. The obligation scale multiplies the parent’s obligation, one year of scheduled debt service at scale 1. Range across loss given default 0.25–0.40. The model’s own coverage is 0.95.

### OA7.7 The Two Single-Instrument Designs

Table OA46 isolates guarantee-aware pricing at the baseline deposit, with  $\lambda_m^{co}$  left at 0.90, and Table OA47 the deposit relief to  $\lambda_m^{co} = 0.975$  with the price schedule left exactly as it is. Both are differenced against the same baseline equilibrium as the headline design of Appendix E.1, solved once and reused, so re-solve drift nets out across the three tables as well as within them.

Table OA48 carries the cosigned flows and the split of pool risk under the headline design, on the same three solves as Table 7. The two Bennet rows say that the extra risk the reform puts into the cosigned pool is composition rather than behaviour at every elasticity, and that market clearing shifts the balance towards intensity as the access margin shrinks.

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TABLE OA46. Counterfactual: guarantee pricing at the baseline deposit — moments, the anatomy of the default wedge, and house size

|                                                                             | Baseline | Counterfactual | $\Delta$ |
|-----------------------------------------------------------------------------|----------|----------------|----------|
| <i>A. Aggregate moments</i>                                                 |          |                |          |
| Homeownership rate (%)                                                      | 38.40    | 38.54          | +0.14    |
| Default rate, solo (%)                                                      | 0.807    | 0.808          | +0.001   |
| Cosign share (%)                                                            | 4.840    | 6.044          | +1.204   |
| Cash-gift share (%)                                                         | 26.17    | 26.19          | +0.02    |
| Cosign income gradient                                                      | -0.012   | -0.080         | -0.068   |
| Cosign rate premium (pp)                                                    | 0.005    | -0.241         | -0.245   |
| <i>B. Default wedge, Bennet decomposition (pp)</i>                          |          |                |          |
| Observable-state composition                                                | -0.02    | 0.02           | +0.04    |
| Hidden-type composition                                                     | 0.28     | 0.33           | +0.05    |
| need-based adverse selection                                                | 1.48     | 1.56           | +0.09    |
| parental screening correction                                               | -1.20    | -1.23          | -0.03    |
| = Within- $(z, h)$ wedge, pre-cure                                          | 0.26     | 0.35           | +0.09    |
| Invocation margin (deterrence)                                              | 0.02     | -0.12          | -0.14    |
| = Wedge on the invocation basis                                             | 0.28     | 0.23           | -0.05    |
| <i>C. Selection primitives</i>                                              |          |                |          |
| $q_G$ , cosign pool                                                         | 0.721    | 0.713          | -0.008   |
| $q_G$ , solo pool                                                           | 0.733    | 0.733          | +0.000   |
| Typed hazard, Good (%)                                                      | 0.59     | 0.62           | +0.03    |
| Typed hazard, Bad (%)                                                       | 3.31     | 3.50           | +0.18    |
| Guarantee absorption (parental coverage)                                    | 0.949    | 0.948          | -0.002   |
| <i>D. Intensive margin: house size at origination (<math>j = 19</math>)</i> |          |                |          |
| $E[h \mid \text{buy}]$                                                      | 0.6061   | 0.6061         | +0.0000  |
| $E[h \mid \text{cosigned}]$                                                 | 0.6048   | 0.6047         | -0.0002  |
| $E[h \mid \text{not cosigned}]$                                             | 0.6062   | 0.6062         | +0.0001  |
| Share at top occupied rung                                                  | 0.000    | 0.000          | +0.000   |
| Buyer mass at $j = 19$                                                      | 0.4284   | 0.4293         | +0.0009  |

Notes: Paired solved equilibria, post-processed without a re-solve.  $\Delta$  is a level difference. Panel B is equation (27).

TABLE OA47. Counterfactual: deposit relief alone, pricing unchanged — moments, the anatomy of the default wedge, and house size

|                                                                             | Baseline | Counterfactual | $\Delta$ |
|-----------------------------------------------------------------------------|----------|----------------|----------|
| <i>A. Aggregate moments</i>                                                 |          |                |          |
| Homeownership rate (%)                                                      | 38.40    | 39.57          | +1.17    |
| Default rate, solo (%)                                                      | 0.807    | 0.806          | -0.001   |
| Cosign share (%)                                                            | 4.840    | 9.122          | +4.282   |
| Cash-gift share (%)                                                         | 26.17    | 26.20          | +0.03    |
| Cosign income gradient                                                      | -0.012   | -0.496         | -0.484   |
| Cosign rate premium (pp)                                                    | 0.005    | 0.047          | +0.043   |
| <i>B. Default wedge, Bennet decomposition (pp)</i>                          |          |                |          |
| Observable-state composition                                                | -0.02    | 0.13           | +0.14    |
| Hidden-type composition                                                     | 0.28     | 0.65           | +0.37    |
| need-based adverse selection                                                | 1.48     | 2.03           | +0.56    |
| parental screening correction                                               | -1.20    | -1.38          | -0.18    |
| = Within- $(z, h)$ wedge, pre-cure                                          | 0.26     | 0.78           | +0.52    |
| Invocation margin (deterrence)                                              | 0.02     | 0.66           | +0.65    |
| = Wedge on the invocation basis                                             | 0.28     | 1.44           | +1.16    |
| <i>C. Selection primitives</i>                                              |          |                |          |
| $q_G$ , cosign pool                                                         | 0.721    | 0.662          | -0.059   |
| $q_G$ , solo pool                                                           | 0.733    | 0.733          | +0.000   |
| Typed hazard, Good (%)                                                      | 0.59     | 0.77           | +0.18    |
| Typed hazard, Bad (%)                                                       | 3.31     | 4.44           | +1.12    |
| Guarantee absorption (parental coverage)                                    | 0.949    | 0.948          | -0.001   |
| <i>D. Intensive margin: house size at origination (<math>j = 19</math>)</i> |          |                |          |
| $E[h \mid \text{buy}]$                                                      | 0.6061   | 0.6061         | +0.0000  |
| $E[h \mid \text{cosigned}]$                                                 | 0.6048   | 0.6068         | +0.0019  |
| $E[h \mid \text{not cosigned}]$                                             | 0.6062   | 0.6061         | -0.0001  |
| Share at top occupied rung                                                  | 0.000    | 0.000          | +0.000   |
| Buyer mass at $j = 19$                                                      | 0.4284   | 0.4391         | +0.0107  |

Notes: Paired solved equilibria, post-processed without a re-solve.  $\Delta$  is a level difference. Panel B is equation (27).

TABLE OA48. Cosigned flows and the decomposition of pool risk, by supply elasticity

|                                                                 | Fixed<br>prices | Market clearing, by supply elasticity |                     |                     | Share<br>surviving |
|-----------------------------------------------------------------|-----------------|---------------------------------------|---------------------|---------------------|--------------------|
|                                                                 |                 | $\varepsilon = 4.0$                   | $\varepsilon = 2.2$ | $\varepsilon = 1.5$ |                    |
| Cosign volume                                                   | +0.287          | +0.270                                | +0.267              | +0.265              | 93%                |
| Cosign losses                                                   | +0.0139         | +0.0131                               | +0.0130             | +0.0129             | 93%                |
| <i>Bennet decomposition of the change in cosigned pool risk</i> |                 |                                       |                     |                     |                    |
| Default intensity (pp)                                          | +0.445          | +0.484                                | +0.516              | +0.530              |                    |
| Pool composition (pp)                                           | +0.896          | +0.840                                | +0.796              | +0.775              |                    |

*Notes:* The headline design against the estimated economy, on the same three solves as Table 7. The two Bennet rows sum to the change in cosigned pool risk reported there; they are levels, the decomposition being already a change.

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